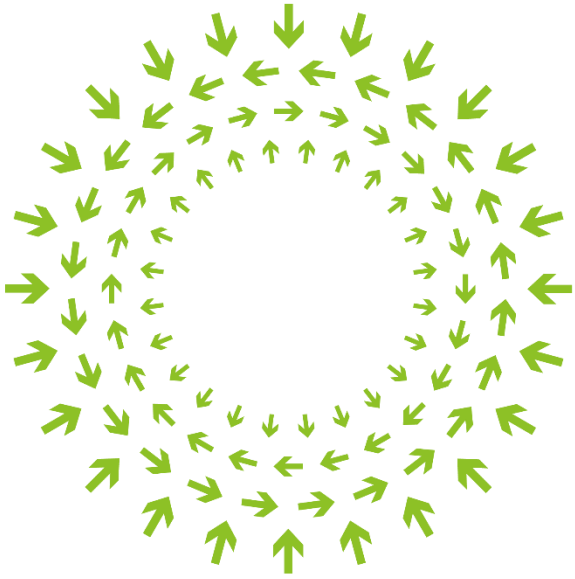


OPEN
Compute Project

Datacenter SAS-SATA Device Specification

Version 1.5 (062725)

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1 License OWF Option

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NONE

2 Overview

This document is to define the requirements for HDDs and SSDs for use in data centers.

The requirements in this document are in addition to the requirements in the underlying standards and may override some requirements in the underlying standards.

If functionality in an underlying standard (e.g., a command) is not covered in this document, then the requirements in the underlying standard apply. For example, if an optional command in an underlying standard is not covered in this document, then that command remains optional.

Customers may impose additional requirements beyond those specified in this document.

3 Scope

This document covers HDDs and SSDs using the SAS and SATA interface.

4 Common Interface Requirements

Requirement ID	Description
COMM-1	From the factory, the firmware revision field shall match the firmware revision on the label (see LABL-1).
COMM-2	The device shall have volatile write cache disabled by default from the factory.
COMM-3	The device shall have non-volatile (e.g., power loss protected) write cache enabled by default from the factory.

Requirement ID	Description
COMM-4	Device vendors shall support standard and vendor unique diagnostic information to be extracted for failure analysis and telemetry monitoring.
COMM-5	The device shall support firmware update without power cycling the device.
COMM-6	The device shall have default maximum I/O command error recovery time out of 6 seconds unless overridden by use of a CDL configuration in which case CDL timeouts shall be followed.
COMM-7	The device shall complete firmware activation within 10 seconds of receiving all the firmware download data.
COMM-8	Obsolete.
COMM-9	HDDs shall support head depopulation based on the appropriate standards.
COMM-10	Any optional features supported by the device not described in this document shall be clearly documented and disclosed to the customer.
COMM-11	Obsolete (see SEC-19).
COMM-12	The vendor shall provide a SAS-SATA compliance report depending on the device type.
COMM-13	The device shall report capacity values with IDEMA established guidelines for devices less than 8TB and SFF-8447 for 8TB and greater.
COMM-14	For HDDs, the vendor shall provide a single firmware binary that includes all internal components (e.g., both microcode and servo firmware).

4.1 SMR Requirements

If the device supports SMR, the following requirements shall be met.

Requirement ID	Description
SMR-1	SMR HDDs shall comply with ZAC-2 SATA / ZBC-2 SAS specifications, and the underlying ACS-6 SATA and SBC-5 SAS specifications.
SMR-2	SMR HDDs shall reserve 0.8% of total device capacity as CMR zone for journaling and logging purposes.
SMR-3	The device shall support a SMR zone size of at least 128MB and shall be a power of two.
SMR-4	Host Managed SAS devices shall always return the last LBA of the Conventional space after successfully processing the READ CAPACITY (16) command i.e., the RC_BASIS field shall always be 00b. SATA devices shall support similar functionality.
SMR-5	A Host Managed SATA devices processing a SECURITY ERASE UNIT command shall comply with the following: • If the ZAC SECURITY OPTION bit=0b, then all write pointer zones shall become EMPTY. • If the ZAC SECURITY OPTION bit=1b, the device shall ABORT the SECURITY ERASE UNIT command. SAS devices shall support similar functionality.

4.2 Device Profile

The following are device profiles. This section is intended to be firmware-based configuration settings configured by device suppliers when manufacturing a device. A device may be configured with a mix of A and/or B settings. Each customer shall provide their A/B preference for each configuration setting.

Requirement ID	Description	Configuration Setting	
		A	B
DP-CFG-1	Default Sector Size	4096 Native	512/512e
DP-CFG-2	TCG Storage Security Subsystem Class (SSC) (see Section 13.4 TCG Implementation Requirements).	Opal	Enterprise

4.3 Volatile Write Cache Enabled Behavior

This section specifies how the device shall behave with regards to Write Cache operations if it is enabled.

Requirement ID	Description
WCH-1	If write cache is enabled and the device has returned a write command completion status, then any cached data that has not yet been written to the medium shall be preserved through any "reset event" and written to the medium. "Reset event" refers to any HW/SW/LINK reset.
WCH-2	Any link associated protocol (valid or invalid), any disparity issue, any PHY operation issue (PHY reset), any elasticity buffer issue, any port mode or port mode operations change (power management or speed change), or any interface signal timing or amplitude problem shall not prevent cached write data of completed write commands from being written to the medium. Assuming power delivery is correct and has been properly maintained with the proper operational environmental conditions, ANY issue with deviations or changes shall not cause data loss or corruption when write caching is enabled.
WCH-3	The device shall flush cached write data to the medium prior to: • Acknowledging a power management command (STANDBY, SLEEP, etc.) that may cause the media to be inaccessible, or • Transitioning into any device maintenance or self-test routine either scheduled or non-scheduled (e.g., idle time functions such as head sweeps, BGMS, etc.), or • Transitioning into any power reduction state or mode, or • Returning successful completion on the disable write cache command.
WCH-4	The device is responsible for: • Invoking auto write reallocation in error recovery to successfully write cached data if needed. • Either temporarily or permanently disabling write-caching if current operating conditions indicate normal writes are at risk.
WCH-5	In the event an error occurs preventing the device from writing any cached data to the medium the device shall either return a deferred error or report a Device Fault error in the Status register on the next (earliest subsequent) command. SAS devices shall support similar functionality.

Requirement ID	Description
WCH-6	The write cache flushing commands shall take no more than 10 seconds at any conditions if write cache is enabled. The write cache flushing commands refer to FLUSH CACHE / FLUSH CACHE EXT for SATA device and SYNCHRONIZE CACHE for SAS device.

5 SATA Requirements

If the device supports SATA, the following requirements shall be met.

5.1 SATA and ATA Requirements

<table>
<tr><th>Requirement ID</th><th>Description</th></tr>
<tr><td>SATA-1</td><td>The device shall comply with Serial ATA Revision 3.5a or newer.</td></tr>
<tr><td>SATA-2</td><td>The device shall comply with ATA Command Set - 5 (ACS-5) or newer. Unless otherwise specified all commands and feature sets shall be based on the most recent version of ACS that defines the command or feature set.</td></tr>
<tr><td>SATA-3</td><td>The device shall follow the table below regarding support for ATA Feature Sets.
M = Mandatory. Command shall be supported.
P = Prohibited. Command shall not be supported (command rejected with ABRT).
ATA Feature Sets that are not listed and are optional in the ATA specifications (see SATA-1 and SATA-2) are optional.
<table>
<tr><th>ATA Feature Set</th><th>Support</th></tr>
<tr><td>48-bit Address feature set</td><td>M</td></tr>
<tr><td>Extended Power Conditions (EPC) feature set</td><td>M[1]</td></tr>
<tr><td>General feature set</td><td>M</td></tr>
<tr><td>General Purpose Logging (GPL) feature set</td><td>M</td></tr>
<tr><td>Native Command Queuing (NCQ) feature set</td><td>M[2]</td></tr>
<tr><td>Power Management feature set</td><td>M</td></tr>
<tr><td>Power-Up In Standby (PUIS) feature set</td><td>M[1]</td></tr>
<tr><td>Sanitize Device feature set</td><td>M[3]</td></tr>
<tr><td>Security feature set</td><td>M</td></tr>
<tr><td>Self-Monitoring, Analysis, and Reporting Technology (SMART) feature set (Include SMART self-test, SMART error logging)</td><td>M</td></tr>
<tr><td>Sense Data Reporting feature set</td><td>M[1]</td></tr>
<tr><td>Software Settings Preservation (SSP) feature set</td><td>M[2]</td></tr>
<tr><td>Trusted Computing feature set</td><td>M
(for devices that support TCG)
P</td></tr>
</table>
</td></tr>
</table>

<table>
<tr><th>Requirement ID</th><th>Description</th></tr>
<tr><td></td><td>
<table>
<tr><td></td><td>(for devices that do not support TCG)</td></tr>
<tr><td>Write-Read-Verify feature set</td><td>M[1]</td></tr>
<tr><td>Read look-ahead feature set</td><td>M[2]</td></tr>
<tr><td>Optional ATA device 28-bit commands</td><td>See SATA-16</td></tr>
<tr><td>NCQ priority information feature set</td><td>M</td></tr>
<tr><td>PHY Event Counters</td><td>M</td></tr>
<tr><td>Storage Element Depopulation</td><td>M</td></tr>
</table>
Notes:
[1]Shall be disabled by default.
[2]Shall be enabled by default.
[3]The SANITIZE FREEZE LOCK EXT and the SANITIZE ANTIFREEZE LOCK EXT commands of the Sanitize Device feature set shall both be supported.
</td></tr>
<tr><td>SATA-4</td><td>The device shall support a negotiated link speed of up to 6Gbps and shall support link speeds of 6Gbps, 3Gbps and 1.5 Gbps.</td></tr>
<tr><td>SATA-5</td><td>Obsolete (see SATA-RLE-3 and Section 7 Telemetry Requirements).</td></tr>
<tr><td>SATA-6</td><td>Obsolete (see SATA-RLE-1).</td></tr>
<tr><td>SATA-7</td><td>Obsolete (see SATA-RLE-3).</td></tr>
<tr><td>SATA-8</td><td>The device shall support receipt of host-initiated power management (HIPM) requests.</td></tr>
<tr><td>SATA-9</td><td>Host-initiated power management (HIPM) shall be disabled by default (see Section 5.4.5.1 Enable/Disable HIPM Feature Set on how to enable HIPM).</td></tr>
<tr><td>SATA-10</td><td>The device shall not support Host Automatic Partial to Slumber transitions.</td></tr>
<tr><td>SATA-11</td><td>The device shall not support device-initiated power management (DIPM).</td></tr>
<tr><td>SATA-12</td><td>The device shall not support Device Automatic Partial to Slumber transitions.</td></tr>
<tr><td>SATA-13</td><td>Obsolete (see GFF-3).</td></tr>
<tr><td>SATA-14</td><td>Obsolete (see Section 5.4.5.4 Maximum I/O Timeout Feature Set).</td></tr>
<tr><td>SATA-15</td><td>AC coupling is required at both physical transmitter and receiver interfaces.</td></tr>
<tr><td>SATA-16</td><td>The device shall follow the table below regarding support for optional T13 commands.

M = Mandatory. Command shall be supported.
P = Prohibited. Command shall not be supported (command rejected with ABRT).

All mandatory commands shall be enabled by default. ATA commands that are not listed and are optional in the ATA specifications (see SATA-1 and SATA-2) are optional.
<table>
<tr><th>Command</th><th>Command / Feature Code</th><th>Support</th></tr>
<tr><td>DATA SET MANAGEMENT</td><td>06h</td><td>M
(SSD only)</td></tr>
</table>
</td></tr>
</table>

Requirement ID	Description		
	DATA SET MANAGEMENT XL	07h	M (SSD only)
	DOWNLOAD MICROCODE	92h	M
	SLEEP	E6h	M
	FLUSH CACHE	E7h	M
	FLUSH CACHE EXT	EAh	M
	NOP	00h	M
	READ BUFFER	E4h	M
	READ DMA (w/retry)	C8h	M
	READ DMA EXT	25h	M
	READ FPDMA QUEUED	60h	M
	READ LOG DMA EXT	47h	M
	READ LOG EXT	2Fh	M
	READ SECTOR(S) (w/retry)	20h	M
	READ SECTOR(S) EXT	24h	M
	READ VERIFY SECTOR(S) (w/retry)	40h	M
	READ VERIFY SECTOR(S) EXT	42h	M
	ZERO EXT	63h/06h	M
	RECEIVE FPDMA QUEUED	65h	M
	REQUEST SENSE DATA EXT	0Bh	M
	BLOCK ERASE EXT	B4h/0012h	M (SSD only)
	CRYPTO SCRAMBLE EXT	B4h/0011h	M
	OVERWRITE EXT	B4h/0014h	M (HDD only)
	SANITIZE ANTIFREEZE LOCK EXT	B4h/0040h	M
	SANITIZE FREEZE LOCK EXT	B4h/0020h	M
	SANITIZE STATUS EXT	B4h/0000h	M
	SET DATE & TIME EXT	77h	M
	SET SECTOR CONFIGURATION EXT	B2h	M
	SMART EXECUTE OFF-LINE IMMEDIATE	B0h/D4h	M
	SMART READ LOG	B0h/D5h	M

Requirement ID	Description		
	SMART RETURN STATUS	B0h/DAh	M
	SMART WRITE LOG	B0h/D6h	M
	TRUSTED NON-DATA	5Bh	M (TCG only)
	TRUSTED RECEIVE	5Ch	M (TCG only)
	TRUSTED RECEIVE DMA	5Dh	M (TCG only)
	TRUSTED SEND	5Eh	M (TCG only)
	TRUSTED SEND DMA	5Fh	M (TCG only)
	WRITE BUFFER	E8h	M
	WRITE DMA (w/retry)	CAh	M
	WRITE DMA EXT	35h	M
	WRITE DMA FUA EXT	3Dh	M
	WRITE FPDMA QUEUED	61h	M
	WRITE LOG DMA EXT	57h	M
	WRITE GATHERED EXT (see ACS-6)	66h	M
	IDLE IMMEDIATE (include Unload feature)	E1h	M
	SCT COMMAND Set		M
	SCT subcommand	**SCT Action Code/Function Code**	**Support**
	SCT WRITE SAME Repeat write pattern in background	0002h/0001h	M
	SCT WRITE SAME Repeat write data block in background	0002h/0002h	M
	SCT WRITE SAME Repeat write pattern in foreground	0002h/0101h	M
	SCT WRITE SAME Repeat write data block in foreground	0002h/0102h	M
	SCT ERROR RECOVERY CONTROL Set Current Timer Value	0003h/0001h	M
	SCT ERROR RECOVERY CONTROL Return Current Timer Value	0003h/0002h	M

<table>
<tr><th>Requirement ID</th><th>Description</th></tr>
<tr><td></td><td>
<table>
<tr><td>SCT ERROR RECOVERY CONTROL
Set Power-on Timer Value</td><td>0003h/0003h</td><td>M</td></tr>
<tr><td>SCT ERROR RECOVERY CONTROL
Return Power-on Timer Value</td><td>0003h/0004h</td><td>M</td></tr>
<tr><td>SCT ERROR RECOVERY CONTROL
Restore Manufacturer’s Default Timer Value</td><td>0003h/0005h</td><td>M</td></tr>
<tr><td>SCT FEATURE CONTROL
Set state and options flags for a feature</td><td>0004h0001h</td><td>M</td></tr>
<tr><td>SCT FEATURE CONTROL
Return the current state of a feature</td><td>0004h/0002h</td><td>M</td></tr>
<tr><td>SCT FEATURE CONTROL
Return feature option flags</td><td>0004h/0003h</td><td>M</td></tr>
<tr><td>SCT DATA TABLE
Read HDA Temperature History Table</td><td>0005h/0001h</td><td>M</td></tr>
<tr><th>Command</th><th>Command / Feature Code</th><th>Support</th></tr>
<tr><td>WRITE SECTOR(S) (w/retry)</td><td>30h</td><td>M</td></tr>
<tr><td>WRITE SECTOR(S) EXT</td><td>34h</td><td>M</td></tr>
<tr><td>WRITE UNCORRECTABLE EXT</td><td>45h</td><td>M</td></tr>
<tr><td>CLOSE ZONE EXT</td><td>9Fh/01h</td><td>M
(SMR only)</td></tr>
<tr><td>FINISH ZONE EXT</td><td>9Fh/02h</td><td>M
(SMR only)</td></tr>
<tr><td>OPEN ZONE EXT</td><td>9Fh/03h</td><td>M
(SMR only)</td></tr>
<tr><td>REPORT ZONES EXT</td><td>4Ah/00h</td><td>M
(SMR only)</td></tr>
<tr><td>RESET WRITE POINTER EXT</td><td>9Fh/04h</td><td>M
(SMR only)</td></tr>
</table>
</td></tr>
<tr><td>SATA-17</td><td>
The device shall support the following features and default settings:
<table>
<tr><th>Feature</th><th>Default Setting</th></tr>
<tr><td>Spread Spectrum Clocking (SSC) receive from host</td><td>Enabled</td></tr>
<tr><td>Spread Spectrum Clocking (SSC) transmit to host at 6Gbps</td><td>Enabled</td></tr>
<tr><td>Spread Spectrum Clocking (SSC) transmit to host at 1.5Gbps or 3.0Gbps</td><td>Disabled</td></tr>
<tr><td>Pin 11 delay spin</td><td>Enabled for HDDs.
Shall be ignored by SSDs.</td></tr>
<tr><td>Pin 11 Device Activity Signal</td><td>Enabled</td></tr>
</table>
</td></tr>
</table>

<table>
<tr><th>Requirement ID</th><th>Description</th></tr>
<tr><td></td><td>
<table>
<tr><td>PHY Asynchronous Signal Recovery</td><td>Enabled</td></tr>
<tr><td>Hot-plug capable</td><td>Enabled</td></tr>
<tr><td>Off-line read scan</td><td>Enabled</td></tr>
<tr><td>SCT</td><td>Enabled</td></tr>
<tr><td>Segmented DOWNLOAD MICROCODE</td><td>Enabled</td></tr>
</table>
</td></tr>
<tr><td>SATA-18</td><td>Download Microcode shall support modes 03h, 0Eh, and 0Fh.</td></tr>
<tr><td>SATA-19</td><td>If the SSD device has Power Loss Protection (PLP) (e.g., battery back-up, capacitor bank, or super cap), then the SSD device shall periodically initiate tests of the PLP, even if power is never removed from the device. The PLP shall have sufficient charge to back up all user data and/or metadata in the cache. If it fails or degrades to a point where it cannot guarantee sufficient charge to back up all user data and/or metadata in the cache, the SSD device shall do the following: Flush all cached writes to NAND. Disable the volatile write cache if the volatile write cache is currently enabled. Generate a SMART trip (threshold exceeded) in response to a SMART RETURN STATUS command (Power Loss Protection Failure). Set WCE to 0b in IDENTIFY DEVICE data word 85, bit 5. Return an error, i.e., ABRT (ST:51h | ER:40h) to host attempts to enable write cache.</td></tr>
<tr><td>SATA-20</td><td>SSD devices shall support the DATA SET MANAGEMENT command with the Trim bit.
Every single LBA specified by the DATA SET MANAGEMENT command with the Trim bit set shall be trimmed (“deterministic zeroes data”). This requirement does not apply to HDDs.</td></tr>
<tr><td>SATA-21</td><td>HDD devices shall support the (Forced Unit Access) FUA bit in WRITE FPDMA QUEUED command. This requirement does not apply to SSDs.</td></tr>
<tr><td>SATA-22</td><td>During the processing of CHECK POWER MODE, IDENTIFY DEVICE, READ LOG EXT (applies to log addresses 03h, 04h, and E0h only), READ LOG DMA EXT (applies to log address 03h, 04h, and E0h only), SMART READ LOG and SMART RETURN STATUS commands, the device shall not exit Idle_B, Idle_C, or Standby Power Condition, and the device shall not reset the Idle_B or Idle_C timers.</td></tr>
<tr><td>SATA-23</td><td>Obsolete (see SATA-37).</td></tr>
<tr><td>SATA-24</td><td>The device shall complete the SET SECTOR CONFIGURATION EXT command within 30s.</td></tr>
<tr><td>SATA-25</td><td>After the SET SECTOR CONFIGURATION EXT command completes, the number of descriptors in the Pending Defects Log (log address 0x0Ch) shall be greater than or equal to the number of descriptors prior to the execution of this command.</td></tr>
<tr><td>SATA-26</td><td>Byte 200 of the SCT status response information shall contain the device’s temperature in degrees Celsius. Care shall be taken to report a value that best represents the internal temperature near the critical components (e.g., read/write interface elements in an HDD device). If the device has multiple temperature sensors, then a composite temperature shall be reported.</td></tr>
</table>

Requirement ID	Description
	Each time the SCT Command/Status log address E0h is read the contents of byte 200 shall be refreshed to reflect the current device temperature. The reported temperature shall be accurate to within ±3 °C from 0 to 70 °C.
SATA-27	When SMART Status is reported, bytes 214 – 215 of the SCT status response information shall contain the SMART status of the device as defined in ACS-5.
SATA-28	The sequential execution time of SCT WRITE SAME command across the entire capacity of the device shall be no greater than 105% of the time for sequential WRITE of the entire device. The reference sequential write time of the entire device shall be measured using a workload of 64kiB writes that are 64kiB aligned and with a queue depth of 32.
SATA-29	The RECOVERY TIME LIMIT field associated with the SCT ERROR RECOVERY CONTROL command shall be programmable via the SCT Error Recovery Control command and shall be set to a default of 0050h (8 seconds).
SATA-30	Reserved for future use.
SATA-31	Device Activity Signal (Pin 11) “LED blink” behavior shall meet the requirements specified in GFF-6 and GFF-7. After DC power is applied, HDD devices shall enable “staggered spin-up” regardless of interface Pin 11 signal level (see SATA-PHY-3). After PHYRDY, all devices shall control the Pin 11 signal (used for activity LED indication) as specified in GFF-6, GFF-7, and the following paragraph. For commands that return command complete without error before all actions of the command are complete (e.g., STANDBY IMMEDIATE, SMART EXECUTE OFF-LINE IMMEDIATE, IDLE IMMEDIATE) command processing continues and the LED blinks (see GFF-6) until all actions of the command are complete.
SATA-32	If the device supports a media pre-scan function that can delay write I/O commands in order to pre-scan media before the media is written, then this function shall be disabled by default.
SATA-33	Unless stated otherwise for specific log parameters, all log parameters that are non-volatile shall be saved at intervals no longer than 10 minutes and shall not impact performance (e.g., spread out saving of log parameters over time).
SATA-34	The device shall reorder NCQ commands for better performance. NCQ command priority specified by the PRIO field shall be implemented as defined in SATA 3.5a. Normal Priority commands may be executed when High Priority commands are in the queue if the Normal Priority commands are able to be completed without impacting the completion time of the High Priority commands.
SATA-35	Obsolete.
SATA-36	The device shall report support for MICROCODE DOWNLOAD subcommands 0Eh and 0Fh in Log Address 2Fh (IDENTIFY DEVICE Data), Page 03h (Supported Capabilities), DOWNLOAD MICROCODE Capabilities QWORD, bit 34 (DM OFFSETS DEFERRED SUPPORTED).
SATA-37	Log Address 13h SATA NCQ Send And Receive Default Settings HDD and SSD devices shall set the following bits: QUEUED READ LOG DMA EXT SUPPORTED

Requirement ID	Description
	SSD devices shall set the following bits: QUEUED DATA SET MANAGEMENT SUPPORTED QUEUED DATA SET MANAGEMENT SUPPORTS TRIM
SATA-38	The device shall not treat SMART trips as transient and shall continue to indicate a SMART trip, regardless of any subsequent change to the condition that caused the SMART trip. The device shall preserve any SMART trip across COMRESET, DEVICE RESET, and power cycles.
SATA-39	The device shall always enable SMART operations. The SMART commands to enable operations (B0h/D8h) and disable operations (B0h/D9h) are prohibited (see SATA-16).
SATA-40	For any new or remanufactured device received from a device vendor, the Extended SMART Self-Test log (Log Address 07h) shall not contain any entries.
SATA-41	Commands that place the device into a reduced power state shall cause all data in any write cache buffer to be written to the media before entering the reduced power state.
SATA-42	For SSD devices, all LBAs that have been cryptographically erased using CRYPTO SCRAMBLE EXT shall be unmapped (or trimmed) as a result of that operation.

5.1.1 SATA Power Conditions Log Requirements

<table>
<tr><th>Requirement ID</th><th>Description</th></tr>
<tr><td>SATA-PCL-1</td><td>Log Address 08h Power Conditions, Log Descriptor Default Settings for Idle_A:
<table>
<tr><th>Log Page</th><th>Byte Offset</th><th>Descriptor</th><th>Supported</th><th>Savable</th><th>Changeable</th><th>Default Timer Enabled</th><th>Saved Timer Enabled</th></tr>
<tr><td>00h</td><td>1</td><td>IDLE_A PWR CONDITION FLAGS</td><td>1b</td><td>0b</td><td>1b</td><td>0b</td><td>0b</td></tr>
</table></td></tr>
<tr><td>SATA-PCL-2</td><td>Log Address 08h Power Conditions, Log Descriptor Default Settings for Idle_B:
<table>
<tr><th>Log Page</th><th>Byte Offset</th><th>Descriptor</th><th>Supported</th><th>Savable</th><th>Changeable</th><th>Default Timer Enabled</th><th>Saved Timer Enabled</th></tr>
<tr><td>00h</td><td>65</td><td>IDLE_B PWR CONDITION FLAGS</td><td>1b</td><td>0b</td><td>1b</td><td>0b</td><td>0b</td></tr>
</table></td></tr>
<tr><td>SATA-PCL-3</td><td>Log Address 08h Power Conditions, Log Descriptor Default Settings for Idle_C:
<table>
<tr><th>Log Page</th><th>Byte Offset</th><th>Descriptor</th><th>Supported</th><th>Savable</th><th>Changeable</th><th>Default Timer Enabled</th><th>Saved Timer Enabled</th></tr>
<tr><td>00h</td><td>129</td><td>IDLE_C PWR CONDITION FLAGS</td><td>1b</td><td>0b</td><td>1b</td><td>0b</td><td>0b</td></tr>
</table></td></tr>
<tr><td>SATA-PCL-4</td><td>Log Address 08h Power Conditions, Log Descriptor Default Settings for Standby_Y:</td></tr>
</table>

<table>
<tr><th>Requirement ID</th><th>Description</th></tr>
<tr><td></td><td>
<table>
<tr><th>Log Page</th><th>Byte Offset</th><th>Descriptor</th><th>Supported</th><th>Savable</th><th>Changeable</th><th>Default Timer Enabled</th><th>Saved Timer Enabled</th></tr>
<tr><td>01h</td><td>385</td><td>STANDBY_Y PWR CONDITION FLAGS</td><td>0b</td><td>0b</td><td>0b</td><td>0b</td><td>0b</td></tr>
</table>
</td></tr>
<tr><td>SATA-PCL-5</td><td>Log Address 08h Power Conditions, Log Descriptor Default Settings for Standby_Z:
<table>
<tr><th>Log Page</th><th>Byte Offset</th><th>Descriptor</th><th>Supported</th><th>Savable</th><th>Changeable</th><th>Default Timer Enabled</th><th>Saved Timer Enabled</th></tr>
<tr><td>01h</td><td>449</td><td>STANDBY_Z PWR CONDITION FLAGS</td><td>1b</td><td>0b</td><td>1b</td><td>0b</td><td>0b</td></tr>
</table>
</td></tr>
</table>

5.2 SATA PHY Layer Requirements

Requirement ID	Description
SATA-PHY-1	The device shall include controls for setting driver strength, pre-emphasis amounts, and (if implemented) receiver equalization gains.
SATA-PHY-2	The HDD device shall complete spin-down within 10 seconds from: • Power removal / power loss. • Any command that causes the device to spin down from an active (spinning at speed) state.
SATA-PHY-3	The HDD device by default shall enable “staggered spin-up” regardless of pin 11 state (i.e., signal input level) after DC power is applied. The HDD device shall only spin-up after receiving a COMWAKE.
SATA-PHY-4	The device shall support all BIST modes (mandatory and optional) for loopback testing with SATA devices with proper support of the BIST Activate FIS bits F, L, T, A, S, and P.
SATA-PHY-5	All mandatory PHY Event Counter identifiers shall be supported as defined by Serial ATA Revision 3.5a, Section 13.9.3. The counter values shall not be retained across power cycles but shall be preserved across COMRESET and software reset events. PHY Event Counter identifiers 003h (R_ERRp response for Device-to-Host Data FIS), 004h (R_ERRp response for Host-to-Device Data FIS), 006h (R_ERRp response for Device-to-Host non-Data FIS), 00Ah (), 00Bh (CRC errors within a host-to-device FIS) and 00Dh (Non-CRC errors within a host-to-device FIS) shall be supported and shall reflect the current total number of instances of that particular event. All supported PHY Event Counter identifiers shall be reported regardless of the event count.
SATA-PHY-6	Obsolete.
SATA-PHY-7	A device that is transmitting a BIST test pattern shall continue transmitting that BIST test pattern after disconnection of the data cable.

Requirement ID	Description
SATA-PHY-8	The device shall not issue two COMINITs within 10 ms of each other.
SATA-PHY-9	The device shall be ready to participate in SATA Out of Band (OOB) signaling (e.g., COMRESET, COMINIT) within a maximum of 1 second after power is applied to the device. Refer to Serial ATA Revision 3.5a or newer.
SATA-PHY-10	An SSD device shall complete SATA PHY initialization from the initial power on reset, COMRESET, or COMINIT (whichever comes first) to transmitting the initial Register Device to Host FIS (that contains the device signature) in less than 1 second, including any link speed changes.

5.3 SATA LINK Layer Requirements

Requirement ID	Description
SATA-LINK-1	If the device receives an invalid Dword (including an R_ERR primitive or a character with bad disparity) inside a frame, it shall treat the frame (if any) as having a CRC error, regardless of whether the CRC is correct. The device shall terminate a WRITE command due to any link-associated error (CRC). The bad data associated with the command shall not be written to the media, and no reduction in normal device performance shall occur due to this type of device error-handling operation.
SATA-LINK-2	The device shall accept a DATA FIS whose data payload size is not a multiple of 512 bytes.
SATA-LINK-3	For device to host data transfers that require multiple DATA FIS, every DATA FIS shall be 2048 Dwords in size except for the last DATA FIS in the transfer.

5.4 SATA Command Layer Requirements

5.4.1 Identify Device (Command code: ECh)

Requirement ID	Description
SATA-ID-1	The device shall set the IDENTIFY DEVICE data words correctly to reflect all requirements listed in this document (e.g., supported features, default enabled/disabled, version/revision number, capacity values, etc.).
SATA-ID-2	Obsolete (see SATA-ID-8).
SATA-ID-3	The device shall be able to process an IDENTIFY DEVICE command at all times when power is applied and the initial FIS has been sent, even if the device is in any standby power management state (e.g., the HDD media is not spinning).
SATA-ID-4	Reserved.
SATA-ID-5	The device shall set the Sanitize command support bits in word 59 as follows: • Bit (15) = 1b for SSD devices (device supports BLOCK ERASE EXT). • Bit (14) = 1b for HDD devices (device supports OVERWRITE EXT). • Bit (13) = 1b (device supports CRYPTO SCRAMBLE EXT). • Bit (12) = 1b (device supports the Sanitize feature set). See SATA-16 for command support requirements.

Requirement ID	Description
SATA-ID-6	The device shall support Word 75: Queue Depth and shall report a Queue Depth of 32.
SATA-ID-7	The device shall support Word 76: Serial ATA Capabilities as follows: • Bit (10) = 1b (Supports the SATA Phy Event Counters log).
SATA-ID-8	The device shall support the following fields in Word 77: Serial ATA Additional Capabilities: • Bits (3:1) Current Negotiated Serial ATA signal speed. • Bit 6 = 1b (device supports RECEIVE FPDMA QUEUED and SEND FPDMA QUEUED commands).
SATA-ID-9	The device shall support Word 78: Serial ATA Features Supported.
SATA-ID-10	The device shall populate words 100–103 with the total number of user addressable sectors, regardless of the setting of IDENTIFY DEVICE data word 69 bit 3.
SATA-ID-11	The device shall support all fields in Word 106: Physical Sector Size / Logical Sector Size.
SATA-ID-12	The device shall populate Words 108–111 with a unique worldwide name that is based on an officially obtained Organization Unique Identifier (OUI). Worldwide names shall not be duplicated across devices.
SATA-ID-13	The device shall support Words 117-118: Logical Sector Size.
SATA-ID-14	The device shall support Word 119 and shall populate it as follows: • Bits (15:14) = 01b (device supports Word 119). • Bit (7) = 1b (device supports Extended Power Conditions (EPC)) (HDD). • Bit (4) = 1b (device supports Download Microcode mode 3). • Bit (3) = 1b (device supports READ LOG DMA EXT and WRITE LOG DMA EXT commands). • Bit (2) = 1b (device supports WRITE UNCORRECTABLE EXT command).
SATA-ID-15	The device shall support Word 120 and shall populate it as follows: • Bits (15:14) = 01b (device supports Word 120). • Bit (7) = 0b (Extended Power Conditions (EPC) is not enabled) (HDD). • Bit (4) = 1b (device supports Download Microcode mode 3). • Bit (3) = 1b (device supports READ LOG DMA EXT and WRITE LOG DMA EXT commands). • Bit (2) = 1b (device supports WRITE UNCORRECTABLE EXT command).
SATA-ID-16	The device shall support and populate these bits in Word 168: • Bits (3:0) - Device Nominal Form Factor.
SATA-ID-17	The device shall populate Words (173:170) with “OCP” followed by five ASCII space characters (20h) and shall replace that with a customer-provided identification string if the customer provides a string.
SATA-ID-18	The device shall support Word 209. For devices that implement 512-byte sector size emulation, this word shall be set to a value of 4000h, “Logical LBA 0 is aligned to the beginning of the first physical sector.”
SATA-ID-19	The device shall report the nominal media rotation rate in Word 217 of the IDENTIFY DEVICE data as described in ACS-4, using the appropriate value (e.g., 7200 rpm = 1C20h, SSD = 0001h).

Requirement ID	Description
SATA-ID-20	The device shall support Words (233:230): Extended Number of User Addressable Sectors. If IDENTIFY data Word 69 bit 3 is set to 1b, then the content of these words shall be identical to that of Words 103:100. If IDENTIFY data Word 69 bit 3 is cleared to 0b these Words are reserved.

5.4.2 Read Log Ext (Command code: 2Fh)

<table>
<tr><th>Requirement ID</th><th>Description</th></tr>
<tr><td>SATA-RLE-1</td><td>
The device shall support Logs and Pages via the Read Log Ext command and the Read Log Ext DMA command per the table below.

M = Mandatory. Log/Page shall be supported.
P = Prohibited. Log/Page shall not be supported (command rejected with ABRT).

ATA logs that are not listed and are optional in the ATA specifications (see SATA-1 and SATA-2) are optional.
<table>
<tr><th>Log Address</th><th>Support</th><th>Logs and Pages</th></tr>
<tr><td>00h</td><td>M</td><td>Log directory</td></tr>
<tr><td>03h</td><td>M</td><td>Extended Comprehensive SMART error log</td></tr>
<tr><td>04h</td><td>M</td><td>Device Statistics logs. See SATA-RLE-3 for specific log page support requirements.</td></tr>
<tr><td>07h</td><td>M</td><td>Extended SMART self-test log</td></tr>
<tr><td>08h</td><td>M</td><td>Extended Power Conditions</td></tr>
<tr><td>0Ah</td><td>M</td><td>Device Statistics Notification</td></tr>
<tr><td>0Ch</td><td>M</td><td>Pending Defects log</td></tr>
<tr><td>0Fh</td><td>M</td><td>Sense Data for Successful NCQ Commands log</td></tr>
<tr><td>10h</td><td>M</td><td>Queued error log. See Serial ATA Revision 3.5a or newer.</td></tr>
<tr><td>11h</td><td>M</td><td>SATA PHY Event Counters log</td></tr>
<tr><td>13h</td><td>M</td><td>SATA NCQ Send and Receive log</td></tr>
<tr><td>19h</td><td>M</td><td>LBA Status</td></tr>
<tr><td>24h</td><td>M[1]</td><td>Current Device Internal Status Data Log</td></tr>
<tr><td>25h</td><td>M[1]</td><td>Saved Device Internal Status Data log</td></tr>
<tr><td>2Fh</td><td>M</td><td>Set Sector Configuration</td></tr>
<tr><td>30h</td><td>M</td><td>IDENTIFY DEVICE data (log pages 00h – 06h, and 08h)</td></tr>
<tr><td>E0h</td><td>M</td><td>SCT Command/Status</td></tr>
<tr><td>E1h</td><td>M</td><td>SCT Data Transfer</td></tr>
</table>
Notes:
[1]The format of these logs shall follow the telemetry description in Section 7 Telemetry Requirements of this specification.
</td></tr>
</table>

<table>
<tr><th>Requirement ID</th><th>Description</th></tr>
<tr><td>SATA-RLE-2</td><td>SATA devices shall return sense data 05/24/00 (INVALID FIELD IN CDB) for unsupported log addresses.</td></tr>
<tr><td>SATA-RLE-3</td><td>The device shall support the following log pages in the Device Statistics Log (log address 04h, see SATA-RLE-1). Any log page that is not listed and is optional in the ATA specifications (see SATA-1 and SATA-2) is optional.
<table>
<tr><th>Log Page</th><th>Description</th></tr>
<tr><td>00h</td><td>List of supported log pages</td></tr>
<tr><td>01h</td><td>General Statistics</td></tr>
<tr><td>03h</td><td>Rotating Media Statistics[1]</td></tr>
<tr><td>04h</td><td>General Error Statistics</td></tr>
<tr><td>05h</td><td>Temperature Statistics</td></tr>
<tr><td>06h</td><td>Transport Statistics</td></tr>
<tr><td>07h</td><td>Solid State Device Statistics[2]</td></tr>
<tr><td>09h</td><td>Command Duration Limits Statistics 0/1[1] (HDD only)</td></tr>
<tr><td>0Ah</td><td>Command Duration Limits Statistics 2/3[1] (HDD only)</td></tr>
</table>
Notes:
[1]Not required for SSD.
[2]Not required for HDD.</td></tr>
</table>

5.4.3 Request Sense Data Ext (Command Code: 0Bh)

<table>
<tr><th>Requirement ID</th><th>Description</th></tr>
<tr><td>SATA-RSDE-1</td><td>This 48-bit command allows the host to obtain the most recent sense data from the device if the Sense Data reporting feature set is supported and sense data reporting is enabled via SET FEATURES command. The sense key, additional sense code, and additional sense code qualifier fields shall be set to values defined in ACS-5 or newer, if any, with the extensions shown in the table below. Otherwise, the sense key, additional sense code, and additional sense code qualifier shall be cleared to zero.
<table>
<tr><th>Condition</th><th>Required Sense Data</th></tr>
<tr><td>Supported command attempting to access unsupported log address or log page.</td><td>05/24/00 (ILLEGAL REQUEST- INVALID FIELD IN CDB)</td></tr>
<tr><td>ATA NOP command w/ subcommand 00h (Return command aborted) .</td><td>0B/00/00 (ABORTED COMMAND-NO ADDITIONAL SENSE INFORMATION)</td></tr>
<tr><td>Supported command with invalid fields (obsolete, reserved values, etc.) .</td><td>05/24/00 (ILLEGAL REQUEST- INVALID FIELD IN CDB)</td></tr>
</table></td></tr>
</table>

5.4.4 Sanitize Device Commands

Sanitize operations are initiated by some Sanitize feature set commands (see SATA-SDC-1) and by some TCG methods (see SEC-52).

<table>
<tr><th>Requirement ID</th><th>Description</th></tr>
<tr><td>SATA-SDC-1</td><td>The device shall support the Sanitize feature set commands in the table below:
<table>
<tr><th>Feature</th><th>Command</th></tr>
<tr><td>0000h</td><td>SANITIZE STATUS EXT</td></tr>
<tr><td>0011h</td><td>CRYPTO SCRAMBLE EXT</td></tr>
<tr><td>0012h</td><td>BLOCK ERASE EXT (SSD only)</td></tr>
<tr><td>0014h</td><td>OVERWRITE EXT (HDD only)</td></tr>
<tr><td>0020h</td><td>SANITIZE FREEZE LOCK EXT</td></tr>
<tr><td>0040h</td><td>SANITIZE ANTIFREEZE LOCK EXT</td></tr>
</table>
The CRYPTO SCRAMBLE EXT, BLOCK ERASE EXT, and OVERWRITE EXT commands initiate Sanitize operations.</td></tr>
<tr><td>SATA-SDC-2</td><td>The device shall service READ LOG EXT, READ LOG DMA EXT, and SMART READ LOG commands during Sanitize operations.</td></tr>
<tr><td>SATA-SDC-3</td><td>The device shall be able to provide any requested log page while a Sanitize operation is under way.</td></tr>
<tr><td>SATA-SDC-4</td><td>Specifically, if a Sanitize operation is in progress and a READ LOG EXT command is received requesting the SCT Command/Status log (E0h) log, the device shall not return an error, but rather, shall service the command by returning all current content for the log page. All the content of the page shall be available and be current, but most critical to the host application, is the current most recent temperature reading (reported in byte 200 (HDA Temp (Current device HDA temperature in deg. C) retrieved by the device’s temperature sensor at the time the command is received from the host. If the device has multiple temperature sensors, then the reported temperature reading shall be a composite temperature.</td></tr>
<tr><td>SATA-SDC-5</td><td>After a Sanitize operation is completed, the device shall enable reads to the sanitized LBAs without generating errors. After any Sanitize operation (e.g., performed by the CRYPTO SCRAMBLE EXT command), an SSD device shall return all zeros for host reads to LBAs that have not been written since the last Sanitize command, and indeterminate data values may be returned by an HDD device for host reads to LBAs that have not been written since the last Sanitize command.

Reading from an LBA (before writing to) that has been cryptographically erased shall not take longer than the time required to read that LBA before it had been cryptographically erased, and the read shall not terminate with an abort condition due to being cryptographically erased.</td></tr>
<tr><td>SATA-SDC-6</td><td>For an HDD device, OVERWRITE EXT command execution time shall be no greater than 105% of the time for sequential WRITE of the entire device. The reference sequential write time of the entire device shall be measured using a workload of 64K writes with queue depth 32.</td></tr>
<tr><td>SATA-SDC-7</td><td>The device’s non-volatile cache shall be included in the scope of data that is sanitized by any Sanitize operation.</td></tr>
<tr><td>SATA-SDC-8</td><td>The CRYPTO SCRAMBLE EXT command shall complete in at most 5 seconds.</td></tr>
</table>

Requirement ID	Description
SATA-SDC-9	If a power loss occurs during any Sanitize operation (see SATA-SDC-1), the operation shall resume after power is restored until the operation completes.
SATA-SDC-10	The results of all Sanitize operations shall meet IEEE Std 2883-2022 Purge requirements.

5.4.5 Set Features (Command code: EFh)

<table>
<tr><th>Requirement ID</th><th>Description</th></tr>
<tr><td>SATA-SF-1</td><td>The device shall support SET FEATURES subcommand (Feature register) values 10h and 90h to facilitate the enabling or disabling of the feature items in the table below.
M = Mandatory. Feature shall be supported.
P = Prohibited Feature shall not be supported (command rejected with ABRT).
Subcommand items that are not listed and are optional in the ATA specifications (see SATA-1 and SATA-2) are optional.
<table>
<tr><th>Subcommand</th><th>Feature</th><th>Support</th></tr>
<tr><td>02h</td><td>DMA Setup FIS Auto-Activate optimization</td><td>M[1]</td></tr>
<tr><td>06h</td><td>Software Settings Preservation</td><td>M</td></tr>
</table>
Notes:
[1]Shall be disabled by default.</td></tr>
<tr><td>SATA-SF-2</td><td>IDENTIFY DEVICE data words 89 and 90 shall report erase time and NOT be affected by this SET FEATURE protocol. In other words, these particular words, 89 and 90, for reporting Normal Erase time and Enhanced Erase time, shall always report the erase time regardless of the activation/deactivation status of this feature.</td></tr>
<tr><td>SATA-SF-3</td><td>The device shall support the subcommand codes for the Feature register of the command Set Feature task file per the table below.
M = Mandatory. Subcommand code shall be supported.
P = Prohibited. Subcommand code shall not be supported (i.e., the command shall be rejected with ABRT).
Subcommand items that are not listed and are optional in the ATA specifications (see SATA-1 and SATA-2) are optional.
<table>
<tr><th>Subcommand</th><th>Feature Description</th><th>Support</th></tr>
<tr><td>02h</td><td>Enable write cache</td><td>M</td></tr>
<tr><td>03h</td><td>Set transfer mode based on value in Sector Count register</td><td>M</td></tr>
<tr><td>06h</td><td>Enable Power-Up In Standby (PUIS) feature set</td><td>M</td></tr>
<tr><td>07h</td><td>Power-Up In Standby feature set device spin-up</td><td>M</td></tr>
<tr><td>10h</td><td>Enable use of Serial ATA feature</td><td>M</td></tr>
<tr><td>4Ah</td><td>Extended Power Conditions</td><td>M (HDD only)</td></tr>
<tr><td>55h</td><td>Disable read look-ahead feature</td><td>M</td></tr>
<tr><td>66h</td><td>Disable reverting to power-on defaults</td><td>M</td></tr>
<tr><td>82h</td><td>Disable volatile write cache</td><td>M</td></tr>
</table></td></tr>
</table>

<table>
<tr><th>Requirement ID</th><th>Description</th></tr>
<tr><td></td><td>
<table>
<tr><td>86h</td><td>Disable Power-Up In Standby feature set</td><td>M</td></tr>
<tr><td>90h</td><td>Disable use of Serial ATA feature</td><td>M</td></tr>
<tr><td>AAh</td><td>Enable read look-ahead feature</td><td>M</td></tr>
<tr><td>C3h</td><td>Sense Data Reporting feature set (enable = 1h in Count field; disable = 0h in Count field)</td><td>M</td></tr>
<tr><td>CCh</td><td>Enable reverting to power-on defaults</td><td>M</td></tr>
</table>
</td></tr>
<tr><td>SATA-SF-4</td><td>Subcommands 06h (Enable Power-Up In Standby (PUIS) feature set), 07h (Power-Up In Standby feature set device spin-up), and 86h (Disable Power-Up In Standby feature set) shall be disabled by default from the factory.</td></tr>
<tr><td>SATA-SF-5</td><td>All Extended Power Conditions (EPC) timers shall be disabled by default from the factory. When the Extended Power Conditions (EPC) timers are disabled, the device shall not transition to different power states based on any EPC timer expiration.</td></tr>
<tr><td>SATA-SF-6</td><td>The device shall only use the SET FEATURES command subcommand code 26h for features defined by this specification.</td></tr>
</table>

5.4.5.1 Enable/Disable HIPM Feature Set

<table>
<tr><th>Requirement ID</th><th>Description</th></tr>
<tr><td>SATA-SF-HIPM-1</td><td>The device shall support the SET FEATURE Subcommand code 26h/01h to allow the host to enable or disable Host Initiated Power Management (HIPM) as follows:
<table>
<tr><th>Field</th><th>Bit Description</th></tr>
<tr><td>COUNT</td><td>01h</td></tr>
<tr><td>LBA (Bits 7:0)</td><td>If set 01h the device shall enable HIPM. If cleared to 00h the device shall disable HIPM.</td></tr>
</table>
</td></tr>
</table>

5.4.5.2 Enable/Disable SSC Feature Set

<table>
<tr><th>Requirement ID</th><th>Description</th></tr>
<tr><td>SATA-SF-SSC-1</td><td>The device shall support SET FEATURE Subcommand code 26h/02h to allow the host to enable or disable Spread Spectrum Clocking (SSC) as follows:
<table>
<tr><th>Field</th><th>Bit Description</th></tr>
<tr><td>COUNT</td><td>02h</td></tr>
<tr><td>LBA (Bit 2)</td><td>If cleared to 0b, then the device shall disable Spread Spectrum Clocking (SSC) transmit to host at 1.5Gbps or 3.0Gbps. If set to 1b, then the device shall enable Spread Spectrum Clocking (SSC) transmit to host at 1.5Gbps or 3.0Gbps.</td></tr>
<tr><td>LBA (Bit 1)</td><td>If cleared to 0b, then the device shall disable Spread Spectrum Clocking (SSC) transmit to host at 6Gbps. If set to 1b, then the device shall enable Spread Spectrum Clocking (SSC) transmit to host at 6Gbps.</td></tr>
</table>
</td></tr>
</table>

Requirement ID	Description
	<table><tr><td>LBA (Bit 0)</td><td>If cleared to 0b, then the device shall disable Spread Spectrum Clocking (SSC) receive from host. If set to 1b, then the device shall enable Spread Spectrum Clocking (SSC) receive from host.</td></tr></table> Upon completion of this command, the device shall drop the link and bring the link back up with the new setting.

5.4.5.3 Enable/Disable IEEE1667

Requirement ID	Description
SATA-SF-1667-1	The device shall support SET FEATURE Subcommand code 26h/03h to allow the host to enable or disable the IEEE1667 silo as follows: <table><tr><th>Field</th><th>Bit Description</th></tr><tr><td>COUNT</td><td>03h</td></tr><tr><td>LBA (Bits 7:0)</td><td>If set 01h the device shall enable IEEE1667. If cleared to 00h the device shall disable IEEE1667.</td></tr></table>

5.4.5.4 Maximum I/O Timeout Feature Set

Requirement ID	Description
SATA-SF-MTO-1	The device shall support the SET FEATURE Subcommand code 26h/04h to allow the host to modify the maximum I/O command error recovery timeout as follows: <table><tr><th>Field</th><th>Bit Description</th></tr><tr><td>COUNT</td><td>04h</td></tr><tr><td>LBA (Bits 7:0)</td><td>Command timeout in seconds. A value of 00h disables the command timeout. The value shall be preserved across power cycles and resets. Supplier shall document the minimum timeout value supported by warranty in the device specification and on the device label (see LABL-22).</td></tr></table>

5.4.6 Sleep (Command code: E6h)

Requirement ID	Description
SATA-SLP-1	Because host systems may not read the Status register and clear the Interrupt Pending, a device may release INTRQ and enter Sleep mode after a vendor specific time period.

5.4.7 Standby (Command code: E2h)

Requirement ID	Description
SATA-STB-1	An SSD device without rotating media shall enter the STANDBY state by transitioning to the low power state and suspending all background activities.

5.4.8 SMART commands (Command code: B0h)

The following requirements apply to supported SMART commands. See SATA-16 for SMART command support requirements.

<table>
<tr><th>Requirement ID</th><th>Description</th></tr>
<tr><td>SATA-SMART-1</td><td>The SMART READ LOG (D5h subcommand), and SMART WRITE LOG (D6h subcommand) commands shall support the same number of logs.</td></tr>
<tr><td>SATA-SMART-2</td><td>The SMART READ LOG (D5h subcommand), and SMART RETURN STATUS (DAh subcommand) commands shall not cause the device to transition out of an Idle_B, Idle_C, or Standby power condition.</td></tr>
<tr><td>SATA-SMART-3</td><td>The device shall support the following self-test actions of the SMART EXECUTE OFF-LINE IMMEDIATE (D4h subcommand) command that are selected by the specified LBA low value for that command:
<table>
<tr><th>Self-Test Action</th><th>LBA Low Value</th></tr>
<tr><td>Execute SMART Short self-test routine (off-line)</td><td>01h</td></tr>
<tr><td>Execute SMART Extended self-test routine (off-line)</td><td>02h</td></tr>
<tr><td>Abort off-line routine</td><td>7Fh</td></tr>
<tr><td>Execute SMART Short self-test routine (captive)</td><td>81h</td></tr>
<tr><td>Execute SMART Extended self-test routine (captive)</td><td>82h</td></tr>
</table></td></tr>
<tr><td>SATA-SMART-4</td><td>The device shall complete any short self-test routine in less than 2 minutes.</td></tr>
<tr><td>SATA-SMART-5</td><td>An SSD device shall complete any extended self-test routine in less than 30 minutes.</td></tr>
<tr><td>SATA-SMART-6</td><td>An HDD device shall complete any extended self-test routine in less than 90 minutes.

If an HDD device needs to skip testing some tracks to meet this requirement, the tracks that are skipped should not be adjacent.</td></tr>
<tr><td>SATA-SMART-7</td><td>An Abort off-line routine action (7Fh LBA Low Value, see SATA-SMART-3) shall abort any self-test in progress.</td></tr>
<tr><td>SATA-SMART-8</td><td>Occurrence of a COMRESET, soft reset, or power cycle event shall abort any short self-test in progress (see SATA-SMART-3) and shall set the status byte for that self-test to 02h.</td></tr>
<tr><td>SATA-SMART-9</td><td>Occurrence of a COMRESET, soft reset, or power cycle event shall not abort any extended self-test in progress (see SATA-SMART-3). The extended self-test shall resume after the event and shall not consume more than 5 minutes of additional execution time to complete as a result of the event.</td></tr>
</table>

5.4.9 Write DMA FUA Ext (Command code: 3Dh)

Requirement ID	Description
SATA-WDFE-1	An SSD device shall cache FUA commands if the Power Loss Protection (PLP) is healthy.

5.4.10 Write Uncorrectable Ext (Command code: 45h)

Requirement ID	Description
SATA-WUE-1	There shall be no limit imposed by the device relative to the number of uncorrectable errors created by the WRITE UNCORRECTABLE EXT command with all supported modes other than the limit imposed by the total device capacity limit. The device shall support WRITE UNCORRECTABLE EXT modes 55h and AAh and be fully functional as defined in ACS-5. It is required that device suppliers validate that the performance of this command complies with the intended command functional result. This shall be validated with every supported mode. The WRITE UNCORRECTABLE EXT command with the Feature field value set to AAh (flagged error) and Count field set to one shall corrupt only one LBA, regardless of number of LBAs per physical sector.

6 SAS Requirements

If the device supports SAS, the following requirements shall be met.

6.1 SAS Requirements

Requirement ID	Description
SAS-1	The device shall comply with the following SCSI specifications: • SCSI primary commands (SPC-5) or newer. • SCSI block commands (SBC-4) or newer. • SAS Protocol Layer (SPL-4) or newer (e.g., SPL-5). • SCSI Architecture Model (SAM-5) or newer. • Serial Attached SCSI (SAS-3) or newer.
SAS-2	At a minimum, HDD devices shall support a SAS3 (12G) interface speed.
SAS-3	Enterprise SAS SSD devices shall support a SAS4 (24G) interface speed. Value SAS SSD devices shall support, at a minimum, a SAS3 (12G) interface speed.
SAS-4	A dual port device shall enable both ports by default, however, for devices that only have single port requirements the secondary SAS port shall be disabled by default to reduce power draw.
SAS-5	Devices shall support sense code reporting, including but not limited to: • 00/0B/xx DST / BGMS related warnings. • 00/5D/xx SMART warnings / thresholds exceeded. • 00/5E/xx Power transitions. • 01/xx/xx Soft errors & faults. • 02/xx/xx Device not ready. • 03/xx/xx Medium errors (Read errors & faults). • 04/xx/xx Miscellaneous failures. • 05/xx/xx Device access/invalid command/illegal request type failures

Requirement ID	Description
	• 06/xx/xx Unit attentions. • 07/xx/xx SMR / Zone related. • 0B/xx/xx Aborted commands.
SAS-6	Telemetry commands involving read activity only shall not block host I/O for more than 100 ms.
SAS-7	Download protection shall be built into the device to recover from an incorrect microcode version being downloaded and/or power fail during download.
SAS-8	The device shall support segmented downloads with segment sizes of 32K and 64K. All other segment sizes are optional. The device shall also support Buffer ID = 0 and use of the Buffer Offset field to segment data.
SAS-9	The device shall support self-test short and extended (long) tests in both foreground (captive) and background modes. When running the short self-test, the device shall complete the test within 2 minutes with and without outstanding I/O workload. When running extended self-test, the device shall take no more than 30 minutes for an SSD device and 90 minutes for an HDD device (see SAS-DST-1). If an HDD device needs to skip testing some tracks to meet the 90-minute maximum time requirement, the tracks that are skipped should not be adjacent.
SAS-10	Device Activity Signal (Pin 11) “LED blink” behavior shall meet the requirements specified in GFF-6 and GFF-7. After power application and prior to SAS PHY_Ready, the device shall control the interface Pin 11 signal (used for activity LED indication) to the inactive state (logic “1” level). After the PHY_Ready state is attained, the device shall control the Pin 11 signal as specified in GFF-6, GFF-7 and the following paragraph. For commands that return GOOD status as soon as the CDB is validated (e.g., FORMAT UNIT or START STOP UNIT with the IMMED bit set), command processing continues and the LED blinks (see GFF-6) until all actions of the command are complete.
SAS-11	All SAS (SCSI) defined data structures (e.g., log pages) shall be in big endian representation.
SAS-12	SMART trips are failure prediction events that are reported or logged with the ASC set to 5Dh. The device shall not treat SMART trips as transient and shall continue to indicate a SMART trip, regardless of any subsequent change to the condition that caused the SMART trip. The device shall preserve any SMART trip across I_T nexus loss, logical unit reset, hard reset, and power cycles. The device shall not report any temperature excursion events as SMART trips. See SAS-13 for temperature related warnings.
SAS-13	Warnings (i.e., events that are reported or logged with the ASC set to 0Bh) shall be reported only while the condition generating the warning exists. Once the condition ceases, the device shall discontinue reporting the warning. The SPECIFIED TEMPERATURE EXCEEDED exception condition shall be handled in this manner by the device.

Requirement ID	Description
SAS-14	The device shall support a queue depth of 128 commands and shall guarantee a minimum queue depth of 1 to each initiator regardless of how many commands are in queue from other initiators. The 128-command queue depth may be reduced by 1 for each initiator connected to the device (e.g., if three initiators are connected to the device, the device shall support a queue depth of 125 commands).
SAS-15	The device shall support the hashed address checking in SSP frame headers and duplicate tag checking.

6.2 SAS PHY Layer Requirements

Requirement ID	Description
SAS-PHY-1	If the device detects a character with bad disparity, it shall count the Dword containing the error in the Protocol Specific Port log page (18h) and shall treat the frame (if any) as having a CRC error, regardless of whether the CRC is correct.
SAS-PHY-2	The device shall not issue two COMINITs within 10 ms of each other.

6.3 SAS LINK Layer Requirements

Requirement ID	Description
SAS-LINK-1	If the device receives an invalid Dword (including an ERROR primitive or a character with bad disparity) inside a frame, it shall treat the frame (if any) as having a CRC error, regardless of whether the CRC is correct.
SAS-LINK-2	The BROADCAST (ASYNCHRONOUS EVENT) primitive shall be supported. It is extremely important for multi-initiator applications.
SAS-LINK-3	The device shall grant a minimum of two frame credits during typical operation.

6.4 SAS Command Layer Requirements

6.4.1 Supported Commands

<table>
<tr><th>Requirement ID</th><th>Description</th></tr>
<tr><td>SAS-CMD-1</td><td>The device shall support the following Commands:
M = Mandatory. Command shall be supported.
P = Prohibited. Command shall not be supported (command rejected with 05h/20h/00h or 05h/24h/00h CHECK CONDITION).
Commands that are not listed and are optional in the SAS specifications (see SAS-1) are optional.
<table>
<tr><th rowspan="2">Command</th><th rowspan="2">Opcode</th><th colspan="2">Support</th></tr>
<tr><th>HDD</th><th>SSD</th></tr>
<tr><td>FORMAT UNIT</td><td>04h</td><td>M</td><td>M</td></tr>
<tr><td>INQUIRY</td><td>12h</td><td>M</td><td>M</td></tr>
</table>
</td></tr>
</table>

LOG SELECT	4Ch	M	M
LOG SENSE	4Dh	M	M
MODE SELECT (6)	15h	M	M
MODE SELECT (10)	55h	M	M
MODE SENSE (6)	1Ah	M	M
MODE SENSE (10)	5Ah	M	M
ORWRITE (16)	8Bh	P	P
ORWRITE (32)	7Fh/000Eh	P	P
PERSISTENT RESERVE IN	5Eh	M	M
PERSISTENT RESERVE OUT	5Fh	M	M
READ (10)	28h	M	M
READ (16)	88h	M	M
READ (32)	7Fh/0009h	M	M
READ BUFFER (10)	3Ch	M	M
READ BUFFER (16)	9Bh	M	M
READ CAPACITY (10)	25h	M	M
READ CAPACITY (16)	9Eh	M	M
READ DEFECT DATA (10)	37h	M	M
READ DEFECT DATA (12)	B7h	M	M
READ LONG (10)	3Eh	M	M
READ LONG (16)	9Eh/11h	M	M
REASSIGN BLOCKS	07h	M	M
RECEIVE DIAGNOSTIC RESULTS	1Ch	M	M
REPORT IDENTIFYING INFORMATION	A3h/05h	M	M
REPORT LUNS	A0h	M	M
REPORT SUPPORTED OPERATION CODES	A3h/0Ch	M	M
REPORT SUPPORTED TASK MANAGEMENT FUNCTIONS	A3h/0Dh	M	M
REPORT TIMESTAMP	A3h/0Fh	M	M
REQUEST SENSE	03h	M	M
SANITIZE	48h	M	M
SECURITY PROTOCOL IN	A2h	M (TCG Only)	M (TCG only)
SECURITY PROTOCOL OUT	B5h	M (TCG Only)	M (TCG Only)

Requirement ID	Description
	<table><tr><td>SEND DIAGNOSTIC</td><td>1Dh</td><td>M</td><td>M</td></tr><tr><td>SET IDENTIFYING INFORMATION</td><td>A4h/06h</td><td>M</td><td>M</td></tr><tr><td>SET TIMESTAMP</td><td>A4/0Fh</td><td>M</td><td>M</td></tr><tr><td>START STOP UNIT</td><td>1Bh</td><td>M</td><td>M</td></tr><tr><td>SYNCHRONIZE CACHE (10)</td><td>35h</td><td>M</td><td>M</td></tr><tr><td>SYNCHRONIZE CACHE (16)</td><td>91h</td><td>M</td><td>M</td></tr><tr><td>TEST UNIT READY</td><td>00h</td><td>M</td><td>M</td></tr><tr><td>UNMAP</td><td>42h</td><td>P</td><td>M</td></tr><tr><td>VERIFY (10)</td><td>2Fh</td><td>M</td><td>M</td></tr><tr><td>VERIFY (16)</td><td>8Fh</td><td>M</td><td>M</td></tr><tr><td>VERIFY (32)</td><td>7Fh/ 000Ah</td><td>M</td><td>M</td></tr><tr><td>WRITE (10)</td><td>2Ah</td><td>M</td><td>M</td></tr><tr><td>WRITE (16)</td><td>8Ah</td><td>M</td><td>M</td></tr><tr><td>WRITE (32)</td><td>7Fh/ 000Bh</td><td>M</td><td>M</td></tr><tr><td>WRITE AND VERIFY (10)</td><td>2Eh</td><td>M</td><td>M</td></tr><tr><td>WRITE AND VERIFY (16)</td><td>8Eh</td><td>M</td><td>M</td></tr><tr><td>WRITE AND VERIFY (32)</td><td>7Fh/ 000Ch</td><td>M</td><td>M</td></tr><tr><td>WRITE BUFFER</td><td>3Bh</td><td>M</td><td>M</td></tr><tr><td>WRITE LONG (10)</td><td>3Fh</td><td>M</td><td>M</td></tr><tr><td>WRITE LONG (16)</td><td>9Fh/11h</td><td>M</td><td>M</td></tr><tr><td>WRITE SAME (10)</td><td>41h</td><td>M</td><td>M</td></tr><tr><td>WRITE SAME (16)</td><td>93h</td><td>M</td><td>M</td></tr><tr><td>WRITE SAME (32)</td><td>7Fh/00h</td><td>M</td><td>M</td></tr></table>
SAS-CMD-2	An INQUIRY command to a device shall return valid information to any initiator port even if a different initiator port has a reservation.
SAS-CMD-3	The EVPD bit shall be supported. An INQUIRY command requesting VPD data that is received by the device to a non-existent LUN shall cause the device to properly report a value of 7Fh (011b - PERIPHERAL QUALIFIER and 1Fh - PERIPHERAL DEVICE TYPE) in byte 0 of the VPD data.
SAS-CMD-4	Obsolete (see SAS-SDC-5).
SAS-CMD-5	Obsolete.
SAS-CMD-6	The UNMAP command shall be implemented by SSD devices. SSD devices shall return zeroes on reads to unmapped LBA(s) that have not been written. Every single LBA specified by the UNMAP command shall be unmapped (or mapped with zeroes as the data content for the LBAs) and clear all uncorrectable errors in the range of the command that were previously caused by a WRITE LONG command.

Requirement ID	Description
SAS-CMD-7	The device microcode shall be field downloadable via the SAS SSP port using the WRITE BUFFER command. All devices shall at least support the WRITE BUFFER command modes 07h, 0Eh, and 0Fh.
SAS-CMD-8	The device supplier shall validate in their design lab that if power is removed from the device precisely at the time the device is programming its firmware code, i.e., "flashing its code" to its non-volatile component, it shall not become inoperable, but rather, recover from that power loss. When power is re-applied it shall become ready for subsequent applicable protocol that normally follows the re-application of power (e.g., COMRESET, DWS, and commands). Recovery may be dependent upon the download mode used as follows. For example: • If mode 7h: Typically, revert to the prior firmware code only if unable to re-flash this new firmware code upon power restoration since it was "saved" prior to the flashing operation which was interrupted by the power loss. • If mode 0Eh: Recommends that the device revert to the prior firmware code since the firmware code was "saved" but not activated (i.e., "considered deferred"). The deferred microcode shall be activated and no longer considered deferred if any one of the following occurs: a power on, a hard reset, a START STOP UNIT command is processed, a FORMAT UNIT command is processed, or a WRITE BUFFER command with the activate deferred microcode mode (0Fh) is processed. This guarantees compliance with the T10 (SPC-5) Specification. • If mode 0Fh: Typically, revert to the prior firmware code only if the WRITE BUFFER mode 0Fh command did not complete with GOOD Status. If the command did complete with GOOD Status, then the device shall re-flash this new firmware code upon power restoration since it was "saved" (via Mode 0Eh) prior to the flashing /activation operation. (NOTE: It is assumed that the power loss did not affect the previously completed mode 0Eh firmware code save operation and GOOD Status has been returned for this WRITE BUFFER mode 0Fh command.) Devices shall be able to reflash and activate the new firmware code under these circumstances.
SAS-CMD-9	The device shall support the WR_UNCOR (1) bit fields of the WRITE LONG command as defined in the SCSI Block Commands – 4 (SBC-4) (T10/INCITS 506) specification document.
SAS-CMD-10	There shall be no limit imposed by the device relative to the number of uncorrectable errors created by the WRITE LONG command with all supported WR_UNCOR (1) bit field settings other than the limit imposed by the total device capacity limit.
SAS-CMD-11	When the Write Long command (with the WR Uncor bit = 1) is issued to a device formatted with a 512e sector size, the device shall only mark the LBA or LBAs identified in the command as uncorrectable (i.e., Not mark the entire physical (4k size) sector as Uncorrectable).
SAS-CMD-12	SSD devices shall support the UNMAP bit and shall ignore the ANCHOR bit for the WRITE SAME command.
SAS-CMD-13	The sequential execution time of the following command across the entire capacity of the device shall be no greater than 105% of the time for sequential WRITE of the entire device. The reference sequential write time of the entire device shall be measured using a workload of 64kiB writes that are 64kiB aligned and with a queue depth of 32: • WRITE SAME (10)

<table>
<tr><th>Requirement ID</th><th>Description</th></tr>
<tr><td></td><td>

WRITE SAME (16)
WRITE SAME (32)
SANITIZE OVERWRITE (48h/01h) (HDD only)
SANITIZE BLOCK ERASE (48h/02h) (SSD only)

</td></tr>
<tr><td>SAS-CMD-14</td><td>Execution time of these commands shall be no greater than 105% of the time for sequential READ of the entire disk:

VERIFY (10)
VERIFY (16)
VERIFY (32)

</td></tr>
<tr><td>SAS-CMD-15</td><td>The following commands shall be supported even when:
a. HDD: the spindle motor is stopped, or in the process of stopping or coming up to speed; or
b. SSD: the device is in the process of stopping or becoming ready.

INQUIRY
READ BUFFER (10)
REPORT LUNS
REQUEST SENSE
START STOP UNIT

</td></tr>
<tr><td>SAS-CMD-16</td><td>For HDD devices, after the initial spin-up, these commands shall be supported even when the spindle motor is stopped, or in the process of stopping or coming up to speed:

READ CAPACITY (10)
READ CAPACITY (16)
REPORT LUNS
REQUEST SENSE
TEST UNIT READY

</td></tr>
<tr><td>SAS-CMD-17</td><td>SSD devices shall set the following in the READ CAPACITY (16) parameter data:

Logical Block Provisioning Management Enabled (LBPME) = 1b.
Logical Block Provisioning Read Zeros (LBPRZ) = 1b.

HDD devices shall clear the LBPME bit to 0b.</td></tr>
<tr><td>SAS-CMD-18</td><td>For HDD devices, the following commands shall not bring the device out of an Idle_B, Idle_C, Standby_Z, or Stopped power condition, nor shall they cause a reset of the Idle_B or Idle_C timers:
<table>
<tr><th>Command Name</th><th>Operation Code / Service Action</th><th>Comments</th></tr>
<tr><td>INQUIRY</td><td>12h</td><td>Standard inquiry.</td></tr>
<tr><td>LOG SENSE</td><td>4Dh</td><td>LOG SENSE for the Temperature (0Dh) and Informational Exceptions (2Fh) log pages.</td></tr>
<tr><td>READ CAPACITY (10)</td><td>25h</td><td></td></tr>
<tr><td>READ CAPACITY (16)</td><td>9Eh/10h</td><td></td></tr>
<tr><td>REPORT LUNS</td><td>A0h</td><td></td></tr>
<tr><td>REQUEST SENSE</td><td>03h</td><td>Returns power condition in Sense Data (i.e., 0h/5Eh/XXh) when the device is in a power condition other than active or stopped.</td></tr>
</table>
</td></tr>
</table>

<table>
<tr><th>Requirement ID</th><th>Description</th></tr>
<tr><td></td><td>
<table>
<tr><td></td><td></td><td>Returns 2h/04h/02h when the device is in the Stopped power condition.</td></tr>
<tr><td>TEST UNIT READY</td><td>00h</td><td>Returns good status when in the Idle_C power condition.</td></tr>
</table>
</td></tr>
</table>

6.4.2 Sanitize Device Commands

Sanitize operations are initiated by some service actions of the SANITIZE command (see SAS-SDC-1) and by some TCG methods (see SEC-52).

<table>
<tr><th>Requirement ID</th><th>Description</th></tr>
<tr><td>SAS-SDC-1</td><td>The device shall support the following SANITIZE command service action codes:
<table>
<tr><th>Code</th><th>Name</th></tr>
<tr><td>01h</td><td>OVERWRITE (HDD only)</td></tr>
<tr><td>02h</td><td>BLOCK ERASE (SSD only)</td></tr>
<tr><td>03h</td><td>CRYPTOGRAPHIC ERASE</td></tr>
<tr><td>1Fh</td><td>EXIT FAILURE MODE</td></tr>
</table>
The OVERWRITE, BLOCK ERASE, and CRYPTOGRAPHIC ERASE service actions initiate Sanitize operations.</td></tr>
<tr><td>SAS-SDC-2</td><td>Obsolete (see SAS-SDC-6).</td></tr>
<tr><td>SAS-SDC-3</td><td>The device shall be able to provide any supported log page that is requested while a Sanitize operation is in progress.</td></tr>
<tr><td>SAS-SDC-4</td><td>Obsolete (see SAS-SDC-3).</td></tr>
<tr><td>SAS-SDC-5</td><td>After a Sanitize operation is completed, the device shall enable reads to the sanitized LBAs without generating errors. After any Sanitize operation (e.g., performed by the CRYPTOGRAPHIC ERASE command per SEC-52), the device shall return all zeros for host reads to LBAs that have not been written since the last Sanitize command.

Reading from an LBA (before writing to) that has been cryptographically erased shall not take longer than the time required to read that LBA before it had been cryptographically erased, and the read shall not terminate with an abort condition (e.g. Sense Key 0Bh – Aborted Command) due to being cryptographically erased.</td></tr>
<tr><td>SAS-SDC-6</td><td>The device shall accept and service the following commands at a minimum while any Sanitize operation is in progress:
INQUIRY (standard INQUIRY data and VPD pages)
LOG SENSE (see SAS-SDC-3)
REPORT LUNS
REQUEST SENSE</td></tr>
<tr><td>SAS-SDC-7</td><td>The device shall support the use of the AUSE (Allow Unrestricted Sanitize Exit) bit in the SANITIZE command with all SANITIZE command service actions that initiate Sanitize operations (see SAS-SDC-1).</td></tr>
<tr><td>SAS-SDC-8</td><td>The device’s non-volatile cache shall be included in the scope of data that is sanitized by any Sanitize operation.</td></tr>
</table>

Requirement ID	Description
SAS-SDC-9	If a power loss occurs during any Sanitize operation (see SAS-SDC-1), the operation shall resume after power is restored until the operation completes.
SAS-SDC-10	The results of all Sanitize operations shall meet IEEE Std 2883-2022 Purge requirements.

6.4.3 VPD Pages

<table>
<tr><th>Requirement ID</th><th>Description</th></tr>
<tr><td>SAS-VPD-1</td><td>The device shall support the following VPD pages:
<table>
<tr><th>Page</th><th>Description</th><th>Standard</th><th>Shall be accessible while NOT READY?</th></tr>
<tr><td>00h</td><td>Supported VPD Pages</td><td>Primary</td><td>Yes</td></tr>
<tr><td>80h</td><td>Unit Serial Number VPD Page</td><td>Primary</td><td>Yes</td></tr>
<tr><td>83h</td><td>Device Identification VPD Page</td><td>Primary</td><td>Yes</td></tr>
<tr><td>86h</td><td>Extended INQUIRY Data VPD Page</td><td>Primary</td><td>No</td></tr>
<tr><td>87h</td><td>Mode Page Policy VPD Page</td><td>Primary</td><td>No</td></tr>
<tr><td>88h</td><td>SCSI Ports VPD Page</td><td>Primary</td><td>No</td></tr>
<tr><td>8Ah</td><td>Power Condition VPD Page</td><td>Primary</td><td>No</td></tr>
<tr><td>8Dh</td><td>Power Consumption VPD Page</td><td>Primary</td><td>No</td></tr>
<tr><td>90h</td><td>Protocol-Specific Logical Unit Information</td><td>Primary</td><td>No</td></tr>
<tr><td>91h</td><td>Protocol-Specific Port Information</td><td>Primary</td><td>No</td></tr>
<tr><td>B0h</td><td>Block Limits</td><td>Block</td><td>No</td></tr>
<tr><td>B1h</td><td>Block Device Characteristics</td><td>Block</td><td>Yes</td></tr>
<tr><td>B2h</td><td>Logical Block Provisioning (SSD Only)</td><td>Block</td><td>No</td></tr>
</table></td></tr>
<tr><td>SAS-VPD-2</td><td>Obsolete (see SAS-VPD-6).</td></tr>
<tr><td>SAS-VPD-3</td><td>The Power Condition VPD page (8Ah) shall be supported and shall report which power conditions are supported and how they operate. Power condition timers shall be disabled by default and shall only be enabled by the host.</td></tr>
<tr><td>SAS-VPD-4</td><td>HDDs shall support the Power Condition VPD page (8Ah), with the following settings:
STANDBY_Z = 1b
STANDBY_Y = 1b
IDLE_A = 1b
IDLE_B = 1b
IDLE_C = 1b</td></tr>
<tr><td>SAS-VPD-4</td><td>In the Device Identification VPD Page (83h), if there are multiple designation descriptors associated with the addressed logical unit (ASSOCIATION = 00b), the first designation descriptor with ASSOCIATION = 00b returned by the device shall have the DESIGNATOR TYPE set to 3h (NAA identifier), and NAA set to 5h (IEEE Registered).</td></tr>
</table>

	Designation descriptors associated with the target port that received the request (ASSOCIATION = 01b) shall be returned after all designation descriptors with ASSOCIATION = 00b. All the information in this VPD page shall be returned whether the media is ready for access or not.
SAS-VPD-5	SSDs shall report actual values in the following fields in the Block Limits VPD Page (B0h): • OPTIMAL TRANSFER LENGTH GRANULARITY (bytes 6:7) • OPTIMAL TRANSFER LENGTH (bytes 12:15) • MAXIMUM UNMAP LBA COUNT (bytes 20:23) • MAXIMUM UNMAP BLOCK DESCRIPTOR COUNT (bytes 24:27) • OPTIMAL UNMAP GRANULARITY (bytes 28:31) • UGAVALID bit (byte 32, bit 7) • UNMAP GRANULARITY ALIGNMENT (bytes 32:35)
SAS-VPD-6	The device shall support the Write Uncorrectable Supported (WU_SUP) bit in the Extended INQUIRY Data VPD page (86h). The device shall set the following by default: • ACTIVATE MICROCODE field to 01b, and • MULTI I_T NEXUS MICROCODE DOWNLOAD field to 2h
SAS-VPD-7	All devices shall support the Block Device Characteristics VPD page (B1h), with the following settings and associated behaviors: • MEDIUM ROTATION RATE = 0001h for SSD; for HDD use the nominal rate and not the exact rate (e.g., 15 000 rpm = 3A98h, 10 000 rpm = 2710h, 7200 rpm = 1C20h, etc.). • WABEREQ = 01b, if the device supports BLOCK ERASE, else 00b. • WACEREQ = 01b, if the device supports CRYPTOGRAPHIC ERASE, else 00b. • NOMINAL FORM FACTOR, set per SBC-3.
SAS-VPD-8	SSDs shall support the Logical Block Provisioning VPD page (B2h), with the following settings and associated behaviors: • LBPU = 1b, indicating support for the UNMAP command. • LBPWS = 1b, indicating support for unmapping LBAs with the WRITE SAME (16) command. • LBPWS10 = 1b, indicating support for unmapping LBAs with the WRITE SAME (10) command. • LBPRZ set to the same value as the LBPRZ bit in the READ CAPACITY (16) parameter data. • ANC SUP = 1b, indicating support for anchored LBAs. • PROVISIONING TYPE = 001b, indicating the device is resource provisioned.

6.4.4 Mode Pages

<table>
<tr><th>Requirement ID</th><th>Description</th></tr>
<tr><td>SAS-MOD-1</td><td>SAS devices shall support the following Mode pages and parameters:

M = Mandatory. Mode Page shall be supported.
P = Prohibited. Mode Page shall not be supported (command rejected with ILLEGAL_FIELD_IN_CDB).
O = Optional. Mode Page support is optional.

<table>
<tr><th>Page Code</th><th>Page Description</th><th>Standard</th><th>Support</th></tr>
<tr><td>00h</td><td>Vendor Specific (Does not require page format)</td><td>SPC-5</td><td>O</td></tr>
<tr><td>01h</td><td>Read-Write Error Recovery</td><td>SBC-5</td><td>M</td></tr>
<tr><td>02h</td><td>Disconnect-Reconnect (SAS specific)</td><td>SPC-5 and SPL-5</td><td>M</td></tr>
<tr><td>07h</td><td>Verify Error Recovery</td><td>SBC-5</td><td>M</td></tr>
<tr><td>08h</td><td>Caching</td><td>SBC-5</td><td>M</td></tr>
<tr><td>0Ah</td><td>Control</td><td>SPC-5</td><td>M</td></tr>
<tr><td>0Ah/01h</td><td>Control Extensions</td><td>SPC-5</td><td>M</td></tr>
<tr><td>0Ah/02h</td><td>Application Tag</td><td>SBC-5</td><td>O</td></tr>
<tr><td>0Ah/FFh</td><td>Return all subpages of page 0Ah (MODE SENSE only)</td><td>SPC-5</td><td>M</td></tr>
<tr><td>18h</td><td>Protocol-Specific Logical Unit</td><td>SPL-5</td><td>M</td></tr>
<tr><td>19h</td><td>Protocol-Specific Port - short format</td><td>SPL-5</td><td>M</td></tr>
<tr><td>19h/01h</td><td>Protocol-Specific Port - PHY Control and Discover</td><td>SPL-5</td><td>M</td></tr>
<tr><td>19h/02h</td><td>Shared Port Control</td><td>SPL-5</td><td>M</td></tr>
<tr><td>19h/03h</td><td>Enhanced PHY Control</td><td>SPL-5</td><td>M</td></tr>
<tr><td>19h/FFh</td><td>Return all subpages of page 19h (MODE SENSE only)</td><td>SPC-5</td><td>M</td></tr>
<tr><td>1Ah</td><td>Power Condition</td><td>SPC-5</td><td>M</td></tr>
<tr><td>1Ah/01h</td><td>Power Consumption</td><td>SPC-5</td><td>M</td></tr>
<tr><td>1Ch</td><td>Informational Exceptions Control</td><td>SBC-5</td><td>M</td></tr>
<tr><td>1Ch/01h</td><td>Background Control</td><td>SBC-5</td><td>M</td></tr>
<tr><td>1Ch/02h</td><td>Logical Block Provisioning</td><td>SBC-5</td><td>M</td></tr>
<tr><td>3Fh</td><td>Return All Pages (MODE SENSE only)</td><td>SPC-5</td><td>M</td></tr>
</table></td></tr>
<tr><td>SAS-MOD-2</td><td>If the host sends a MODE SENSE (6) command with SUBPAGE CODE field value of 00h, the device shall NOT terminate the command with CHECK CONDITION status due to more data to return than is allowed by the one-byte allocation length of a MODE SENSE (6) command. The amount of data returned shall be equal to or less than the amount allowed by that one-byte allocation length. If the</td></tr>
</table>

<table>
<tr><th>Requirement ID</th><th>Description</th></tr>
<tr><td></td><td>amount of data in the mode page exceeds the amount allowed by that one-byte allocation length, then the device shall not return that mode page instead of returning a partial or truncated mode page. This requirement is specific to the MODE SENSE (6) command.

If a host sends a MODE SENSE (10) command with SUBPAGE CODE field value of 00h, and the device has more data to return than is allowed by the two-byte allocation length then (as specified in the SPC-5 standard) the device shall NOT terminate the command with CHECK CONDITION status for this reason and the device shall return a partial mode page truncated to the length allowed by that two-byte allocation length.

The device shall return ALL supported subpage 00h mode pages in page_0 format.</td></tr>
<tr><td>SAS-MOD-3</td><td>The device shall have the following default parameters from the factory for the Read-Write Error Recovery mode page (01h):
<table>
<tr><th>Byte</th><th>Bit</th><th>Description</th><th>Default</th></tr>
<tr><td>2</td><td>7</td><td>Automatic Write Reassignment Enabled (AWRE)</td><td>1b</td></tr>
<tr><td></td><td>6</td><td>Automatic Read Reassignment Enabled (ARRE)</td><td>1b</td></tr>
<tr><td></td><td>5</td><td>Transfer Block (TB)</td><td>0b</td></tr>
<tr><td></td><td>4</td><td>Read Continuous (RC)</td><td>0b</td></tr>
<tr><td></td><td>2</td><td>Post Error (PER)</td><td>1b</td></tr>
<tr><td></td><td>1</td><td>Data Terminate on Error (DTE)</td><td>0b</td></tr>
<tr><td>10:11</td><td>-</td><td>Recovery Time Limit</td><td>1F40h</td></tr>
</table></td></tr>
<tr><td>SAS-MOD-4</td><td>The device shall have the following default parameters for the Verify Error Recovery mode page (07h):
<table>
<tr><th>Byte</th><th>Bit</th><th>Description</th><th>Default</th></tr>
<tr><td>2</td><td>2</td><td>Post Error (PER)</td><td>1b</td></tr>
<tr><td></td><td>1</td><td>Data Terminate on Error (DTE)</td><td>0b</td></tr>
<tr><td>10:11</td><td>-</td><td>Verify Recovery Time Limit</td><td>1F40h</td></tr>
</table></td></tr>
<tr><td>SAS-MOD-5</td><td>The device shall implement the Caching mode page with its full 20-byte length, not the 12-byte length defined in SCSI-2. A MODE SELECT with the SCSI-2 length shall be accepted by the device and shall not modify the bits not sent with the data.

The device shall have the following default parameters for the Caching mode page (08h) for byte 2 from the factory:
<table>
<tr><th>Bit</th><th>Description</th><th>Default</th></tr>
<tr><td>7</td><td>Initiator Control (IC)</td><td>0b</td></tr>
<tr><td>6</td><td>Abort Pre-Fetch (ABPF)</td><td>0b</td></tr>
<tr><td>5</td><td>Caching Analysis Permitted (CAP)</td><td>0b</td></tr>
</table></td></tr>
</table>

<table>
<tr><th>Requirement ID</th><th>Description</th></tr>
<tr><td></td><td>
<table>
<tr><td>2</td><td>Writeback Cache Enable (WCE)</td><td>1b[1]</td></tr>
<tr><td>1</td><td>Multiplication Factor (MF)</td><td>0b</td></tr>
<tr><td>0</td><td>Read Cache Disable (RCD)</td><td>0b</td></tr>
</table>
Notes:
[1]Set to 1b if the device is power loss protected. Clear to 0b if device is not power loss protected (see COMM-2 and COMM-3). WCE shall have no negative performance or effect on device functionality.
</td></tr>
<tr><td>SAS-MOD-6</td><td>The device shall support minimum of two IDLE modes described below as defined in power condition mode page (1Ah):
Idle_A (active idle)
Idle_B (heads loaded to ramp)</td></tr>
<tr><td>SAS-MOD-7</td><td>The device shall support MRIE (Method of Reporting Informational Exceptions) to control reporting methods or warnings and exceptions. The device shall have the following parameters set in the Informational Exceptions mode page (1Ch) by default:
MRIE = 00h or equivalent to disable all asynchronous warnings and exceptions.
E_{WASC} = 01h, enable warnings reporting.
D_{EXCPT} = 00h, enable exception reporting.</td></tr>
<tr><td>SAS-MOD-8</td><td>HDD devices shall allow user/host to control BGMS activity via background control mode page (page code 1Ch, subpage code 01h).</td></tr>
</table>

6.4.5 Log Pages

<table>
<tr><th>Requirement ID</th><th>Description</th></tr>
<tr><td>SAS-LOG-1</td><td>The device shall support the following log pages for reporting purposes.
M = Mandatory. Log Page shall be supported.
P = Prohibited. Log Page shall not be supported (command rejected with 05h/24h/00h CHECK CONDITION).
Log pages that are not listed and are optional in the SAS specifications (see SAS-1) are considered optional.
<table>
<tr><th>Page / Subpage Code</th><th>Log Page Name</th><th>Support</th></tr>
<tr><td>00h</td><td>Supported Log Pages</td><td>M</td></tr>
<tr><td>02h</td><td>Write Errors Log Page</td><td>M</td></tr>
<tr><td>03h</td><td>Read Errors Log Page</td><td>M</td></tr>
<tr><td>05h</td><td>Verify Errors Log Page</td><td>M</td></tr>
<tr><td>06h</td><td>Non-Medium Error Log Page</td><td>M</td></tr>
<tr><td>0Dh</td><td>Temperature Log Page</td><td>M</td></tr>
<tr><td>0Dh / 01h</td><td>Environmental Reporting Log Page</td><td>M</td></tr>
<tr><td>0Dh / 02h</td><td>Environmental Limits Log Page</td><td>M</td></tr>
<tr><td>0Eh</td><td>Start/Stop Cycle Log Page</td><td>M</td></tr>
</table>
</td></tr>
</table>

<table>
<tr><th>Requirement ID</th><th>Description</th></tr>
<tr><td></td><td>
<table>
<tr><td>0Fh</td><td>Application Client Log Page</td><td>M</td></tr>
<tr><td>10h</td><td>Drive Self-Test Log Page</td><td>M</td></tr>
<tr><td>11h</td><td>Solid State Media Log Page</td><td>M
(SSD only)</td></tr>
<tr><td>15h</td><td>Background Media Scan Log Page</td><td>M</td></tr>
<tr><td>18h</td><td>Protocol-specific Log Page (SAS)</td><td>M</td></tr>
<tr><td>19h</td><td>Cache Memory Statistics / General Statistics and Performance Log Page</td><td>M</td></tr>
<tr><td>1Ah</td><td>Power Transition Log Page</td><td>M</td></tr>
<tr><td>2Fh</td><td>Informational Exceptions Log Page</td><td>M</td></tr>
</table>
</td></tr>
<tr><td>SAS-LOG-2</td><td>Unless stated otherwise for specific log parameters, log parameters shall be saved at 10-minute intervals. Parameters thus saved shall persist across power cycles and all types of resets. The interval for saving logs shall not include any time that the device is in an Idle_B, Idle_C, or standby power condition, nor shall the device leave such a power condition just for the purpose of saving a log.</td></tr>
<tr><td>SAS-LOG-3</td><td>Log pages 02h, 03h, and 05h shall implement the parameter codes listed in the table below. Counters are incremented each time a logical block requires correction. Two different blocks corrected during the same command count as two events.
<table>
<tr><th rowspan="2">Code / Parameter</th><th colspan="3">Required in Log Page?</th><th rowspan="2">Definition</th></tr>
<tr><th>02h</th><th>03h</th><th>05h</th></tr>
<tr><td>0001h / Errors corrected with possible delays</td><td>Yes</td><td>Yes</td><td>Yes</td><td>HDD: Any recovered error that uses 4 or more additional revolutions to successfully recover a logical block. This counter is incremented once for each block that requires correction.

SSD: Errors corrected with substantial delay.</td></tr>
<tr><td>0002h / Total</td><td>Yes</td><td>Yes</td><td>Supplier’s option</td><td>As determined by the supplier.</td></tr>
<tr><td>0003h / Total errors corrected</td><td>Yes</td><td>Yes</td><td>Supplier’s option</td><td>As determined by the supplier.</td></tr>
<tr><td>0004h / Total times error correction</td><td>Yes</td><td>Yes</td><td>Yes</td><td>The number of retries used to correct the errors logged by parameter 0001h as well as the number of retries used in the attempt to correct errors logged by parameter 0006h.</td></tr>
<tr><td>0005h / Total bytes processed</td><td>Yes</td><td>Yes</td><td>Yes</td><td>Every byte that was either successfully or unsuccessfully read, written, or verified (depending on the log page).</td></tr>
</table>
</td></tr>
</table>

<table>
<tr><th>Requirement ID</th><th>Description</th></tr>
<tr><td></td><td>
<table>
<tr><td>0006h / Total uncorrected errors</td><td>Yes</td><td>Yes</td><td>Yes</td><td>Any error that is not corrected within the retry steps allowed by mode page 01h settings (see SAS-MOD-3). This counter is incremented once and only once for each block on which an uncorrected data error has occurred.</td></tr>
</table>
</td></tr>
<tr><td>SAS-LOG-4</td><td>The following parameters shall be supported in the Temperature Log Page (0Dh):

Temperature Log Parameter (0000h)
Byte 5 (TEMPERATURE) of this parameter shall contain the temperature of the device, in degrees Celsius. The reported temperature shall be accurate to within ±2 °C from 0 to 70 °C. For HDDs, care shall be taken to report a value that best represents the internal air temperature near the critical components (e.g., read/write interface elements in an HDD device). If the device has multiple temperature sensors, then a composite temperature shall be reported.

Each time the Temperature log page is read, the contents of byte 5 shall be refreshed to reflect the current device temperature.

Temperature Log Parameter (0001h)
For HDDs, byte 5 (REFERENCE TEMPERATURE) of this parameter shall be set to 50 °C.
For SSDs, byte 5 (REFERENCE TEMPERATURE) of this parameter shall be set to 65 °C.</td></tr>
<tr><td>SAS-LOG-5</td><td>The following parameters in the Start-Stop Cycle Counter Log Page (0Eh) shall be implemented:
<table>
<tr><th>Parameter Code</th><th>Description</th><th>HDD Support Required?</th><th>SSD Support Required?</th><th>Content</th></tr>
<tr><td>0001h</td><td>Date of Manufacture</td><td>Yes</td><td>Yes</td><td>As per SPC-5</td></tr>
<tr><td>0002h</td><td>Accounting Date</td><td>Yes</td><td>Yes</td><td>As per SPC-5</td></tr>
<tr><td>0003h</td><td>Specified Cycle Count Over Device Lifetime</td><td>Yes</td><td>Yes</td><td>As per SPC-5</td></tr>
<tr><td>0004h</td><td>Accumulated Start-Stop Cycles</td><td>Yes</td><td>Yes</td><td>As per SPC-5</td></tr>
<tr><td>0005h</td><td>Specified Load-Unload Count Over Device Lifetime</td><td>Yes</td><td>Yes</td><td>As per SPC-5</td></tr>
<tr><td>0006h</td><td>Accumulated Load-Unload Cycles</td><td>Yes</td><td>Yes</td><td>As per SPC-5</td></tr>
</table>
For devices that do not have a head-ramp design, Parameters 0005h and 0006h must always be 0000h.</td></tr>
<tr><td>SAS-LOG-6</td><td>For SSDs, the Solid-State Media Log Page (11h) shall implement the parameter code 0001h (Percentage Used Endurance Indicator). The method of determining remaining life shall include such factors as remaining reserved flash blocks and P/E cycles used versus maximum P/E cycles flash parts specification.</td></tr>
<tr><td>SAS-LOG-7</td><td>The following shall be supported in the Background Scan Results Log Page (15h):</td></tr>
</table>

Requirement ID	Description
	Background Scan Results Log Page (HDD) The Background Scan Results log page stores the BACKGROUND SCAN STATUS parameter and zero or more BACKGROUND MEDIUM SCAN parameters. The NUMBER OF BACKGROUND SCANS PERFORMED field shall be initialized to 0h. A minimum of 512 BACKGROUND MEDIUM SCAN parameters shall be preserved at any one time. Whenever the log page is read, bytes 4:7 of the BACKGROUND SCAN STATUS parameter shall always return the current value of ACCUMULATED POWER ON MINUTES as of the time the log page is read. **Background Scan Results Log Page (SSD)** Whether a particular SSD supports BMS or not, the Background Scan Results Log Page (15h) shall be present for at least the purpose of reporting power-on time. Whenever the log page is read, bytes 4:7 of the BACKGROUND SCAN STATUS parameter shall always return the current value of ACCUMULATED POWER ON MINUTES as of the time the log page is read.
SAS-LOG-8	For Dual port SAS devices, the Protocol Specific Port Log Page (18h) shall include two protocol specific port log parameter entries, one for each port. Each protocol specific port log entry shall include a SAS Phy log descriptor for the Phy associated with that port, which shall include the following: • The PHY IDENTIFIER (byte 1 of the SAS Phy log descriptor) shall be set to 00h for the first Phy and 01h for the second Phy. • The PARAMETER CODE (relative target port identifier) at bytes 0–1 of the Protocol-specific log parameter for SAS shall be set equal to the PHY IDENTIFIER + 1 (from the SAS Phy log). • The NUMBER OF PHYS at byte 7 of the Protocol-Specific log parameter for SAS shall be set to 01h. Counters shall only be cleared upon power up (or by use of the LOG SELECT command with PCR = 1); they shall not be cleared by a firmware download operation, a logical unit reset, or any other “hard” reset. All cleared counters shall be set to 0h. If a counter reaches the maximum value for that field, the field will stay at the maximum value. No error will be reported. No performance degradation shall occur because of tracking these counts.
SAS-LOG-9	The MOST RECENT TEMPERATURE READING field (byte 6h) of the Informational Exceptions General log within the Informational Exceptions Log Page (2Fh) shall contain the temperature of the device in degrees Celsius. For HDDs, care shall be taken to report a value that best represents the internal air temperature near the critical components (e.g., read/write interface components). Each time this log page is read, the contents of byte 10 shall be refreshed to reflect the current device temperature. The reported temperature shall be accurate to within ±2 °C from 0 to 70 °C. For SSDs, the reported value shall be a composite temperature based on all the temperature sensors in the device.

6.5 Dual Port Requirements

6.5.1 General Requirements

If Dual port is supported the following requirements shall be supported.

Requirement ID	Description
SAS-DP-1	Obsolete (see SAS-DP-CMD-1).
SAS-DP-2	A dual port (two-PHY) physical connector is recommended with the primary PHY (i.e., the PHY for the primary port) located in the “single port” location.
SAS-DP-3	Each port shall support OOB, link resets, indiscriminately and asynchronously, totally without regard to the other port. Link resets shall be able to occur and complete simultaneously on both ports during the discovery process and any time thereafter.
SAS-DP-4	Each port shall have a separate, unique, SAS address (device port identifier).
SAS-DP-5	Each port shall have a different relative device port identifier.
SAS-DP-6	Each port has the same logical unit identifier and device name (for direct-access devices).
SAS-DP-7	Each port shall be able to support a minimum of 10 separate, independent initiators.
SAS-DP-8	Each port shall support all the SCSI command set recommended.
SAS-DP-9	Each port shall support the total set of mode pages, log pages, VPD pages, and diagnostic pages as recommended. Values and data reporting content are different for dual port devices.
SAS-DP-10	Each port shall be able to support indiscriminately, independently, full duplex, concurrent command(s) reception while returning status.
SAS-DP-11	Each port shall support indiscriminately, independently, full duplex, concurrent data reception and data transmission.
SAS-DP-12	Each port shall support the BROADCAST (ASYNCHRONOUS EVENT) primitive.
SAS-DP-13	Each port shall support and report INQUIRY data and VPD for that port.
SAS-DP-14	Each port shall support and report mode page settings for that port.
SAS-DP-15	Each port shall support and report log page settings for that port.
SAS-DP-16	Dual port implementations shall operate and perform exactly as how a single port device operates and performs with the benefit of providing an additional “connection point” (port) to the device. The additional identification and reporting data structures shall be implemented as well.
SAS-DP-17	A dual port device shall enable both ports by default (see SAS-4).
SAS-DP-18	Dual port implementations shall be capable of simultaneous dual-port operation and must be able to handle a data frame on one port simultaneously with a non-data frame on the other port.

6.5.2 Command Requirements

For Dual Port devices the following command requirements shall be supported.

<table>
<tr><th>Requirement ID</th><th>Description</th></tr>
<tr><td>SAS-DP-CMD-1</td><td>A SAS device may have more than one port. The device shall report all port associated data structures accordingly. A single port device shall report the single port configuration information as described in # of Ports 1. A dual port device shall report the dual port configuration information as described in # of Ports 2.
<table>
<tr><th rowspan="2">Data Structure Description</th><th colspan="2"># of Ports</th></tr>
<tr><th>1</th><th>2</th></tr>
<tr><td>Standard INQUIRY “MULTIP” bit = 0b</td><td>x</td><td></td></tr>
<tr><td>Standard INQUIRY “MULTIP” bit = 1b</td><td></td><td>x</td></tr>
<tr><td></td><td></td><td></td></tr>
<tr><td>INQUIRY Device Identification VPD page (83h):</td><td>x</td><td>x</td></tr>
<tr><td>- Logical Unit identifier</td><td>x</td><td>x</td></tr>
<tr><td>- Target port identifier for the port being used by INQUIRY</td><td>x</td><td>x</td></tr>
<tr><td>- Relative target port identifier for the target port being used by INQUIRY</td><td>x</td><td>x</td></tr>
<tr><td>- Target device name</td><td>x</td><td>x</td></tr>
<tr><td></td><td></td><td></td></tr>
<tr><td>INQUIRY SCSI Ports VPD page (88h):</td><td>x</td><td>x</td></tr>
<tr><td>- Primary port descriptor (only)</td><td>x</td><td></td></tr>
<tr><td>- Primary and secondary port descriptors available</td><td></td><td>x</td></tr>
<tr><td></td><td></td><td></td></tr>
<tr><td>Log Pages:</td><td></td><td></td></tr>
<tr><td>Protocol-Specific log page (18h):</td><td>x</td><td>x</td></tr>
<tr><td>- One log parameter (only); parameter code 0001h for primary port</td><td>x</td><td></td></tr>
<tr><td>- Two log parameters available; both parameter code 0001h and parameter code 0002h</td><td></td><td>x</td></tr>
<tr><td>- One PHY IDENTIFIER (00h for primary port, only)</td><td>x</td><td></td></tr>
<tr><td>- Two PHY IDENTIFIERs available; (00h primary port and 01h secondary port)</td><td></td><td>x</td></tr>
<tr><td>- SAS ADDRESS (for primary port, only)</td><td>x</td><td></td></tr>
<tr><td>- SAS ADDRESS (both available for primary port and secondary port)</td><td></td><td>x</td></tr>
<tr><td></td><td></td><td></td></tr>
<tr><td>Mode Pages:</td><td></td><td></td></tr>
<tr><td>Protocol-Specific Port – PHY Control And Discover subpage (19h/01h):</td><td>x</td><td>x</td></tr>
<tr><td>- NUMBER OF PHYS field (01h for one PHY)</td><td>x</td><td></td></tr>
<tr><td>- NUMBER OF PHYS field (02h for dual-port device)</td><td></td><td>x</td></tr>
<tr><td>- One PHY mode descriptor (only)</td><td>x</td><td></td></tr>
<tr><td>- Two PHY mode descriptors available</td><td></td><td>x</td></tr>
</table></td></tr>
</table>

Requirement ID	Description		
	- PHY IDENTIFIER (00h for primary port, only)	x	
	- PHY IDENTIFIER (00h for primary port and 01h for secondary port)		x
	- SAS ADDRESS (for primary port, only)	x	
	- SAS ADDRESS (both available for primary port and secondary port)		x

6.5.3 Persistent Reservation

For dual port devices the following Persistent Reservation requirements shall be supported.

Requirement ID	Description
SAS-DP-PR-1	PERSISTENT RESERVE IN and PERSISTENT RESERVE OUT commands shall have the capabilities for reservations that apply to the full logical unit (LU_SCOPE = 0h) and at least support WRITE EXCLUSIVE (01h), EXCLUSIVE ACCESS (03h), WRITE EXCLUSIVE REGISTRANTS ONLY (05h) and EXCLUSIVE ACCESS REGISTRANTS ONLY (06h). Support of all other reservation type codes is optional.
SAS-DP-PR-2	Devices shall implement the following PERSISTENT RESERVE IN service action codes: • READ KEYS (00h) • READ RESERVATION (01h) • REPORT CAPABILITIES (02h) • READ FULL STATUS (03h)
SAS-DP-PR-3	Devices shall implement the following PERSISTENT RESERVE OUT service action codes: • REGISTER (00h) • RESERVE (01h) • RELEASE (02h) • CLEAR (03h) • PREEMPT (04h) • PREEMPT AND ABORT (05h) • REGISTER AND IGNORE EXISTING KEY (06h) Additionally, the SPEC_I_PT (Specify Initiator Ports), ALL_TG_PT (All Target Ports), and APTPL (Activate Persist Through Power Loss) bits along with the feature functionality associated with these bits of the parameter list shall be supported.
SAS-DP-PR-4	Persistent Reservations shall persist across firmware download and activation.
SAS-DP-PR-5	The SIP_C (Specify Initiator Ports Capable), ATP_C (All Target Ports Capable), PTPL_C (Persist Through Power Loss Capable), PTPL_A (Persist Through Power Loss Activated), and TMV (Type Mask Valid) bits along with the feature functionality associated with these bits of the parameter data for REPORT CAPABILITIES shall be supported.

6.6 Drive Self-Test (DST)

The following Drive Self-Test (DST) requirements shall apply:

Requirement ID	Description
SAS-DST-1	DST shall be implemented by using the SEND DIAGNOSTIC Command Descriptor Block (CDB). The SELFTEST bit and the following SELF-TEST CODEs shall be supported:

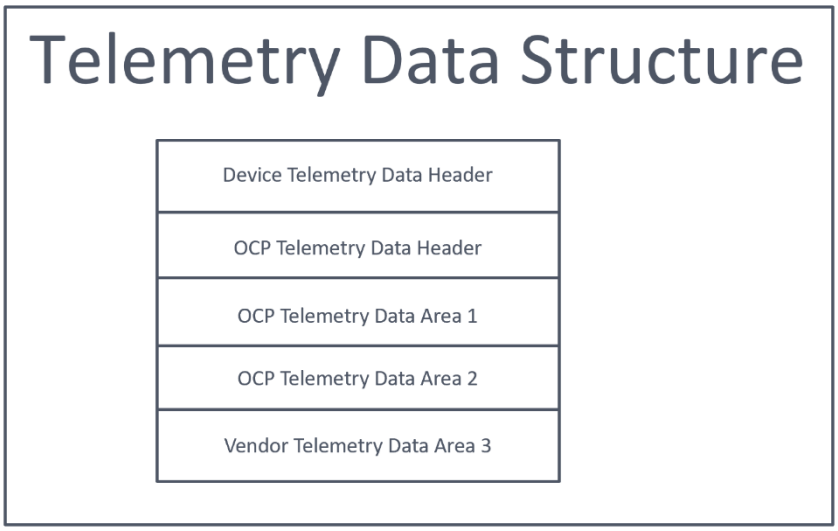
<table>
<tr><th>Requirement ID</th><th>Description</th></tr>
<tr><td></td><td>
<table>
<tr><th>CODE</th><th>Description</th><th>Maximum Execution Time</th></tr>
<tr><td>000b</td><td>Default Self-Test (when SELFTEST bit = 1b)</td><td>10 seconds</td></tr>
<tr><td>001b</td><td>Background short Drive Self-Test (DST)</td><td>120 seconds</td></tr>
<tr><td>010b</td><td>Background extended DST</td><td>SSDs - 30 minutes
HDDs - 90 minutes
See SAS-9</td></tr>
<tr><td>100b</td><td>Abort background DST</td><td>N/A</td></tr>
</table>
</td></tr>
<tr><td>SAS-DST-2</td><td>While a background DST is in progress the device shall supply completion status via REQUEST SENSE. Note that the current settings of ARRE, and AWRE apply during DST.</td></tr>
<tr><td>SAS-DST-3</td><td>The table below describes the device’s behavior when background DST is interrupted.
<table>
<tr><th>Interruptive Event</th><th>Background Short DST Behavior</th><th>Background Extended DST Behavior</th></tr>
<tr><td>Receipt of a SEND DIAGNOSTIC command with the SELF TEST CODE set to 100b (Abort background self-test)</td><td>Abort</td><td>Abort</td></tr>
<tr><td>Receipt of a FORMAT UNIT command</td><td>Abort</td><td>Abort</td></tr>
<tr><td>Receipt of a WRITE BUFFER command with the mode set to any download microcode option</td><td>Abort</td><td>Abort</td></tr>
<tr><td>Receipt of a START STOP UNIT command with START = 0b</td><td>Abort</td><td>Abort</td></tr>
<tr><td>Receipt of a START STOP UNIT command with START = 1b and POWER CONDITION field = 0h</td><td>Abort</td><td>Resume (or continue if running)</td></tr>
<tr><td>Drive reset</td><td>Abort</td><td>Resume after reset</td></tr>
<tr><td>Receipt of NOTIFY (POWER LOSS EXPECTED)</td><td>Abort</td><td>Resume after reset</td></tr>
<tr><td>Power interruption</td><td>Abort</td><td>Resume when power is restored</td></tr>
</table>
</td></tr>
<tr><td>SAS-DST-4</td><td>If background extended DST is interrupted by a device reset or a power interruption, the device shall resume the DST without external intervention, such that no more than 5 minutes is added to the total time required to complete.</td></tr>
</table>

7 Telemetry Requirements

7.1 Device Telemetry Overview

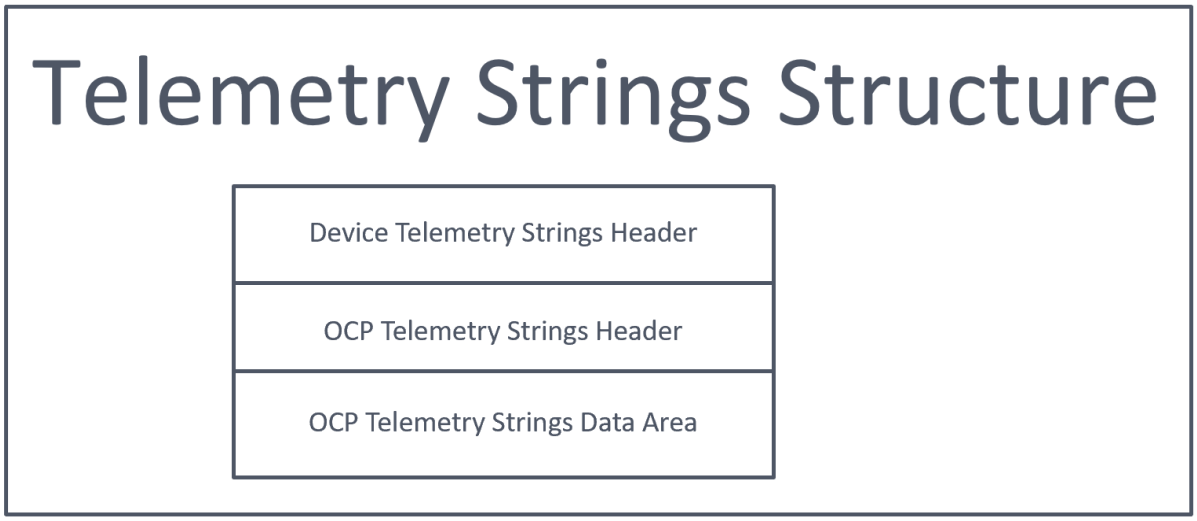
Device telemetry shall be tracked to enable open-source tools to decode human readable information based on two data structures. The two data structures are:

1. Telemetry Data Structure. This contains the device events, counters and data related to the operation of the device. This data structure consists of the Device Telemetry Data Header, OCP Telemetry Data header, OCP Telemetry Data Area 1, 2 and Vendor Telemetry Data Area 3. A figure of this can be seen below.



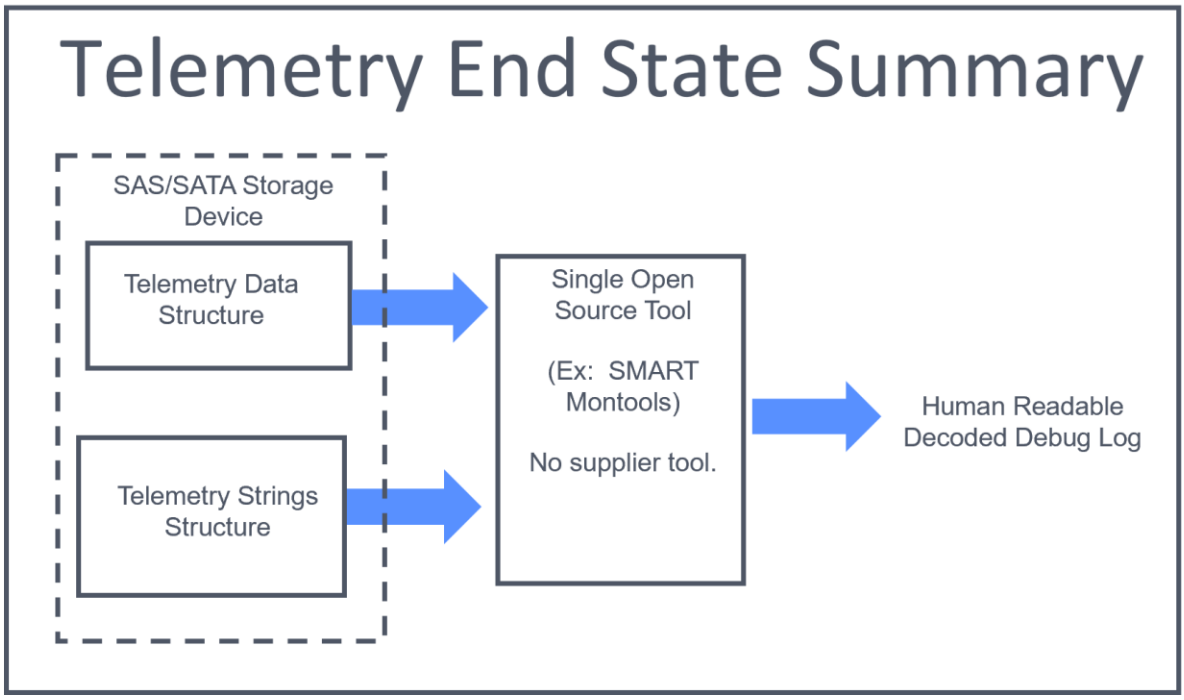
The Device Telemetry Data Header format depends on the protocol (SATA or SAS). The OCP Telemetry Data Header, OCP Telemetry Data Area 1, 2 and Vendor Telemetry Data Area 3 format are protocol independent.

2. The Telemetry String Structure contains the information to decode the Telemetry Data Structure into human readable information. The Telemetry String Structure consists of the Device Telemetry Strings Header, OCP Telemetry Strings Header and OCP Telemetry Strings Data Area. A figure of this can be seen below.



The Device Telemetry Strings header format depends on the protocol (SATA or SAS). The OCP Telemetry Strings Header, OCP Telemetry Strings Data Area are protocol independent.

Devices providing the Telemetry Data Structure and Telemetry String Data Structure have the ability for non-vendor specific open-source tooling to decode the telemetry into a human readable format as seen in the figure below.



7.2 General Telemetry Requirements

7.2.1 Device Telemetry Requirements

<table>
<tr><th>Requirement ID</th><th>Description</th></tr>
<tr><td>TEL-1</td><td>The device shall track the operational/event history and any critical parameters that are required to debug device issues in the Telemetry Data Structure.</td></tr>
<tr><td>TEL-2</td><td>The supplier shall provide a table that categorizes the reason identifiers.</td></tr>
<tr><td>TEL-3</td><td>The Reason Identifier associated with the Telemetry Data Structure shall be the reason identifier for the most recent failure and shall not be cleared by any power cycle or reset.</td></tr>
<tr><td>TEL-4</td><td>In the Telemetry Data Structure, OCP Telemetry Data Area 1, OCP Telemetry Data Area 2 and Vendor Telemetry Data Area 3 shall meet the following requirements. Any vendor-specific data required for root-cause analysis shall be included in Vendor Telemetry Data Area 3.
<table>
<tr><th>Data Structure</th><th>Purpose</th><th>Data Area Size</th><th>Latency Impact to IO</th></tr>
<tr><td>OCP Telemetry Data Area 1</td><td>Periodic logging for monitoring trends/problems</td><td>32,768 bytes</td><td>SSD: < 1 ms typical
< 5 ms max
HDD: < 5% increase in average latency</td></tr>
</table>
</td></tr>
</table>

<table>
<tr><th>Requirement ID</th><th>Description</th></tr>
<tr><td></td><td>
<table>
<tr><td>OCP Telemetry Data Area 2</td><td>Logging for debug items that impact latency</td><td>-</td><td>Greater than OCP Telemetry Data Area 1 Requirement</td></tr>
<tr><td>Vendor Telemetry Data Area 3</td><td>Optional Vendor Unique</td><td>-</td><td>-</td></tr>
</table>
</td></tr>
<tr><td>TEL-5</td><td>Information that is used to populate OCP Telemetry Data Area 1 in the Telemetry Data Structure shall be updated in the background at least once every ten minutes to prevent data older than ten minutes from being returned.</td></tr>
<tr><td>TEL-6</td><td>If the device supports an encrypted and/or vendor unique log for debug this may be contained in Vendor Telemetry Data Area 3 in the Telemetry Data Structure.</td></tr>
<tr><td>TEL-7</td><td>The number of Timestamp Events inserted into the Event FIFOs shall be sufficient to debug.</td></tr>
<tr><td>TEL-8</td><td>The data in the Telemetry Data Structure shall contain all the debug information needed to root cause an issue.</td></tr>
<tr><td>TEL-9</td><td>When the device encounters a failure/error the Reason Identifier (see Section 7.2.2 Reason Identifier Requirements) shall be a non-zero value.</td></tr>
<tr><td>TEL-10</td><td>In the Telemetry Data Structure data in OCP Telemetry Data Area 1 and OCP Telemetry Data Area 2 when combined with the Telemetry Strings Structure shall be decodable by the SMART MON Open-Source Tool that supports this version of this specification.</td></tr>
<tr><td>TEL-11</td><td>On a firmware update the Vendor Specific Statistics Identifier, the Debug Event Class/Events Identifier combinations and the Debug Class/VU Event Identifier combinations shall not be re-purposed or removed from the Telemetry String Structure. This is to ensure on a firmware update that any pre-existing Telemetry Data Structure can be decoded correctly.</td></tr>
<tr><td>TEL-12</td><td>The following behavior types are used to define the functional boundary condition behaviors for saturating or wrapping, under reset conditions and power cycles.
<table>
<tr><th>Behavior Type</th><th>Saturating Counter[1]</th><th>Reset Persistent[2]</th><th>Power Cycle Persistent[3]</th></tr>
<tr><td>0</td><td colspan="3">N/A</td></tr>
<tr><td>1</td><td>No (Runtime Value)</td><td>No (Runtime Value)</td><td>No (Runtime Value)</td></tr>
<tr><td>2</td><td>No</td><td>Yes</td><td>Yes</td></tr>
<tr><td>3</td><td>Yes</td><td>Yes</td><td>No</td></tr>
<tr><td>4</td><td>Yes</td><td>Yes</td><td>Yes</td></tr>
<tr><td>5</td><td>Yes</td><td>No</td><td>No</td></tr>
<tr><td>6</td><td>No</td><td>Yes</td><td>No</td></tr>
<tr><td>15 - 7</td><td colspan="3">Reserved</td></tr>
</table>
Notes:
[1]Saturating counts do not roll over past the max value.
[2]Reset Persistent means the value does not reset on a reset event (e.g., SRST, COMRESET, hard reset, logical unit reset, etc.).
[3]Power cycle persistent means the value does not reset on a power cycle.</td></tr>
<tr><td>TEL-13</td><td>Reserved for future use.</td></tr>
<tr><td>TEL-14</td><td>Reserved for future use.</td></tr>
</table>

Requirement ID	Description
TEL-15	All Telemetry Data Structures and Telemetry Strings Structures defined by this specification are little endian unless otherwise specified by this specification. This requirement does not apply to data structures defined by other specifications, e.g., for SAS devices, SPC-5 and subsequent versions of SPC define the Device Telemetry Data Header (see Section 7.4.2 Telemetry Data Structure Mapping to Current Internal Status Data) and the Device Telemetry Strings Header (see Section 7.4.3 Telemetry Strings Structure Mapping to Saved Internal Status Data) and specify that these data structures are big endian.
TEL-16	The scope of all telemetry data (see TEL-4) shall be the entire device.
TEL-17	Statistics that are listed in Section 7.2.4.2 Common (HDD/SSD) Statistic Identifiers , Section 7.2.4.3 SSD Specific Statistic Identifiers, and Section 7.2.4.4 HDD Specific Identifiers as applicable to the device (HDD or SSD) and as replacing a SMART attribute shall be implemented, unless otherwise specified, and shall be in Data Area 1.
TEL-18	For the Statistics in TEL-17, the device shall populate the Telemetry String Log with strings that exactly match the text in the Statistic Specific Data Field column in the table in Section 7.2.4 Statistic Identifiers Definitions.
TEL-19	For the statistics in TEL-17, in the Statistic Information Field, the Host Hint Type shall be set to a value of Host Hint Type 1. For statistics not in TEL-17, the Host Hint Type shall be set to a value of No Host Hint.
TEL-20	Telemetry Data Area 1 and Telemetry Data Area 2 shall be supported per this specification.

7.2.2 Reason Identifier Requirements

The format of the Reason Identifier shall be as follows:

Requirement ID	Byte Address	Field	# of Bytes	Field Description
TELRI-1	63:0	Error ID	64	The device shall set a unique vendor-defined Error ID defining the persistent or transient error at the time of log capture (see TEL-9).
TELRI-2	71:64	File ID	8	File name or hash of the firmware source file where the error occurred. Clear to zero if not applicable.
TELRI-3	73:72	Line Number	2	Line Number in the firmware source file where the error occurred. Clear to zero if not applicable.
TELRI-4	74	Valid Flags	1	**Bit** / **Bit Description** 7:4 — Reserved. Shall be cleared to zero. 3 — VU Reason Extension Field is valid. 2 — Shall be set to 1b if the Error ID is valid. Shall be cleared to 0b if Error ID is not valid. 1 — Shall be set to 1b if the File ID is valid. Shall be cleared to 0b if File ID is not valid.

Requirement ID	Byte Address	Field	# of Bytes	Field Description
				0: Shall be set to 1b if the Line Number is valid. Shall be cleared to 0b if Line Number is not valid.
TELRI-5	95:75	Reserved	21	Shall be cleared to zero.
TELRI-6	127:96	VU Reason Extension	32	Vendor Unique reason code.

7.2.3 Statistic Descriptor Format

The format of the Statistic Descriptor is in the following table:

Requirement ID	Byte Address	Field	# of Bytes	Field Description
STATF-1	1:0	Statistic Identifier	2	Shall contain the identifier that defines the data format of this entry.
STATF-2	2	Statistic Information 1	1	**Bit / Statistic Information 1 Bit Description** 7:4 — Statistic Type. This is the statistic type that is used to decode the format of the Statistic Specific Field. **Statistic Type / Description** 0000b: Single 0001b: Array 0010b: Custom 0011b:1111b: Reserved 3:0 — Behavior Type (see TEL-12).
STATF-3	3	Statistic Information 2	1	Statistic Information 2: Data Unit. This provides data units used in the Statistic Specific Field. **Value (h) / Data Unit** 00: No Data Unit Information provided 01: Milliseconds 02: Seconds 03: Hours 04: Days 05: MB 06: GB 07: TB 08: PB 09: Celsius

<table>
<tr><th>Requirement ID</th><th>Byte Address</th><th>Field</th><th># of Bytes</th><th>Field Description</th></tr>
<tr><td></td><td></td><td></td><td></td><td>
<table>
<tr><td>0A</td><td>Kelvin</td></tr>
<tr><td>0B</td><td>Fahrenheit</td></tr>
<tr><td>0C</td><td>Voltage (mV)</td></tr>
<tr><td>0D</td><td>Current (mA)</td></tr>
<tr><td>0E</td><td>Ohm</td></tr>
<tr><td>0F</td><td>RPM</td></tr>
<tr><td>10</td><td>Micrometer</td></tr>
<tr><td>11</td><td>Nanometer</td></tr>
<tr><td>12</td><td>Angstroms</td></tr>
<tr><td>13-FF</td><td>Reserved</td></tr>
</table>
</td></tr>
<tr><td>STATF-4</td><td>4</td><td>Statistic Information 3</td><td>1</td><td>
Statistic Information 3
<table>
<tr><th>Bit</th><th>Statistic Information 3
Bit Description</th></tr>
<tr><td>7:6</td><td>Reserved. Shall be cleared to zero.</td></tr>
<tr><td>5:4</td><td>Host Hint Type. The decode for the Host Hint Type is as follows.
<table>
<tr><th>Value (h)</th><th>Description</th></tr>
<tr><td>00</td><td>No Host Hint</td></tr>
<tr><td>01</td><td>Host Hint Type 1</td></tr>
<tr><td>02-03</td><td>Reserved</td></tr>
</table>
This field is used for host decoding information. TEL-17 describes the requirements of what value shall be in this field.</td></tr>
<tr><td>3:0</td><td>Data Type. This provides the data in the statistic specific field.
<table>
<tr><th>Data Type (h)</th><th>Description</th></tr>
<tr><td>00</td><td>No Data Type Information provided in this field</td></tr>
<tr><td>01</td><td>Signed Integer</td></tr>
<tr><td>02</td><td>Unsigned Integer</td></tr>
<tr><td>03</td><td>Floating Point</td></tr>
<tr><td>04</td><td>ASCII (7-bit)</td></tr>
<tr><td>05-F</td><td>Reserved</td></tr>
</table>
</td></tr>
</table>
</td></tr>
<tr><td>STATF-5</td><td>5</td><td>Reserved</td><td>1</td><td>Shall be cleared to 0h.</td></tr>
<tr><td>STATF-6</td><td>7:6</td><td>Statistic Data Size</td><td>2</td><td>Shall contain the number of Dwords that is used by the Statistic Specific Field. Valid values for this field are 0h to FFFFh.</td></tr>
</table>

Requirement ID	Byte Address	Field	# of Bytes	Field Description
STATF-7	(Statistic Data Size * 4) + 7:8	Statistic Specific Data	Statistic Data Size *4	Shall contain data for this Statistic. This data shall be a single element value.
STATF-8	8	Element Size	1	Element Size. This is the number of bytes in a single array element. This is a zero-based value (i.e. a value of 0h represents 1 byte).
STATF-9	9	Reserved	1	Shall be cleared to zero.
STATF-10	11:10	Number Elements	2	This is the number of elements in the array. This is a zero-based value (i.e. a value of 0h represents 1 element).
STATF-11	((Statistic Data Size - 4) * 4) + 11:12	Statistic Specific Data	(Statistic Data Size-4) *4	This contains the array element data where the first entry is array element [0], and the array elements are in ascending order with no gaps between elements. If padding is needed at the end of this data structure the data shall be padded with zeros.
STATF-12	(Statistic Data Size * 4) + 7:8	Statistic Specific Data	Statistic Data Size *4	The format of this data is defined by the statistic identifier and that custom format is specific to the individual Statistic Identifier value.

7.2.4 Statistic Identifiers Definitions

The definitions of the Statistic Identifiers are in the following sections. Statistic Identifiers that provide value to the vendor in debugging the device should be implemented.

7.2.4.1 Statistic Identifier Usage

The usage of Statistics Identifier values is described in the following table.

Requirement ID	Statistic Identifier	Statistic Specific Data Field	Description
STATI-GEN-1	0000h	-	Reserved.
STATI-GEN-2	0001h	-	Reserved.
STATI-GEN-3	0002h	ATA Log	This enables ATA Logs to be mapped into the Data Area (see Section 7.2.7.1 OCP ATA Log Statistic Log Descriptor Format (02h)).
STATI-GEN-4	0003h	SCSI Log	This enables SCSI Logs to be mapped into the Data Area (see Section 7.2.7.2 OCP SCSI Log Page Statistic Log Descriptor Format (03h)). The SCSI log page contents remain in their original SCSI big endian representation (see STAT3LG-8).
STATI-GEN-5	0004h - 1FFFh	Reserved	Statistic Identifiers between 0004h and 1FFFh are reserved for future expansion.

Requirement ID	Statistic Identifier	Statistic Specific Data Field	Description
STATI-GEN-6	2000h - 3FFFh	Common (HDD/SSD) Statistics	See Section 7.2.4.2 Common (HDD/SSD) Statistics Identifiers.
STATI-GEN-7	4000h - 5FFFh	SSD Specific Statistics	See Section 7.2.4.3 SSD Specific Statistics Identifiers.
STATI-GEN-8	6000h - 7FFFh	HDD Specific Statistics	See Section 7.2.4.4 HDD Specific Statistics Identifiers.
STATI-GEN-9	8000h - FFFFh	Vendor Unique	Statistic Identifiers between 8000h and FFFFh are vendor unique.

7.2.4.2 Common (HDD/SSD) Statistics Identifiers

The definitions of Statistics Identifiers that are common to HDD and SSD devices are in the following table.

Requirement ID	Statistic Identifier	Statistic Type (see STATF-2)	Description
	Recommended Behavior Type (see TEL-12)	Statistic Information	
STATI-COM-1	2000h	N/A	**Reserved.**
	N/A	N/A	
STATI-COM-2	2001h	0h (Single)	**Reallocated Block Count:**
	4h	Statistic Data Size: 2h	Shall contain the number of logical blocks that have been reallocated since the device was manufactured. This statistic is the same as the Number of Reallocated Logical Sectors statistic in the ATA Device Statistics Log (04h) Rotating Media Statistics log page (03h) when that statistic reports a raw block count (i.e., does not report a percentage). This statistic replaces the SMART Reallocated Sectors Count attribute (ID 5).
STATI-COM-3	2002h	0h (Single)	**Pending Defects Count:**
	1h	Statistic Data Size: 2h	Shall contain the number of pending defects currently listed in the Pending Defects log (ATA log 0Ch, SCSI log page/subpage 15h/01h). This statistic is the same as the Pending Error Count statistic in the ATA Device Statistics Log (04h), General Statistics log page (01h) and the

Requirement ID	Statistic Identifier	Statistic Type (see STATF-2)	Description
	Recommended Behavior Type (see TEL-12)	Statistic Information	
			Pending Defect Count parameter in the SCSI Pending Defects log (15h/01h). This statistic replaces the SMART Current Pending Sector Count attribute (ID 197).
STATI-COM-4	2003h	0h (Single)	**Power-On Hours Count:** Shall contain the amount of time in hours that the device has been operational since the device was manufactured. This statistic is the same as the Power-on Hours statistic in the ATA Device Statistics Log (04h) General Statistics log page (01h). This statistic replaces the SMART Power-On Hours count attribute (ID 9).
	4h	Statistic Data Size: 1h	
STATI-COM-5	2004h	0h (Single)	**Power Cycle Count:** Shall contain the number of times that power has been applied to the device since the device was manufactured. For SATA, this statistic counts power-on reset events. For SAS, this statistic counts power on events. This statistic is the same as the Lifetime Power-On Resets statistic in the ATA Device Statistics Log (04h) General Statistics log page (01h). This statistic replaces the SMART Power Cycle Count attribute (ID 12).
	4h	Statistic Data Size: 1h	
STATI-COM-6	2005h	0h (Single)	**Spare Blocks Used:** Shall contain the number of spare flash blocks (SSD) or physical blocks (HDD) that have been consumed since the device was manufactured. This statistic replaces the SMART Used Reserved Block Count Total attribute (ID 179).
	1h	Statistic Data Size: 2h	
	2006h	0h (Single)	**Spare Blocks Remaining:**

Requirement ID	Statistic Identifier Recommended Behavior Type (see TEL-12)	Statistic Type (see STATF-2) Statistic Information	Description
STATI-COM-7	1h	Statistic Data Size: 2h	Shall contain the number of spare flash blocks (SSD) or physical blocks (HDD) that are currently available for use. This statistic replaces the SMART Spare Blocks Available attribute (ID 16) when used for this purpose, the SMART Remaining Spare Blocks attribute (ID 17) when used for this purpose, and the SMART Unused Reserved Block Count Total attribute (ID 180).
STATI-COM-8	2007h	0h (Single)	**Unexpected Power Loss Count:** Shall contain the number of unexpected power loss events since the device was manufactured. This statistic replaces the SMART Unexpected Power Loss Count attribute (ID 174).
	4h	Statistic Data Size: 1h	
STATI-COM-9	2008h	0h (Single)	**Current Temperature:** Shall contain the current device temperature in degrees Celsius measured at the time this statistic is read. The value is represented as a signed (2's complement) integer. If the device has multiple temperature sensors, then the measured temperature shall be a composite temperature based on the multiple temperature sensors in the device. This statistic is the same as the Current Temperature statistic in the ATA Device Statistics Log (04h) Temperature Statistics log page (05h) and the Temperature log parameter (0000h) in the SCSI Temperature log page (0Dh/00h). This statistic replaces the SMART Temperature attribute (ID 194).
	1h	Statistic Data Size: 1h	
STATI-COM-10	2009h	0h (Single)	**Minimum Lifetime Temperature:**
	1h	Statistic Data Size: 1h	Shall contain the minimum powered-on device temperature in degrees Celsius experienced over the lifetime of the device. The value is represented as a signed (2's complement) integer.

Requirement ID	Statistic Identifier Recommended Behavior Type (see TEL-12)	Statistic Type (see STATF-2) Statistic Information	Description
			If the device has multiple temperature sensors, then the measured temperature shall be a composite temperature based on the multiple temperature sensors in the device.
STATI-COM-11	200Ah	0h (Single)	**Maximum Lifetime Temperature:** Shall contain the maximum powered-on device temperature in degrees Celsius experienced over the lifetime of the device. The value is represented as a signed (2's complement) integer. If the device has multiple temperature sensors, then the measured temperature shall be a composite temperature based on the multiple temperature sensors in the device.
	1h	Statistic Data Size: 1h	
STATI-COM-12	200Bh	0h (Single)	**Uncorrectable Read Error Count:** Shall contain the number of uncorrectable read errors (ATA) or unrecovered read errors (SCSI) that have occurred since the device was manufactured. This count excludes errors that occur during background activity and excludes any errors caused by reading a block that has been marked or flagged as containing an uncorrectable error (ATA) or a pseudo unrecoverable error (SCSI) by a command (e.g., WRITE UNCORRECTABLE EXT, WRITE LONG (10)). This statistic is the same as the Number of Reported Uncorrectable Errors statistic in the ATA Device Statistics Log (04h) General Errors Statistics log page (04h) except that exclusion of blocks that a command has marked or flagged as containing an error is required (shall) not just recommended (should). This statistic replaces the SMART Reported Uncorrectable Errors attribute (ID 187).
	4h	Statistic Data Size: 2h	
	200Ch	0h (Single)	**Background Uncorrectable Read Error Count:**

Requirement ID	Statistic Identifier Recommended Behavior Type (see TEL-12)	Statistic Type (see STATF-2) Statistic Information	Description
STATI-COM-13	4h	Statistic Data Size: 2h	Shall contain the number of uncorrectable read errors (ATA) or unrecovered errors (SCSI) that have been detected by background activity since the device was manufactured. These errors are excluded from the Uncorrectable Error Count (see STATI-COM-12). This statistic replaces the SMART Offline scan Uncorrectable Sector Count attribute (ID 198).
STATI-COM-14	200Dh	0h (Single)	**Interface CRC Error Count:**
	4h	Statistic Data Size: 2h	Shall contain the number of CRC errors that have occurred at the device interface since the device was manufactured. This statistic replaces the SMART CRC Error Count attribute (ID 199) and clarifies that the scope of this statistic is CRC errors at the device interface.
STATI-COM-15	200Eh	0h (Single)	**Volatile Memory Backup Source Failure:**
	1h	Statistic Data Size: 1h	Shall be set to 1h if the volatile memory backup solution has failed (e.g., SSD capacitor failure, HDD back EMF circuitry failure). If the device does not have a volatile memory backup solution, then this statistic shall be cleared to 0h. This statistic replaces the Volatile memory backup source failure SMART attribute (ID 201 when used for this purpose).
STATI-COM-16	200Fh	0h (Single)	**Read-Only Mode:**
	1h	Statistic Data Size: 1h	Shall be set to 1h if the media has been placed in read only mode because the device is unable to reliably write data to media or is completely unable to write data to media. This statistic replaces the Exception mode status SMART attribute (ID 202 when used for this purpose).
	2010h	0h (Single)	**Host Write Commands:**

Requirement ID	Statistic Identifier Recommended Behavior Type (see TEL-12)	Statistic Type (see STATF-2) Statistic Information	Description
STATI-COM-17	4h	Statistic Data Size: 2h	Shall contain the number of Write commands that have completed successfully since the device was manufactured. This statistic is incremented by one for each Write command that completes without error. A write command that invokes deallocation functionality (e.g., a WRITE SAME command with the UNMAP bit set to 1b and an all-zero data buffer) is a trim/unmap command and shall not be counted as a write command (see STATI-COM-24). This statistic is the same as the Number of Write Commands statistic in the ATA Device Statistics Log (04h), General Statistics log page (01h) and the NUMBER OF WRITE COMMANDS field in the General Access Statistics and Performance log parameter (0001h) in the SCSI General Statistics and Performance log page (19h/00h). This statistic replaces the SMART Write Commands Count attribute (ID 233 when used for this purpose).
STATI-COM-18	2011h	0h (Single)	**Host Read Commands:**
	4h	Statistic Data Size: 2h	Shall contain the number of read commands that have been completed successfully since the device was manufactured. This statistic is incremented by one for each read command that completes without error. This statistic is the same as the Number of Read Commands statistic in the ATA Device Statistics Log (04h) General Statistics log page (01h) and the NUMBER OF READ COMMANDS field in the General Access Statistics and Performance log parameter (0001h) in the SCSI General Statistics and Performance log page (19h/00h).
STATI-COM-19	2012h	0h (Single)	**Logical Blocks Read:**
	4h	Statistic Data Size: 2h	Shall contain the number of logical blocks sent to the host since the device was manufactured. This

Requirement ID	Statistic Identifier	Statistic Type (see STATF-2)	Description
	Recommended Behavior Type (see TEL-12)	Statistic Information	
			statistic is incremented by one for each logical block that was sent to the host without an error. This statistic is the same as the Logical Sectors Read statistic in the ATA Device Statistics Log (04h) General Statistics log page (01h) and the NUMBER OF LOGICAL BLOCKS TRANSMITTED field in the General Access Statistics and Performance log parameter (0001h) in the SCSI General Statistics and Performance log page (19h/00h). This statistic replaces the SMART Write Commands Count attribute (ID 234 when used for this purpose).
STATI-COM-20	2013h	0h (Single)	**Logical Blocks Written:** Shall contains the number of logical blocks received from the host since the device was manufactured. This statistic is incremented by one for each logical sector that was received from the host without an error. This statistic is the same as the Logical Sectors Written statistic in the ATA Device Statistics Log (04h), General Statistics log page (01h) and the NUMBER OF LOGICAL BLOCKS RECEIVED field in the General Access Statistics and Performance log parameter (0001h) in the SCSI General Statistics and Performance log page (19h/00h). This statistic replaces the SMART Write Commands Count attribute (ID 235 when used for this purpose).
	4h	Statistic Data Size: 2h	
	2014h	0h (Single)	**Total Media Writes:**

Requirement ID	Statistic Identifier Recommended Behavior Type (see TEL-12)	Statistic Type (see STATF-2) Statistic Information	Description
STATI-COM-21	4h	Statistic Data Size: 2h	Shall contain the number of bytes written to the media since the device was manufactured; this value includes both user and metadata written to the user and system areas. For an SSD device, it shall be possible to use this attribute to calculate the Write Amplification Factor (WAF). This statistic replaces the SMART Total Host Writes attribute (ID 241).
STATI-COM-22	2015h	0h (Single)	**Total Media Reads:**
	4h	Statistic Data Size: 2h	Shall contain the number of bytes read from the media since the device was manufactured; this value includes both user and metadata read from the user and system areas. This statistic replaces the SMART Total Host Reads attribute (ID 242).
STATI-COM-23	2016h	0h (Single)	**Soft ECC Error Count:**
	4h	Statistic Data Size: 2h	Shall contain the total count of media reads that were not correctable by first level ECC since the device was manufactured. This shall cover all media read accesses. Data recovery may have succeeded or failed. If the device has more than one recovery level, then this counter only increments once per recovery action. This statistic replaces the SMART Soft Read Error Rate attribute (ID 13) by providing an error counter from which a rate can be calculated over multiple values.
STATI-COM-24	2017h	0h (Single)	**Host Trim/Unmap Commands:**
	4h	Statistic Data Size: 2h	Shall contain the number of trim/unmap commands (e.g., DATA SET MANAGEMENT with the TRIM bit set to 1b, UNMAP) that have been completed successfully since the device was manufactured. This statistic is incremented by one

Requirement ID	Statistic Identifier Recommended Behavior Type (see TEL-12)	Statistic Type (see STATF-2) Statistic Information	Description
			for each trim/unmap command that completes without error. A write command that invokes trim or unmap functionality is a trim/unmap command and shall not be counted as a Write command (see STATI-COM-17). For SATA, trim/unmap commands are DATA SET MANAGEMENT commands that have the TRIM bit set to 1b, DATA SET MANAGEMENT XL commands that have the TRIM bit set to 1b, and ZERO EXT commands that have the TRIM bit set to 1b. For SAS, trim/unmap commands are UNMAP commands and WRITE SAME commands (i.e., WRITE SAME (10), WRITE SAME (16) and WRITE SAME (32) commands) that have the UNMAP bit set to 1b.
STATI-COM-25	2018h 4h	0h (Single) Statistic Data Size: 1h	**End-to-End Detected Errors:** Shall contain the number of errors detected by internal end-to-end error protection mechanisms, which include DRAM, SRAM, or other storage element ECC/CRC protection mechanism, but excludes media protection mechanisms (e.g., media parity, media ECC or media RAID).
STATI-COM-26	2019h 4h	0h (Single) Statistic Data Size: 1h	**End-to-End Corrected Errors:** Shall contain the number of errors detected and corrected by internal end-to-end error protection mechanisms which includes DRAM, SRAM, or other storage element ECC/CRC protection mechanism, but excludes media protection mechanisms (e.g., media parity, ECC or RAID).
STATI-COM-27	201Ah 3h	0h (Single) Statistic Data Size: 2h	**Unaligned I/O Count:** Shall contain the number of write I/Os performed by the device that are not aligned to the Indirection Unit (IU) size or physical block size of the device.

<table>
<tr><th rowspan="2">Requirement ID</th><th>Statistic Identifier</th><th>Statistic Type (see STATF-2)</th><th rowspan="2">Description</th></tr>
<tr><th>Recommended Behavior Type (see TEL-12)</th><th>Statistic Information</th></tr>
<tr><td></td><td></td><td></td><td>Alignment indicates only the start and end of each IO.

This statistic shall be cleared to zero on factory exit.</td></tr>
<tr><td rowspan="2">STATI-COM-28</td><td>201Bh</td><td>0h (Single)</td><td rowspan="2">Security Version Number:
Shall contain the Security Version Number of the currently running firmware image. The value of this field shall be 0h for the initial release of firmware. The supplier shall increment this field by one any time a firmware includes a fix for a security issue or critical firmware fix that customer agrees rollback prevention is required.</td></tr>
<tr><td>0h</td><td>Statistic Data Size: 1h</td></tr>
<tr><td rowspan="2">STATI-COM-29</td><td>201Ch</td><td>0h (Single)</td><td>Thermal Throttling Status:</td></tr>
<tr><td>1h</td><td>Statistic Data Size: 1h</td><td>Shall contain the status of device thermal throttling as follows:
<table>
<tr><th>Value (h)</th><th>Description</th></tr>
<tr><td>00</td><td>Unthrottled</td></tr>
<tr><td>01</td><td>1st level throttle</td></tr>
<tr><td>02</td><td>2nd level throttle</td></tr>
<tr><td>03</td><td>3rd level throttle</td></tr>
<tr><td>All other values</td><td>Reserved</td></tr>
</table></td></tr>
<tr><td rowspan="2">STATI-COM-30</td><td>201Dh</td><td>0h (Single)</td><td rowspan="2">Thermal Throttling Count:
Shall contain a count of the number of thermal throttling initiation events independent of level. This statistic shall be cleared to zero on factory exit.</td></tr>
<tr><td>4h</td><td>Statistic Data Size: 1h</td></tr>
<tr><td rowspan="2">STATI-COM-31</td><td>201Eh</td><td>0h (Single)</td><td>DSS Specification Version:</td></tr>
<tr><td>0h</td><td>Statistic Data Size: 1h</td><td>Shall contain the version of the Datacenter SAS-SATA Specification that this device conforms to formatted as 4 bytes:
<table>
<tr><th>Byte Offset</th><th>Description</th></tr>
<tr><td>00</td><td>Major Version: shall be 1h</td></tr>
<tr><td>01</td><td>Minor Version: shall be 5h</td></tr>
<tr><td>02</td><td>Point Version: shall be 0h</td></tr>
<tr><td>03</td><td>Errata version: shall be 0h</td></tr>
</table></td></tr>
</table>

Requirement ID	Statistic Identifier	Statistic Type (see STATF-2)	Description
	Recommended Behavior Type (see TEL-12)	Statistic Information	
STATI-COM-32	201Fh	0h (Single)	**Incomplete Shutdown Count:** Shall contain the number of shutdowns and power loss events that have occurred and that did not completely flush all required user data and metadata to non-volatile memory for any reason. This statistic shall be cleared to zero on factory exit.
	4h	Statistic Data Size: 2h	
STATI-COM-33	2020h	0h (Single)	**Percent Free Blocks:** Shall contain the normalized count of the number of NAND blocks which are currently erased and ready to be written with user data, as a percentage of the total number of NAND blocks in the spare area. i.e., this is the percentage of erased NAND blocks in spare area, where the spare area is defined as the NAND blocks that contain no valid user data, no valid system data, and are: • waiting to be erased for use to write user data; or • already erased and ready to be used to write user data. (i.e., Percent Free Blocks = A / (A + B) expressed as an integer percentage where A is the number of currently erased NAND blocks and B is the number of NAND blocks waiting to be erased.)
	1h	Statistic Data Size: 1h	
STATI-COM-34	2021h	0h (Single)	**Lowest Permitted Firmware Revision:** This field indicates the value of the Firmware Revision field of the lowest firmware revision that is permitted to be rolled back to.
	0h	Statistic Data Size: 2h	
STATI-COM-35	2022h	0h (Single)	**Maximum Peak Power Capability:** The maximum peak power that the device can measure in Watts over a 100 µS window with a sampling interval of 4 µS or smaller (see PWR-3). This is a static value based on the worst case.
	0h	Statistic Data Size: 1h	
STATI-COM-36	2023h	0h (Single)	**Current Maximum Average Power:** The current device Max Average Power in Watts measured over a 1 second window with a sampling rate of 2ms or smaller (see PWR-2).
	1h	Statistic Data Size: 1h	
	2024h	0h (Single)	**Lifetime Power Consumed:**

Requirement ID	Statistic Identifier Recommended Behavior Type (see TEL-12)	Statistic Type (see STATF-2) Statistic Information	Description
STATI-COM-37	4h	Statistic Data Size: 2h	The total cumulative power consumed by the device in kWh based on the Current Max Average Power. This field shall be cleared to 0h from the factory.
	2025h	0h (Single)	**Power Changes:**
STATI-COM-38	4h	Statistic Data Size: 2h	Shall contain a count of the number of changes to power condition (SAS and SATA devices) or power mode (SATA devices only), whether host or device initiated. This counter shall only increment during run-time. This statistic shall be cleared to zero on factory exit.
STATI-COM-39	2026h	0h (Single)	**Phy Reinitialization Count:**
	4h	Statistic Data Size: 2h	Shall contain a count of the number of times a link Phy is initialized excluding any initialization caused by application of power. For a dual-port SAS device, this statistic is for the primary port. A Phy initialization begins with a COMRESET signal or a COMINIT signal (whichever occurs first) and ends when the Phy becomes ready and able to transmit data. Each link Phy initialization that is not caused by a power cycle or initial application of power is counted once even if COMRESET and/or COMINIT occur again before the Phy becomes ready.
STATI-COM-40	2027h	0h (Single)	**Secondary Phy Reinitialization Count:**
	4h	Statistic Data Size: 2h	Shall contain the Phy Reinitialization Count (see STATI-COM-39) for the secondary port of a dual port SAS Device (see Section 6.5 Dual Port Requirements). This statistic shall not be reported by a single port device.
STATI-COM-41	2028h	0h (Single)	**Command Timeouts:**
	4h	Statistic Data Size: 2h	For a SATA device, this shall contain a count of SATA commands that were completed with error because a Command Timer expired. For a SAS device, this shall contain a count of SAS commands that were terminated with CHECK

Requirement ID	Statistic Identifier	Statistic Type (see STATF-2)	Description
	Recommended Behavior Type (see TEL-12)	Statistic Information	
			CONDITION STATUS because a recovery time limit was exceeded.
STATI-COM-42	2029h	0h	**Hardware Revision:** This is the hardware revision of the device. This is a 1s based value so a value of 0h means this field is invalid. The hardware revision shall start at 1h and increment in ascending order.
	4h	Statistic Data Size: 1h	
STATI-COM-43	202Ah	0h	**Firmware Revision:** This is the firmware revision of the device. A value of 0h means this field is invalid.
	4h	Statistic Data Size: 1h	
STATI-COM-44	202Bh - 3FFFh	-	**Reserved:** Statistic Identifiers for future expansion.
	-	-	

7.2.4.3 SSD Specific Statistics Identifiers

The definitions of Statistics Identifiers that are specific to SSD devices are in the following table.

Requirement ID	Statistic Identifier	Statistic Type (see STATF-2)	Description
	Recommended Behavior Type (see TEL-12)	Statistic Information	
STATI-SSD-1	4000h	-	**Reserved.**
	-	-	
STATI-SSD-2	4001h	0h (Single)	**Raw Capacity:** Shall contain the raw device capacity (in GB or TB) of the SSD. Raw device capacity is the total NAND capacity without any reductions (e.g., for ECC, internal RAID, or overprovisioning). This statistic replaces the SMART Raw flash capacity attribute (ID 14) when used for this purpose.
	0h	Statistic Data Size: 1h	
STATI-SSD-3	4002h	0h (Single)	**User Capacity:** Shall contain the maximum capacity addressable at the device interface.
	0h	Statistic Data Size: 1h	

Requirement ID	Statistic Identifier Recommended Behavior Type (see TEL-12)	Statistic Type (see STATF-2) Statistic Information	Description
			This statistic reports the same capacity as the Device Capacity statistic in the ATA IDENTIFY DEVICE data log (30h) Capacity log page (01h). This statistic is reported as capacity instead of a logical block count (used by the ATA statistic). This statistic replaces the SMART User Capacity attribute (ID 15) when used for this purpose.
STATI-SSD-4	4003h 4h	0h (Single) Statistic Data Size: 2h	**Erase Count:** Shall contain the total number of flash block erase operations across the entire device since the device was manufactured. This statistic replaces the SMART Erase Count attribute (ID 100) when used for this purpose.
STATI-SSD-5	4004h 4h	0h (Single) Statistic Data Size: 2h	**Erase Fail Count:** Shall contain the total number of flash block erase failures for the entire device since the device was manufactured. This statistic replaces the SMART Erase Fail Count (total) attribute (ID 172 and ID 182).
STATI-SSD-6	4005h 1h	0h (Single) Statistic Data Size: 1h	**Maximum Erase Count:** Shall contain the maximum erase count of all flash blocks across the entire device. If a block goes bad, any subsequent attempts to recover the block shall not increment this statistic. This statistic replaces the SMART Wear Leveling Count attribute (ID 173).
STATI-SSD-7	4006h 4h	0h (Single) Statistic Data Size: 1h	**Average Erase Count:** Shall contain the average number of Erase operations across all flash blocks

Requirement ID	Statistic Identifier	Statistic Type (see STATF-2)	Description
	Recommended Behavior Type (see TEL-12)	Statistic Information	
			across in the entire device since the device was manufactured. This statistic replaces the SMART Average Erase Count attribute (ID 175) when used for this purpose.
STATI-SSD-8	4007h	0h (Single)	**Program Fail Count:** Shall contain the number of flash programming failures across the entire device since the device was manufactured. This statistic replaces the SMART Program Fail Count (total) attribute (ID 171, ID 181, and ID 175 when used for this purpose).
	4h	Statistic Data Size: 2h	
STATI-SSD-9	4008h	0h (Single)	**XOR Recovery Count:** For SSD devices that contain internal RAID or similar redundancy mechanisms that can rebuild data when a read fails due to read corruption that is not corrected by ECC, this shall contain the number of times that an internal rebuild has been activated for both successful rebuilds and failed rebuilds since the device was manufactured. This statistic replaces the SMART UECC rebuild event count attribute (ID 200 when used for this purpose). This statistic shall be cleared to zero on factory exit.
	4h	Statistic Data Size: 2h	
STATI-SSD-10	4009h	0h (Single)	**Percent Device Life Remaining:** Shall contain a vendor specific estimate of the percentage device life remaining based on the actual usage and the manufacturer's prediction of media life. A zero value indicates that the estimated endurance of the media has been reached but may not indicate a device
	1h	Statistic Data Size: 1h	

Requirement ID	Statistic Identifier Recommended Behavior Type (see TEL-12)	Statistic Type (see STATF-2) Statistic Information	Description
			failure. This value is allowed to exceed 100. Percentages greater than 254 shall be represented as 255. This value shall be updated once per power-on hour (when the controller is not in a sleep state). This statistic replaces the SMART Remaining Rated Write Endurance attribute (ID 245).
STATI-SSD-11	400Ah 0h	0h (Single) Statistic Data Size: 1h	**Lifetime Erase Count:** Shall contain the rated number of Program-Erase cycles for the lifetime of a flash block. An individual flash block has reached the end of its rated life after this number of Program-Erase cycles.
STATI-SSD-12	400Bh 4h	0h (Single) Statistic Data Size: 2h	**Bad User NAND Blocks:** Shall contain the number of User NAND blocks that have been retired for any reason (e.g., program fails, erase fails, other events). If the Bad System NAND Blocks statistic is not reported, this statistic represents all NAND on the device. This statistic shall be cleared to zero on factory exit.
STATI-SSD-13	400Ch 4h	0h (Single) Statistic Data Size: 2h	**Bad System NAND Blocks:** Shall contain the number of System NAND blocks that have been retired for any reason (e.g., program fails, erase fails, other events). If this statistic is not reported, then the Bad User NAND Blocks statistic (see STATI-SSD-12) represents all NAND on the device. This statistic shall be cleared to zero on factory exit.

Requirement ID	Statistic Identifier	Statistic Type (see STATF-2)	Description
	Recommended Behavior Type (see TEL-12)	Statistic Information	
STATI-SSD-14	400Dh	0h (Single)	**Minimum Erase Count:**
	1h	Statistic Data Size: 1h	Shall contain the minimum erase count of all flash blocks across the entire device. If a block goes bad, any subsequent attempts to recover the block shall not increment this statistic.
STATI-SSD-15	400Eh	0h (Single)	**Power Loss Protection Start Count:**
	4h	Statistic Data Size: 4h	Shall contain a count of the number of times the device has initiated its power loss protection process due to supply voltage drop, independent of whether power was actually lost or not. This counter shall be incremented on the initial detection of the power loss condition. This does not include PLP health check operations.
STATI-SSD-16	400Fh	0h (Single)	**System Data Percent Used:**
	1h	Statistic Data Size: 1h	Shall contain a normalized cumulative count of the number of Erase cycles per block since leaving the factory for the system (firmware and metadata) area. Starts at 0 and increments. 100 indicates that the estimated endurance has been consumed. The value may exceed 100 up to 255. This count shall increment regardless of what the backing media of the blocks are (e.g., SLC and TLC). If system data is split between media types, then this statistic shall report the worst-case count so that the device wear out is clearly understood. For this counter, 100% (64h) indicates that the device may no longer function reliably as the max erase cycles has been hit – this differs from the implications of the Percent Device Life Remaining statistic (see STATI-SSD-10) reaching 0%.
	4010h	0h (Single)	**Power Loss Protection Health:**

Requirement ID	Statistic Identifier	Statistic Type (see STATF-2)	Description
	Recommended Behavior Type (see TEL-12)	Statistic Information	
STATI-SSD-17	1h	Statistic Data Size: 1h	Shall contain an indicator of the capacitor health that represents the capacitor holdup energy margin during operation. If no PLP protection is present a value of FFFFh shall be reported. 100% represents the minimum acceptable hold up energy threshold when a device leaves manufacturing. Thus, a device will typically report greater than 100% in this field after leaving manufacturing at the beginning of life. 1% is the minimum hold up energy required to conduct a proper shutdown reliably. A value of 0% may or may not result in a device failing to shut down properly. This value shall never go negative. Zero is the minimum.
STATI-SSD-18	4011h	0h (Single)	**Endurance Estimate:**
	0h	Statistic Data Size: 4h	Shall contain an estimate of the total number of data bytes that may be written to the device over its lifetime assuming a write amplification of 1 (i.e., no increase in the number of write operations performed by the device beyond the number of write operations requested by a host).
STATI-SSD-19	4012h	0h (Single)	**Percent User Spare Available:**
	1h	Statistic Data Size: 1h	The ratio of available user spare blocks to the original number of user spare blocks expressed as a percentage. This statistic shall be set to 64h (i.e., 100) on factory exit.
STATI-SSD-20	4013h	0h (Single)	**Percent System Spare Available:**
	1h	Statistic Data Size: 1h	The ratio of available system spare blocks to the original number of system spare blocks expressed as a percentage. This statistic shall be set to 64h (i.e., 100) on factory exit. If the Bad System NAND Blocks statistic (see STATI-SSD-13) is not reported, then this statistic shall not be reported.

Requirement ID	Statistic Identifier Recommended Behavior Type (see TEL-12)	Statistic Type (see STATF-2) Statistic Information	Description
STATI-SSD-21	4014h 0h	0h (Single) Statistic Data Size: 1h	**Total Media Dies:** Shall contain the total number of media dies in the device.
STATI-SSD-22	4015h 0h	0h (Single) Statistic Data Size: 1h	**Media Die Failure Tolerance:** Shall contain the maximum number of NAND dies that are allowed to fail and the device will continue to operate.
STATI-SSD-23	4016h 4h	0h (Single) Statistic Data Size: 1h	**Media Dies Offline:** Shall contain the number of media dies permanently offline (retired).
STATI-SSD-24	4017h 4h	0h (Single) Statistic Data Size: 1h	**System Area Program Fail Count:** Shall contain the number of program failures in the system area. If this statistic is not reported, then the Program Fail Count statistic (see STATI-SSD-8) represents all NAND on the device. This statistic shall be cleared to zero on factory exit.
STATI-SSD-25	4018h 1h	0h (Single) Statistic Data Size: 1h	**System Area Program Fail Percentage Remaining:** Shall contain the number of allowed system area program failures remaining expressed as a percentage of the failures allowed. This statistic shall be set to 64h (i.e., 100) on factory exit. If the System Area Program Fail Count statistic (see STATI-SSD-24) is not reported, then this statistic shall not be reported.
STATI-SSD-26	4019h 4h	0h (Single) Statistic Data Size: 1h	**System Area Uncorrectable Read Error Count:** Shall contain the number of read failures (Uncorrectable Reads) in the system area. If this statistic is not reported, then the Uncorrectable Read Error Count (see STATI-COM-12) and Background Uncorrectable Read Error Count (see

Requirement ID	Statistic Identifier Recommended Behavior Type (see TEL-12)	Statistic Type (see STATF-2) Statistic Information	Description
			STATI-COM-13) statistics represent all NAND on the device. This statistic shall be cleared to zero on factory exit.
STATI-SSD-27	401Ah 4h	0h (Single) Statistic Data Size: 1h	**System Area Uncorrectable Read Percentage Remaining:** Shall contain the number of allowed system area read failures (Uncorrectable Reads) remaining expressed as a percentage of the failures allowed. This statistic shall be set to 64h (i.e., 100) on factory exit. If the System Area Uncorrectable Read Count statistic (see STATI-SSD-26) is not reported, then this statistic shall not be reported.
STATI-SSD-28	401Bh 0h	0h (Single) Statistic Data Size: 1h	**System Area Erase Fail Count:** Shall contain the number of Erase failures in the system area. If this statistic is not reported, then the Erase Fail Count statistic (see STATI-SSD-5) represents all NAND on the device. This statistic shall be cleared to zero on factory exit.
STATI-SSD-29	401Ch 4h	0h (Single) Statistic Data Size: 1h	**System Area Erase Fail Percentage Remaining:** Shall contain the number of allowed system area erase failures remaining expressed as a percentage of the failures allowed. This statistic shall be set to 64h (i.e., 100) on factory exit. If the System Area Erase Faile Count statistic (see STATI-SSD-28) is not reported, then this statistic shall not be reported.
STATI-SSD-30	401Dh-5FFFh	-	**Reserved:**

Requirement ID	Statistic Identifier Recommended Behavior Type (see TEL-12)	Statistic Type (see STATF-2) Statistic Information	Description
	-	-	Statistic Identifiers for future expansion.

7.2.4.4 HDD Specific Statistics Identifiers

The definitions of Statistics Identifiers that are specific to HDD devices are in the following table.

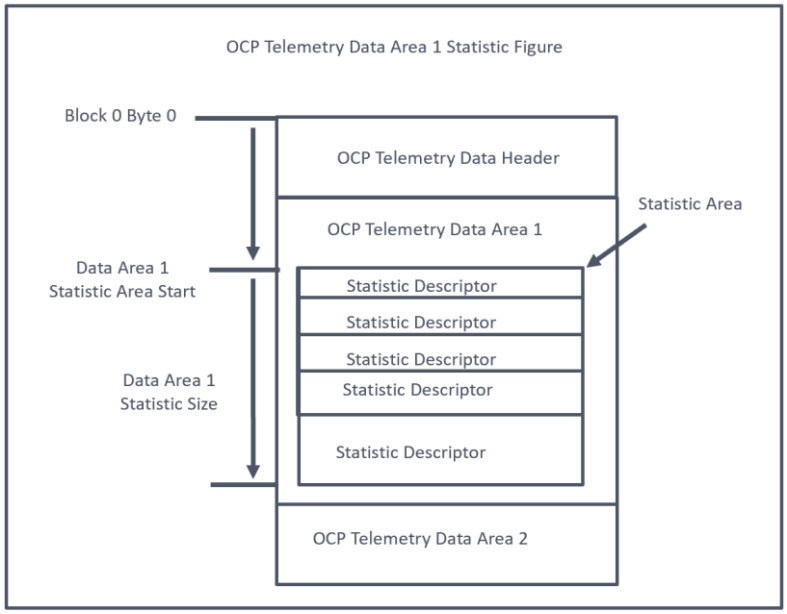
Requirement ID	Statistic Identifier Recommended Behavior Type (see TEL-12)	Statistic Type (see STATF-2) Statistic Information	Description
STATI-HDD-1	6000h -	- -	**Reserved.**
STATI-HDD-2	6001h 4h	0h (Single) Statistic Data Size: 2h	**Start/Stop Count:** Shall contain the number of start-stop cycles (complete device spin-up followed by complete device spin-down) since the device was manufactured. This statistic is the same as the Accumulated Start-Stop Cycles parameter in the SCSI Start-Stop Cycle Counter log page (0Eh/00h). This statistic replaces the SMART Start/Stop Count attribute (ID 4).
STATI-HDD-3	6002h 4h	0h (Single) Statistic Data Size: 2h	**Load Cycle Count:** Shall contain the number of head load events (head load from ramp or landing zone) since the device was manufactured. This count is incremented when the heads are moved, independent of whether the device fully spins up and becomes operational. This statistic is the same as the Head Load Events statistic in the ATA Device Statistics Log (04h) Rotating Media Statistics log page (03h). This statistic replaces the SMART Load/Unload Cycle Count attribute (ID 193).

Requirement ID	Statistic Identifier Recommended Behavior Type (see TEL-12)	Statistic Type (see STATF-2) Statistic Information	Description
STATI-HDD-4	6003h 4h	0h (Single) Statistic Data Size: 1h	**Shock Overlimit Count:** Shall contain the number of shock events detected by the device that have a magnitude higher than the maximum shock rating of the device since the device was manufactured. This statistic shall not be reported by a device that does not have a shock, free fall, or G-force sensor. This statistic is the same as the Overlimit Shock Events statistic in the ATA Device Statistics Log (04h) Free Fall Statistics log page (02h). This statistic replaces the SMART G-sense Error Rate attribute (ID 191).
STATI-HDD-5	6004h 4h	0h (Single) Statistic Data Size: 1h	**Head Flying Hours:** Shall contain number of hours that the device heads have been flying over the surface of the media since the device was manufactured. This statistic is the same as the Head Flying Hours statistic in the ATA Device Statistics Log (04h) Rotating Media Statistics log page (03h). This statistic replaces the SMART Head Flying Hours attribute (ID 240).
STATI-HDD-6	6005h 4h	0h (Single) Statistic Data Size: 1h	**Free Fall Events Count:** Shall contain the number of free-fall events that have been detected by the device since the device was manufactured. This statistic shall not be reported by a device that does not have a shock, free fall, or G-force sensor. This statistic is the same as the Number of Free-Fall Events Detected statistic in the

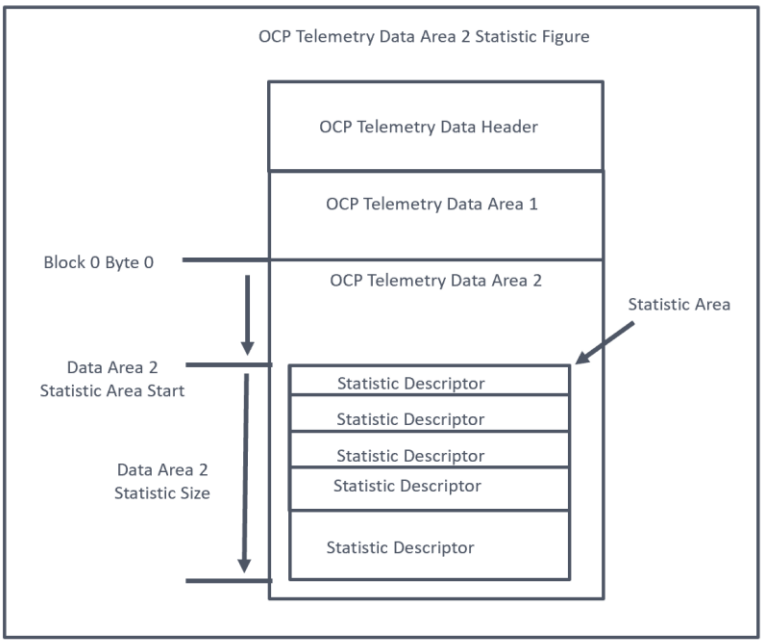
Requirement ID	Statistic Identifier Recommended Behavior Type (see TEL-12)	Statistic Type (see STATF-2) Statistic Information	Description
			ATA Device Statistics Log (04h) Free Fall Statistics log page (02h). This statistic replaces the SMART Free Fall Events Count attribute (ID 243).
STATI-HDD-7	6006h 4h	2h (Custom) -	**Spinup Times:** This contains statistics related to device spinup times (see Section 7.2.7.3 HDD Spinup Statistic Format (6006h)).
STATI-HDD-8	6007h-7FFFh -	- -	**Reserved:** Statistic Identifiers for future expansion.

7.2.5 Statistic Descriptor Location

The Statistic Area location in the OCP Telemetry Data 1 Area in the Telemetry Data Structure can be found by using the Data Area 1 Statistic Area Start and Data Area 1 Statistic Area Size. This can be seen in the following OCP Data Area 1 Statistics Figure:



The Statistic Area location in the OCP Telemetry Data 2 Area in the Telemetry Data Structure can be found by using the Data Area 2 Statistic Area Start and Data Area 2 Statistic Area Size. This can be seen in the following OCP Data Area 2 Statistics Figure:



7.2.6 Statistic Requirements

Requirement ID	Description
SDL-1	In the Telemetry Data Structure, Statistic Area 1 shall not cross from OCP Telemetry Data Area 1 to OCP Telemetry Data Area 2.
SDL-2	Bytes in the Statistics Areas that are not contained in a Statistic Descriptor shall be zero.
SDL-3	The decode of the Statistic Identifiers shall be provided to the customer via the Telemetry Strings Structure.

7.2.7 Custom Statistics

7.2.7.1 OCP ATA Log Statistic Log Descriptor Format (02h)

This enables mapping one or more log pages in an ATA Log to a Statistic Log Descriptor. An ATA Log has multiple pages; each page uses a separate instance of this descriptor. The format of the Statistic Log Descriptor is in the following table:

Requirement ID	Byte Address	Field	# of Bytes	Field Description
STAT2LG-1	1:0	Statistic Identifier	2	This shall be set to 02h.
STAT2LG-2	2	Statistic Information 1	1	The Statistic Type shall be set to 2 (Custom). The Behavior Type shall be set to 1 (Runtime Value) (see STATF-2 and TEL-12).
STAT2LG-3	3	Statistic Information 2 (Data Unit)	1	This shall be cleared to 0h (No Data Unit Information) (see STATF-3).

Requirement ID	Byte Address	Field	# of Bytes	Field Description
STAT2LG-4	4	Statistic Information 3	1	The Host Hint Type shall be cleared to 0h (No Host Hint). The Data Type shall be cleared to 0h (No Data Type Information).
STAT2LG-5	5	Reserved	1	Shall be cleared to 0h.
STAT2LG-6	7:6	Statistic Data Size	2	Shall contain the number of Dwords that are used by the Statistic Specific Field. This statistic contains one or more 512-byte ATA log pages and hence this field shall be set to 1 + 128 * Log Page Count (see STAT2LG-8).
Statistic Specific Field				
STAT2LG-7	8	Log Address	1	This is the ATA Log Address (see ACS-5).
STAT2LG-8	9	Log Page Count	1	This is the number of log pages (i.e., the COUNT input to an ATA command that reads log pages) (see ACS-5).
STAT2LG-9	11:10	Initial Log Page	2	This is the ATA log page number of the initial log page. Log Pages are numbered starting with Log Page 0h (see ACS-5).
STAT2LG-10	(512 * Log Page Count) + 11:12	Log Page Data	(512 * Log Page Count)	This contains the ATA log page data.

7.2.7.2 OCP SCSI Log Page Statistic Log Descriptor Format (03h)

This enables mapping an SCSI Log page to a Statistic Log Descriptor. The format of the Statistic Log Descriptor is in the following table:

Requirement ID	Byte Address	Field	# of Bytes	Field Description
STAT3LG-1	1:0	Statistic Identifier	2	This shall be set to 03h.
STAT3LG-2	2	Statistic Information 1	1	The Statistic Type shall be set to 2 (Custom). The Behavior Type shall be set to 1 (Runtime Value) (see STATF-2 and TEL-12).
STAT3LG-3	3	Statistic Information 2 (Data Unit)	1	This shall be cleared to 0h No Data Unit Information) (see STATF-3).
STAT2LG-4	4	Statistic Information 3	1	The Host Hint Type shall be cleared to 0h (No Host Hint). The Data Type shall be cleared to 0h (No Data Type Information).
STAT2LG-5	5	Reserved	1	Shall be cleared to 0h.

Requirement ID	Byte Address	Field	# of Bytes	Field Description
STAT3LG-6	7:6	Statistic Data Size	2	Shall contain the number of Dwords that are used by the Statistic Specific Field. SCSI log pages are variable size and use a 4-byte header that is excluded from the reported log page size, e.g., if that SCSI log page header reports a 512-byte log page size, then there are 516 bytes of log page data plus 4 bytes of Statistic Specific Field header (see STAT3LG-7 through STAT3LG-10) for a total of 520 bytes, hence this field is set to 82h.
Statistic Specific Field				
STAT3LG-7	8	Log Page	1	This is the SCSI log page code (see SPC-5).
STAT3LG-8	9	Log Subpage	1	This is the SCSI log subpage code (see SPC-5). If the log specified by the log page code (see STAT3LG-7) does not use log subpages (e.g., the SPF bit in the SCSI log page data is cleared to 0b), then this field shall be cleared to 0h.
STAT3LG-9	11:10	Reserved	2	This shall be cleared to 0.
STAT3LG-10	(Statistic Data Size * 4) + 7:12	Log Page	(Statistic Data Size * 4) - 4	This contains the SCSI log page data. This log page data shall remain in its original SCSI big endian representation (e.g., shall not be byte-swapped to little endian representation).

7.2.7.3 HDD Spinup Statistic Format (6006h)

The format of the HDD Spinup Statistic Format is in the following table:

<table>
<tr><th>Requirement ID</th><th>Byte Address</th><th>Field</th><th># of Bytes</th><th>Field Description</th></tr>
<tr><td>STATSPIN-1</td><td>1:0</td><td>Statistic Identifier</td><td>2</td><td>This shall be set to 6006h.</td></tr>
<tr><td>STATSPIN-2</td><td>2</td><td>Statistic Information 1</td><td>1</td><td><table><tr><th>Bit</th><th>Statistic Information 1 Bit Description</th></tr><tr><td>7:4</td><td>Statistic Type. This shall be set to 2h (Custom).</td></tr><tr><td>3:0</td><td>Behavior Type (see TEL-12). This shall be set to 4h.</td></tr></table></td></tr>
<tr><td>STATSPIN-3</td><td>3</td><td>Statistic Information 2</td><td>1</td><td>Statistic Information 2: Data Unit. This provides the data units used in the Statistic Specific Field. This shall be set to 01h (Milliseconds).</td></tr>
<tr><td>STATSPIN-4</td><td>4</td><td>Statistic Information 3</td><td>1</td><td>Statistic Information 3<table><tr><th>Bit</th><th>Statistic Information 3 Bit Description</th></tr><tr><td>7:6</td><td>Reserved. Shall be cleared to zero.</td></tr><tr><td>5:4</td><td>Host Hint Type. This shall be set to 00b (No Host Hint).</td></tr></table></td></tr>
</table>

Requirement ID	Byte Address	Field	# of Bytes	Field Description
				3:0 Data Type. This shall be set to 02h (Unsigned Integer).
STATSPIN-5	5	Reserved	1	Shall be cleared to 0h.
STATSPIN-6	7:6	Statistic Data Size	2	Shall contain the number of Dwords that is used by the Statistic Specific Field. This shall be set to 18h.
Statistic Specific Field (Statistic Type = 2h)				
STATSPIN-7	9:8	Lifetime Spinup Max	2	This is the maximum spin up time in the lifetime of the device. For values greater than 0xFFFF a value of 0xFFFF shall be used. A value of 0h means that this field is not valid.
STATSPIN-8	11:10	Lifetime Spinup Min	2	This is the minimum spin up time in the lifetime of the device. For values greater than 0xFFFF a value of 0xFFFF shall be used. A value of 0h means that this field is not valid.
STATSPIN-9	13:12	Spinup History[0]	2	Spinup history. This is an array of the 10 most recent spinup times. Spinup History[0] is the most recent spinup time and Spinup History[9] is the oldest spinup up time. The array shall be ordered from the most recent to the oldest spinup times with the entries being consecutive spinup times. For values greater than 0xFFFF a value of 0xFFFF shall be used. A value of 0h means that a field is not valid.
STATSPIN-10	15:14	Spinup History[1]	2	
STATSPIN-11	17:16	Spinup History[2]	2	
STATSPIN-12	19:18	Spinup History[3]	2	
STATSPIN-13	21:20	Spinup History[4]	2	
STATSPIN-14	23:22	Spinup History[5]	2	
STATSPIN-15	25:24	Spinup History[6]	2	
STATSPIN-16	27:26	Spinup History[7]	2	
STATSPIN-17	29:28	Spinup History[8]	2	
STATSPIN-18	31:30	Spinup History[9]	2	

7.2.8 Event FIFO Requirements

7.2.8.1 Event Descriptor

The Event Descriptor is used to log debug events in the Event FIFOs in the Device Telemetry Data Structure.

Requirement ID	Byte Address	Field	# of Bytes	Field Description
EVENTD-1	0	Debug Event Class Type	1	Shall contain the corresponding Event Class Type for this event.
EVENTD-2	2:1	Event Identifier	2	Shall contain the corresponding Event Identifier for this event.
EVENTD-3	3	Event Data Size	1	Shall contain the number of Dwords used by the Event Specific Data field.
EVENTD-4	(Event Data Size * 4) + 3:4	Event Specific Data	Event Data Size * 4	Shall contain the event specific extended information for this event (if any).

7.2.8.2 Event Classes

This section defines the Event Class information that can be used to debug issues with the device.

7.2.8.2.1 Debug Event Class Types

Debug Event Class Code	Description
00h	Reserved
01h	Timestamp Debug Class
02h	Reserved
03h	Reserved
04h	Reset Debug Class
05h	Boot Sequence Debug Class
06h	Firmware Assert Debug Class
07h	Temperature Debug Class
08h	Media Debug Class
09h	Media Wear Class
0Ah	Statistic Snapshot Class
0Bh	Virtual FIFO Event Class
0Ch	SATA Phy/Link Debug Class
0Dh	SATA Transport Debug Class
0Eh	SAS Phy/Link Debug Class
0Fh	SAS Transport Debug Class
7Fh - 10h	Reserved
FFh - 80h	Vendor Unique Class

7.2.8.2.2 Timestamp Debug Event Class Format

Requirement ID	Byte Address	Field	# of Bytes	Field Description
EVC-TS-1	0	Timestamp Debug Class	1	Shall be set to 01h.
EVC-TS-2	2:1	Event Identifier	2	<table><tr><th>Event ID</th><th>Description</th></tr><tr><td>0000h</td><td>Host issued an ATA SET DATE & TIME EXT Command or a SCSI SET TIMESTAMP Command.</td></tr><tr><td>0001h</td><td>Firmware initiated Timestamp Snapshot.</td></tr><tr><td>0002h</td><td>Obsolete.</td></tr><tr><td>0003h - 7FFFh</td><td>Reserved.</td></tr><tr><td>8000h - FFFFh</td><td>Vendor Unique.</td></tr></table>
EVC-TS-3	3	Event Data Size	1	This field indicates the number of Dwords in the Event Specific Data field. If the VU Event Identifier field and VU data field do not exist then, then this field shall be set to 2h.
Event Specific Data				
EVC-TS-4	11:4	Timestamp	8	Timestamp value based on Event ID, in the format defined in the NVM Express Base Specification.
EVC-TS-5	13:12	VU Event Identifier	2	Vendor Unique Event Identifier. The VU Event Identifier that enables the VU Data field to be decoded.
EVC-TS-6	(Event Data Size * 4) + 3:14	VU Data	(Event Data Size * 4) - 10	Vendor Unique Data. The size of this data is determined based on the Event Data Size if this field exists.

7.2.8.2.3 Reset Debug Event Class Format

Requirement ID	Byte Address	Field	# of Bytes	Field Description
EVC-RST-1	0	Reset Debug Class	1	Shall be set to 04h.
EVC-RST-2	2:1	Event Identifier	2	The device shall log Reset Debug Events as follows:<table><tr><th>Event ID</th><th>Description</th></tr><tr><td>0000h</td><td>Main Power Cycle.</td></tr><tr><td>0001h</td><td>SATA - SRST.</td></tr><tr><td>0002h</td><td>SATA - COMRESET.</td></tr><tr><td>0003h</td><td>SAS - Hard Reset.</td></tr><tr><td>0004h</td><td>SAS - COMINIT.</td></tr></table>

<table>
<tr><th>Requirement ID</th><th>Byte Address</th><th>Field</th><th># of Bytes</th><th>Field Description</th></tr>
<tr><td></td><td></td><td></td><td></td><td><table>
<tr><td>0005h</td><td>SAS - DWORD synchronization loss.</td></tr>
<tr><td>0006h</td><td>SAS - SPL packet synchronization loss.</td></tr>
<tr><td>0007h</td><td>SAS - Receive Identify Timeout timer expired.</td></tr>
<tr><td>0008h</td><td>SAS - hot-plug timeout.</td></tr>
<tr><td>0009h - 7FFFh</td><td>Reserved.</td></tr>
<tr><td>8000h - FFFFh</td><td>Vendor Unique.</td></tr>
</table></td></tr>
<tr><td>EVC-RST-3</td><td>3</td><td>Event Data Size</td><td>1</td><td>This field indicates the number of Dwords in the Event Specific Data field. If the VU Event Identifier field and VU data field do not exist then, then this field shall be set to 0h.</td></tr>
<tr><td colspan="5">Event Specific Data</td></tr>
<tr><td>EVC-RST-4</td><td>5:4</td><td>VU Event Identifier</td><td>2</td><td>Vendor Unique Event Identifier. The VU Event Identifier that enables the VU data to be decoded.</td></tr>
<tr><td>EVC-RST-5</td><td>(Event Data Size * 4) + 3:6</td><td>VU Data</td><td>(Event Data Size * 4) - 2</td><td>Vendor Unique Data. The size of this data is determined based on the Event Data Size if this field exists.</td></tr>
</table>

7.2.8.2.4 Boot Sequence Debug Event Class Format

<table>
<tr><th>Requirement ID</th><th>Byte Address</th><th>Field</th><th># of Bytes</th><th>Field Description</th></tr>
<tr><td>EVC-BS-1</td><td>0</td><td>Boot Sequence Debug Class</td><td>1</td><td>Shall be set to 05h.</td></tr>
<tr><td>EVC-BS-2</td><td>2:1</td><td>Event ID</td><td>2</td><td>An SSD device shall log Boot Sequence Debug Events as follows:<table>
<tr><th>Event ID</th><th>Description</th></tr>
<tr><td>0000h</td><td>Main Firmware Boot Complete. For a single port device, this is when all boot activities required to allow I/O commands to be processed are completed. This is always after the FTL load from NVM Complete, FTL Rebuild Started and FTL Rebuild Completes events.</td></tr>
<tr><td>0001h</td><td>FTL Load from NVM Complete. This means the SSD downloaded the entire FTL table from media to internal</td></tr>
</table></td></tr>
</table>

<table>
<tr><th>Requirement ID</th><th>Byte Address</th><th>Field</th><th># of Bytes</th><th>Field Description</th></tr>
<tr><td></td><td></td><td></td><td></td><td>
<table>
<tr><td></td><td>memory (i.e., no journaling data is required to be loaded).</td></tr>
<tr><td>0002h</td><td>FTL Rebuild Started. This is after FTL Load from media has completed, if the FTL table is required to be rebuilt from journaling data, then mark the start.</td></tr>
<tr><td>0003h</td><td>FTL Ready. This means the FTL table is ready to fully support I/O commands (e.g., any necessary rebuild is complete).</td></tr>
<tr><td>0004h - 7FFFh</td><td>Reserved.</td></tr>
<tr><td>8000h - FFFFh</td><td>Vendor Unique.</td></tr>
</table>

An HDD device shall log Boot Sequence Debug Events as follows:
<table>
<tr><th>Event ID</th><th>Description</th></tr>
<tr><td>0000h - 00FFh</td><td>Reserved.</td></tr>
<tr><td>0100h</td><td>Main Firmware Boot Complete. For a single port device, this is when all boot activities required to allow I/O commands to be processed are completed excluding spin-up.</td></tr>
<tr><td>0101h</td><td>Spin-up Start. This means that the HDD device has started to spin-up.</td></tr>
<tr><td>0102h</td><td>Spin-up Complete. This means that the HDD device has completed spin-up and reached a stable rotation rate.</td></tr>
<tr><td>0103h</td><td>Device Ready. This means that HDD device is ready to fully support I/O commands.</td></tr>
<tr><td>0104h - 7FFFh</td><td>Reserved.</td></tr>
<tr><td>8000h - FFFFh</td><td>Vendor Unique.</td></tr>
</table>
</td></tr>
<tr><td>EVC-BS-3</td><td>3</td><td>Event Data Size</td><td>1</td><td>This field indicates the number of Dwords in the Event Specific Data field. If the VU Event Identifier field and VU data field do not exist then, then this field shall be set to 0h.</td></tr>
</table>

Requirement ID	Byte Address	Field	# of Bytes	Field Description
Event Specific Data				
EVC-BS-4	5:4	VU Event Identifier	2	Vendor Unique Event Identifier. The VU Event Identifier that enables the VU data to be decoded.
EVC-BS-5	(Event Data Size * 4) + 3:6	VU Data	(Event Data Size * 4) - 2	Vendor Unique Data. The size of this data is determined based on the Event Data Size if this field exists.

7.2.8.2.5 Firmware Assert Debug Event Class Format

Requirement ID	Byte Address	Field	# of Bytes	Field Description
EVC-ASRT-1	0	Firmware Assert Debug Class	1	Shall be set to 06h.
EVC-ASRT-2	2:1	Event Identifier	2	The device shall log Firmware Assert Debug Events as follows: **Event ID** / **Description** 0000h: Assert in SAS, SCSI, SATA or ATA Processing Code. 0001h: Assert in Media Code. 0002h: Assert in Security Code. 0003h: Assert in Background Services Code. 0004h: FTL Rebuild Failed. (SSD devices only.) 0005h: FTL Data Mismatch. (SSD devices only.) 0006h: Assert in Bad Block Relocation Code (HDD devices only). 0007h: Assert in Other Code. 0008h - 7FFFh: Reserved. 8000h - FFFFh: Vendor Unique.
EVC-ASRT-3	3	Event Data Size	1	This field indicates the number of Dwords in the Event Specific Data field. If the VU Event Identifier field and VU data field do not exist then, then this field shall be set to 0h.
Event Specific Data				
EVC-ASRT-4	5:4	VU Event Identifier	2	Vendor Unique Event Identifier. The VU Event Identifier that enables the VU data to be decoded.
EVC-ASRT-5	(Event Data Size * 4) + 3:6	VU Data	(Event Data Size * 4) - 2	Vendor Unique Data. The size of this data is determined based on the Event Data Size if this field exists.

7.2.8.2.6 Temperature Debug Event Class Format

<table>
<tr><th>Requirement ID</th><th>Byte Address</th><th>Field</th><th># of Bytes</th><th>Field Description</th></tr>
<tr><td>EVC-TEMP-1</td><td>0</td><td>Temperature Debug Class</td><td>1</td><td>Shall be set to 07h.</td></tr>
<tr><td>EVC-TEMP-2</td><td>2:1</td><td>Event Identifier</td><td>2</td><td>The device shall log Temperature Debug Events as follows:
<table>
<tr><th>Event ID</th><th>Description</th></tr>
<tr><td>0000h</td><td>Reported Temperature decreases to a value that causes thermal throttling to cease.</td></tr>
<tr><td>0001h</td><td>Reported Temperature increases to a value that reduces device performance due to thermal throttling.</td></tr>
<tr><td>0002h</td><td>Reported Temperature increases to a value that causes thermal shutdown.</td></tr>
<tr><td>0003h - 7FFFh</td><td>Reserved.</td></tr>
<tr><td>8000h - FFFFh</td><td>Vendor Unique.</td></tr>
</table>
Once the event is logged, at least 2 degrees of hysteresis shall be applied to avoid additional entries being generated.</td></tr>
<tr><td>EVC-TEMP-3</td><td>3</td><td>Event Data Size</td><td>1</td><td>This field indicates the number of Dwords in the Event Specific Data field. If the VU Event Identifier field and VU data field do not exist then, then this field shall be set to 0h.</td></tr>
<tr><th colspan="5">Event Specific Data</th></tr>
<tr><td>EVC-TEMP-4</td><td>5:4</td><td>VU Event Identifier</td><td>2</td><td>Vendor Unique Event Identifier. The VU Event Identifier that enables the VU data to be decoded.</td></tr>
<tr><td>EVC-TEMP-5</td><td>(Event Data Size * 4) + 3:6</td><td>VU Data</td><td>(Event Data Size * 4) - 2</td><td>Vendor Unique Data. The size of this data is determined based on the Event Data Size if this field exists.</td></tr>
</table>

7.2.8.2.7 Media Debug Event Class Format

<table>
<tr><th>Requirement ID</th><th>Byte Address</th><th>Field</th><th># of Bytes</th><th>Field Description</th></tr>
<tr><td>EVC-MEDIA-1</td><td>0</td><td>Media Debug Class</td><td>1</td><td>Shall be set to 08h.</td></tr>
<tr><td>EVC-MEDIA-2</td><td>2:1</td><td>Event Identifier</td><td>2</td><td>The device shall log Media Debug Events as follows:
<table>
<tr><th>Event ID</th><th>Description</th></tr>
<tr><td>0000h</td><td>XOR (or equivalent) Recovery Invoked.</td></tr>
</table></td></tr>
</table>

<table>
<tr><th>Requirement ID</th><th>Byte Address</th><th>Field</th><th># of Bytes</th><th>Field Description</th></tr>
<tr><td></td><td></td><td></td><td></td><td><table>
<tr><td>0001h</td><td>Uncorrectable Media Error</td></tr>
<tr><td>0002h</td><td>Block Marked Bad Due to SSD Media Program Error. (SSD only)</td></tr>
<tr><td>0003h</td><td>Block Marked Bad Due to SSD Media Erase Error. (SSD only)</td></tr>
<tr><td>0004h</td><td>Block Marked Bad Due to Read Error. This includes both foreground and background reads.</td></tr>
<tr><td>0005h</td><td>SSD Media Plane Failure. (SSD only)</td></tr>
<tr><td>0006h</td><td>SSD Media Die Failure. (SSD only)</td></tr>
<tr><td>0007h</td><td>HDD Head or Surface Failure. (HDD only)</td></tr>
<tr><td>0008h - 7FFFh</td><td>Reserved</td></tr>
<tr><td>8000h - FFFFh</td><td>Vendor Unique</td></tr>
</table></td></tr>
<tr><td>EVC-MEDIA-3</td><td>3</td><td>Event Data Size</td><td>1</td><td>This field indicates the number of Dwords in the Event Specific Data field. If the VU Event Identifier field and VU data field do not exist then, then this field shall be set to 0h.</td></tr>
<tr><th colspan="5">Event Specific Data</th></tr>
<tr><td>EVC-MEDIA-4</td><td>5:4</td><td>VU Event Identifier</td><td>2</td><td>Vendor Unique Event Identifier. The VU Event Identifier that enables the VU data to be decoded.</td></tr>
<tr><td>EVC-MEDIA-5</td><td>(Event Data Size * 4) + 3:6</td><td>VU Data</td><td>(Event Data Size * 4) - 2</td><td>Vendor Unique Data. The size of this data is determined based on the Event Data Size if this field exists.</td></tr>
</table>

7.2.8.2.8 Media Wear Event Class Format

<table>
<tr><th>Requirement ID</th><th>Byte Address</th><th>Field</th><th># of Bytes</th><th>Field Description</th></tr>
<tr><td>EVC-WEAR-1</td><td>0</td><td>Media Wear Class</td><td>1</td><td>Shall be set to 09h. This event shall be logged once every 24 hours.</td></tr>
<tr><td>EVC-WEAR-2</td><td>2:1</td><td>Event Identifier</td><td>2</td><td>The device shall log Wear Debug Events as follows:<table>
<tr><th>Event ID</th><th>Description</th></tr>
<tr><td>0000h</td><td>Media Wear.</td></tr>
<tr><td>0001h - 7FFFh</td><td>Reserved.</td></tr>
<tr><td>8000h - FFFFh</td><td>Vendor Unique.</td></tr>
</table></td></tr>
</table>

<table>
<tr><th>Requirement ID</th><th>Byte Address</th><th>Field</th><th># of Bytes</th><th>Field Description</th></tr>
<tr><td>EVC-WEAR-3</td><td>3</td><td>Event Data Size</td><td>1</td><td>This field indicates the number of Dwords in the Event Specific Data field. If the VU Event Identifier field and VU data field do not exist then, then this field shall be set to 3h.</td></tr>
<tr><td colspan="5">Event Specific Data</td></tr>
<tr><td>EVC-WEAR-4</td><td>15:4</td><td>Current Media Wear</td><td>12</td><td>For Wear Debug Event ID 00h, the device shall log the current wear as follows:<table><tr><th>Byte Address</th><th>Description</th></tr><tr><td>7:4</td><td>Host Terabytes Written.</td></tr><tr><td>11:8</td><td>Media Terabytes Written.</td></tr><tr><td>15:12</td><td>SSD Media Terabytes Erased. (SSD only) Shall be cleared to 0h by HDD devices.</td></tr></table>For all other Wear Event IDs, this field shall be cleared to zero.</td></tr>
<tr><td>EVC-WEAR-4</td><td>17:16</td><td>VU Event Identifier</td><td>2</td><td>Vendor Unique Event Identifier. The VU Event Identifier that enables the VU data to be decoded.</td></tr>
<tr><td>EVC-WEAR-5</td><td>(Event Data Size * 4) + 3:18</td><td>VU Data</td><td>(Event Data Size * 4) - 14</td><td>Vendor Unique Data. The size of this data is determined based on the Event Data Size if this field exists.</td></tr>
</table>

7.2.8.2.9 Statistic Snapshot Class

For the Statistic Snapshot Class. This enables the device to take a snapshot of a statistic descriptor and store this value in the Event FIFO.

Requirement ID	Byte Address	Field	# of Bytes	Field Description
EVC-SNP-1	0	Statistic Snapshot Class	1	Shall be set to 0Ah. This is a statistic snapshot of a statistics descriptor.
EVC-SNP-2	3:1	Reserved	3	Shall be cleared to 0h.
EVC-SNP-3	5:4	Statistic Identifier	2	This field shall contain the value in STATF-1.
EVC-SNP-4	6	Statistic Information 1	1	This field shall contain the value in STATF-2.
EVC-SNP-5	7	Statistic Information 2	1	This field shall contain the value in STATF-3.
EVC-SNP-6	8	Statistic Information 3	1	This field shall contain the value in STATF-4.
EVC-SNP-7	9	Reserved	1	This field shall contain the value in STATF-5.

Requirement ID	Byte Address	Field	# of Bytes	Field Description
EVC-SNP-8	11:10	Statistic Data Size	2	This field shall contain the value in STATF-6.
Event Specific Data (Statistic Type = 0h, see EVC-SNP-4 and STATF-2)				
EVC-SNP-9	(Statistic Data Size * 4) + 11:12	Statistic Specific Data	Statistic Data Size * 4	This field shall contain the value in STATF-7.
Event Specific Data (Statistic Type = 1h, see EVC-SNP-4 and STATF-2)				
EVC-SNP-10	12	Element Size	1	This field shall contain the value in STATF-8.
EVC-SNP-11	13	Reserved	1	This field shall contain the value in STATF-9.
EVC-SNP-12	15:14	Number Elements	2	This field shall contain the value in STATF-10.
EVC-SNP-13	((Statistic Data Size-4) * 4) + 15:16	Statistic Specific Data	(Statistic Data Size-4) * 4	This field shall contain the value in STATF-11.

7.2.8.2.10 Virtual FIFO Event Class Format

For the Virtual FIFO Class the format of the data is:

Requirement ID	Byte Address	Field	# of Bytes	Field Description
EVC-VF-1	0	Virtual FIFO Class	1	Shall be set to 0Bh.
EVC-VF-2	2:1	Virtual FIFO Event Identifier	2	Virtual FIFO Event Identifier: **Event ID** / **Description** 0000h – Virtual FIFO Start. 0001h – Virtual FIFO End. 0002 - FFFFh – Reserved.
EVC-VF-3	3	Virtual FIFO Data Size	1	This shall be set to 1h.
EVC-VF-4	5:4	Virtual FIFO Marker	2	This is the required format of the Virtual FIFO Marker. **Bit** / **Virtual FIFO Marker Bit Description** 15:14 – Reserved. 13:11 – Data Area the FIFO is in. Valid values are 1-4. All other values are reserved. 10:0 – The virtual FIFO number within the Data Area FIO.

Requirement ID	Byte Address	Field	# of Bytes	Field Description
				Note: Bits 7:0 are the LSB which is in byte 4.
EVC-VF-5	7:6	Reserved	2	This shall be set to 0h.

This indicates the start/end of a virtual FIFO within a FIFO in a Data Area. The name of a virtual FIFO can be found in the String Log using the Virtual FIFO Identifier.

7.2.8.2.11 SATA Phy/Link Debug Event Class Format

Requirement ID	Byte Address	Field	# of Bytes	Field Description
EVC-SLINK-1	0	SATA Phy Link Debug Class	1	Shall be set to 0Ch.
EVC-SLINK-2	2:1	Event Identifier	2	SATA Phy/Link Debug Events:

Event ID	Description
0000h	DR_Reset entered due to unexpected COMRESET
0001h	DR_Reset entered due to phy signal not detected
0002h	Device dropped link while host link is up.
0003h	DR_Ready entered at Gen 3
0004h	DR_Ready entered at Gen 2
0005h	DR_Ready entered at Gen 1
0006h	DR_Partial Entered
0007h	DR_Partial Exited
0008h	DR_Reduce Speed Entered
0009h	DR Error_Entered
000Ah	Transmitting HOLD
000Bh	Receiving HOLD
000Ch	PMNAK Received
000Dh	PMNAK Transmitted
000Eh	R_ERR Received
000Fh	R_ERR Transmitted
0010h	Set Device Bits Transmitted with Error bit set.
0011h - 7FFFh	Reserved.
8000h - FFFFh	Vendor Unique.

EVC-SLINK-3	3	Event Data Size	1	This field indicates the number of Dwords in the Event Specific Data field. If the VU Event Identifier

Requirement ID	Byte Address	Field	# of Bytes	Field Description
				field and VU data field do not exist then, then this field shall be set to 0h.
Event Specific Data				
EVC-SLINK-4	5:4	VU Event Identifier	2	Vendor Unique Event Identifier. The VU Event Identifier that enables the VU data to be decoded.
EVC-SLINK-5	(Event Data Size * 4) + 3:6	VU Data	(Event Data Size * 4) - 2	Vendor Unique Data. The size of this data is determined based on the Event Data Size if this field exists.

7.2.8.2.12 SATA Transport Debug Event Class Format

These events are intended for device debug and should not be logged during normal operation.

Requirement ID	Byte Address	Field	# of Bytes	Field Description
EVC-STRAN-1	0	SATA Phy Link Debug Class	1	Shall be set to 0Dh.
EVC-STRAN-2	2:1	Event Identifier	2	Transport Debug Events as follows: **Event ID** / **Description** 0000h: Non-Data FIS Received. 0001h: Non-Data FIS Transmitted. 0002h: Data FIS Received. 0003h: Data FIS Transmitted. 0004h - 7FFFh: Reserved. 8000h - FFFFh: Vendor Unique.
EVC-STRAN-3	3	Event Data Size	1	This field indicates the number of Dwords in the Event Specific Data field. If the VU Event Identifier field and VU data field do not exist then, then this field shall be set to 7h.
Event Specific Data				
EVC-STRAN-4	7:4	FIS0	4	This is Dword 0 of the FIS.
EVC-STRAN-5	11:8	FIS1	4	If the FIS Type contains Dword 1 this is Dword 1 otherwise this is 0h.
EVC-STRAN-6	15:12	FIS2	4	If the FIS Type contains Dword 2 this is Dword 2 otherwise this is 0h.
EVC-STRAN-7	19:16	FIS3	4	If the FIS Type contains Dword 3 this is Dword 3 otherwise this is 0h.
EVC-STRAN-8	23:20	FIS4	4	If the FIS Type contains Dword 4 this is Dword 4 otherwise this is 0h.

Requirement ID	Byte Address	Field	# of Bytes	Field Description
EVC-STRAN-9	27:24	FIS5	4	If the FIS Type contains Dword 5 this is Dword 5 otherwise this is 0h.
EVC-STRAN-10	31:28	FIS6	4	If the FIS Type contains Dword 6 this is Dword 6 otherwise this is 0h.
EVC-STRAN-11	33:32	VU Event Identifier	2	Vendor Unique Event Identifier. The VU Event Identifier that enables the VU data to be decoded.
EVC-STRAN-12	(Event Data Size * 4) + 3:10	VU Data	(Event Data Size * 4) - 6	Vendor Unique Data. The size of this data is determined based on the Event Data Size if this field exists.

7.2.8.2.13 SAS Phy/Link Debug Event Class Format

<table>
<tr><th>Requirement ID</th><th>Byte Address</th><th>Field</th><th># of Bytes</th><th>Field Description</th></tr>
<tr><td>EVC-SASLINK-1</td><td>0</td><td>SAS Phy Link Debug Class</td><td>1</td><td>Shall be set to 0Eh.</td></tr>
<tr><td>EVC-SASLINK-2</td><td>2:1</td><td>Event Identifier</td><td>2</td><td>SAS Phy/Link Debug Events:
<table>
<tr><th>Event ID</th><th>Description</th></tr>
<tr><td>0000h</td><td>Link up - 1.5 Gbps[1].</td></tr>
<tr><td>0001h</td><td>Link up - 3.0 Gbps[1].</td></tr>
<tr><td>0002h</td><td>Link up - 6.0 Gbps[1].</td></tr>
<tr><td>0003h</td><td>Link up - 12.0 Gbps[1].</td></tr>
<tr><td>0004h</td><td>Link up - 22.5 Gbps[1].</td></tr>
<tr><td>0005h</td><td>Identify received (data).</td></tr>
<tr><td>0006h</td><td>HARD_RESET received.</td></tr>
<tr><td>0007h</td><td>Link loss.</td></tr>
<tr><td>0008h</td><td>DWORD synchronization loss.</td></tr>
<tr><td>0009h</td><td>SPL packet synchronization loss.</td></tr>
<tr><td>000Ah</td><td>Identify receive timeout.</td></tr>
<tr><td>000Bh</td><td>BREAK received.</td></tr>
<tr><td>000Ch</td><td>BREAK_REPLY received.</td></tr>
<tr><td>000Dh - 7FFFh</td><td>Reserved.</td></tr>
<tr><td>8000h - FFFFh</td><td>Vendor Unique.</td></tr>
</table>
Notes:
[1]Including Identify frame exchange.</td></tr>
<tr><td>EVC-SASLINK-3</td><td>3</td><td>Event Data Size</td><td>1</td><td>This field indicates the number of Dwords in the Event Specific Data field. If the VU Event Identifier</td></tr>
</table>

Requirement ID	Byte Address	Field	# of Bytes	Field Description
				field and VU data field do not exist then, then this field shall be set to 0h.
Event Specific Data				
EVC-SASLINK-4	5:4	VU Event Identifier	2	Vendor Unique Event Identifier. The VU Event Identifier that enables the VU data to be decoded.
EVC-SASLINK-5	(Event Data Size * 4) + 3:6	VU Data	(Event Data Size * 4) - 2	Vendor Unique Data. The size of this data is determined based on the Event Data Size if this field exists.

7.2.8.2.14 SAS Transport Debug Event Class Format

These events are intended for device debug and should not be logged during normal operation.

Requirement ID	Byte Address	Field	# of Bytes	Field Description
EVC-SASTRAN-1	0	SAS Phy Link Debug Class	1	Shall be set to 0Fh.
EVC-SASTRAN-2	2:1	Event Identifier	2	Transport Debug Events as follows: **Event ID** / **Description** 0000h: DATA frame received. 0001h: DATA frame sent. 0002h: XFER_RDY frame sent. 0003h: COMMAND frame received. 0004h: RESPONSE frame sent. 0005h: TASK frame received. 0006h: SSP frame received. 0007h: SSP frame sent. 0008h: NAK received. 0009h - 7FFFh: Reserved. 8000h - FFFFh: Vendor Unique.
EVC-SASTRAN-3	3	Event Data Size	1	This field indicates the number of Dwords in the Event Specific Data field. If the VU Event Identifier field and VU data field do not exist then, then this field shall be set to 0h.
Event Specific Data				
EVC-SASTRAN-4	5:4	VU Event Identifier	2	Vendor Unique Event Identifier. The VU Event Identifier that enables the VU data to be decoded.

Requirement ID	Byte Address	Field	# of Bytes	Field Description
EVC-SASTRAN-5	(Event Data Size * 4) + 3:6	VU Data	(Event Data Size * 4) - 2	Vendor Unique Data. The size of this data is determined based on the Event Data Size if this field exists.

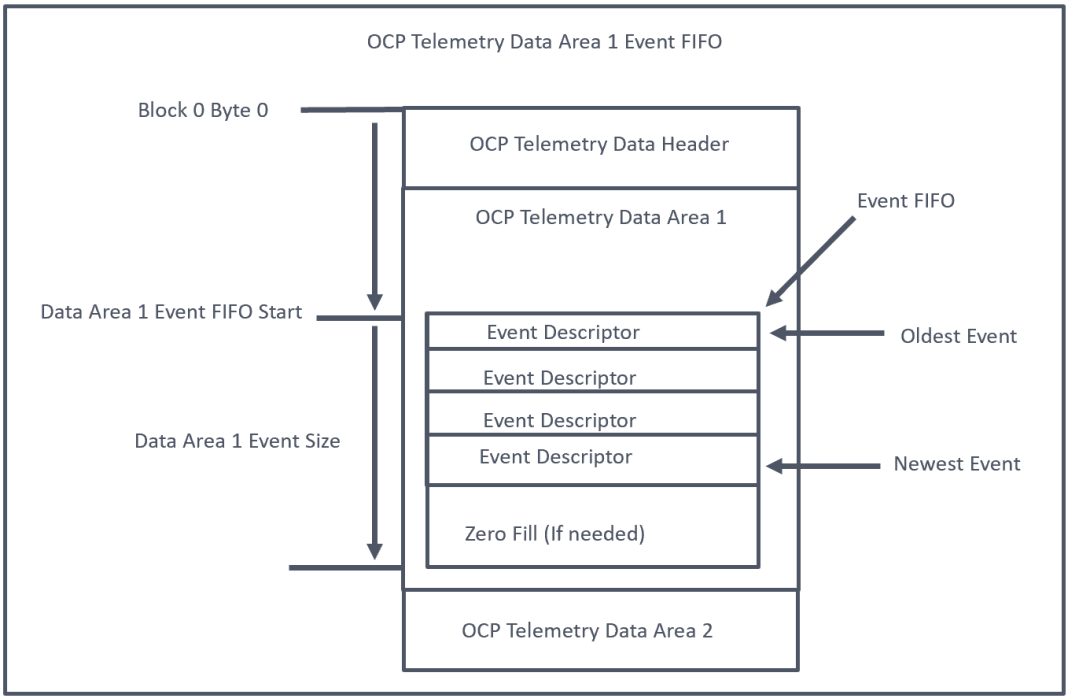
7.2.8.2.15 Vendor Unique Event Class Format

For the Vendor Unique debug class.

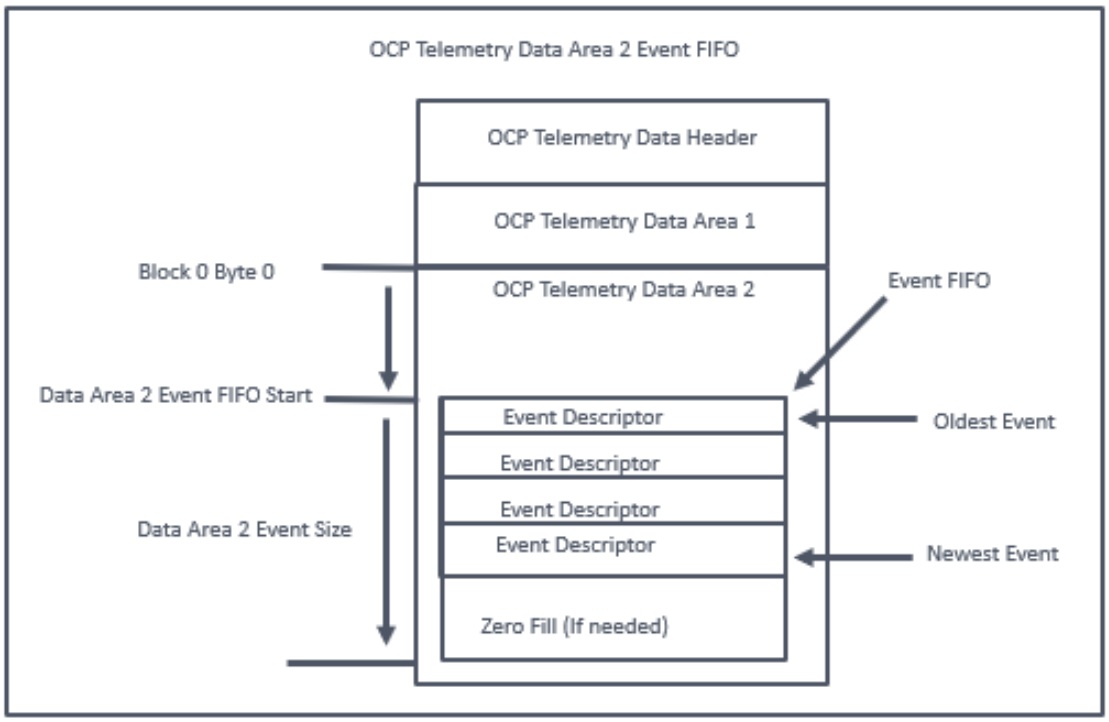
Requirement ID	Byte Address	Field	Bytes	Field Description
EVC-VU-1	0	Debug VU Class	1	Shall be set to a value in the range 80h to FFh.
EVC-VU-2	2:1	VU Event Identifier	2	Vendor Unique Event Identifier. The Vendor Unique event identifiers under the debug VU Class shall be in the range 0h to FFFFh.
EVC-VU-3	3	VU Event Data Size	1	This field indicates the number of Dwords in the Event Specific Data field. If the VU Specific Data field does not exist then, then this field shall be set to 0h. Valid values for this field are 0 to FF.
Event Specific Data				
EVC-VU-4	(VU vent Data Size * 4) + 3:4	VU Data	VU Event Data Size * 4	Vendor Unique Data. The size of this data is determined based on the Event Data Size if this field exists.

7.2.9 Event FIFO Location

In the Telemetry Data Structure, the Event FIFO location in OCP Telemetry Data Area 1 can be found by using the Data Area 1 Event FIFO Start and Data Area 1 Event FIFO Size. This can be seen in the following OCP Telemetry Data Area 1 Event FIFO Figure:



In the Telemetry Data Structure, the Event FIFO location in OCP Telemetry Data Area 2 can be found by using the Data Area 2 Event FIFO Start and Data Area 2 Event FIFO Size. This can be seen in the following OCP Telemetry Data Area 2 Event FIFO Figure:



7.2.9.1 Event FIFO Requirements

Requirement ID	Description
EVF-1	The oldest event descriptor shall be at the Event FIFO Start Address.
EVF-2	There shall be no gaps between the data in one descriptor and the next descriptor.
EVF-3	The event FIFO shall only have complete descriptors (i.e., no partial descriptors).
EVF-4	All data contained in the event FIFO that is not contained in an event descriptor shall be cleared to zero.
EVF-5	The Event FIFO Start and End shall be contained within a single OCP Telemetry Area.
EVF-6	Event FIFO Start and End locations shall not overlap with the Statistic Descriptor locations.
EVF-7	The decode of the Event Class and Vendor Unique Event Class shall be provided to the customer via the Telemetry String Structure.

7.2.10 OCP Telemetry Data Header Format

The OCP Telemetry Data Header in the Telemetry Data Structure has the following header format. It should be noted in the SAS and SATA sections describe the method to find the start of OCP Telemetry Data Area 1, OCP Telemetry Data Area 2 and Vendor Telemetry Data Area 3 as these are protocol dependent.

Requirement ID	Byte Address	Field	# of Bytes	Field Description
DITELF-1	1:0	Major Version	2	This field shall be set to 0001h. A change of the Major Version number indicates that the changes to the Telemetry Data Area format decode are not host software backward compatible.
DITELF-2	3:2	Minor Version	2	This field shall be set to 0001h. A change of the Minor Version number indicates that the changes to the Telemetry Data Area format decode are host software backward compatible.
DITELF-3	7:4	Reserved	4	Shall be cleared to zero.
DITELF-4	13:8	Timestamp	6	Shall be set to the Timestamp when the read of this log was initiated with Feature cleared to 0h. The format of this field shall be the format defined for the SET DATE & TIME EXT command (ATA) and the SET TIMESTAMP command (SCSI).
DITELF-5	15:14	Additional Info	2	Timestamp Additional Info: **Bit** / **Additional Info** 15:6 — Reserved. 5:4 — Protocol: 00b - Reserved 01b - SAS 10b - SATA 11b - Reserved 3:2 — Device Type:

Requirement ID	Byte Address	Field	# of Bytes	Field Description
				<table><tr><td></td><td>00b - Reserved 01b - HDD 10b - SSD 11b - Reserved</td></tr><tr><td>1</td><td>Timestamp is based on receiving a SET DATE & TIME EXT command (ATA) or a SET TIMESTAMP command (SCSI).</td></tr><tr><td>0</td><td>Timestamp is based on lifetime Power On Hours.</td></tr></table>
DITELF-6	31:16	Log Page GUID	16	This field shall be set to F5DAF2C03433422EB616D11C79F6F9E3h.
DITELF-7	33:32	Device String Data Size (DSDS)	2	Device String Data Size. The description of the value of this field is described in the SAS and SATA specific sections.
DITELF-8	41:34	Firmware Revision	8	SAS: Shall be set to 4 ASCII space (20h) characters followed by the 4-byte ASCII value of the PRODUCT REVISION LEVEL field in standard INQUIRY data. SATA: Shall be set to the value of the FIRMWARE REVISION field in IDENTIFY DEVICE.
DITELF-9	109:42	Reserved	68	Shall be cleared to zero.
DITELF-10	117:110	Data Area 1 Statistic Start	8	This field shall contain the Dword offset of the start of the Statistic Buffer in OCP Telemetry Data Area 1. This offset starts from Dword 0 / byte 0 in the OCP Telemetry Data Header to the Data Area 1 Static Buffer in OCP Telemetry Data Area 1.
DITELF-11	125:118	Data Area 1 Statistic Size	8	This field shall contain the size of the Statistic Buffer in Dwords which is located in OCP Telemetry Data Area 1. If there are no Statistic Descriptors a value of 0h shall be reported.
DITELF-12	133:126	Data Area 2 Statistic Start	8	This field shall contain the Dword offset of the start of the Statistic Buffer in OCP Telemetry Data Area 2. This offset starts from Dword 0 / byte 0 in OCP Telemetry Data Area 2 to the Data Area 2 Static Buffer in OCP Telemetry Data Area 2.
DITELF-13	141:134	Data Area 2 Statistic Size	8	This field shall contain the size of the Statistic Buffer in Dwords which is located in OCP Telemetry Data Area 2. If there are no Statistic Descriptors a value of 0h shall be reported.
DITELF-14	149:142	Data Area 1 Event FIFO Start	8	This field shall contain the Dword offset of the start of the Event FIFO within OCP Telemetry Data Area 1. This offset starts from Dword 0 / byte 0 of the OCP Telemetry Data Header.

Requirement ID	Byte Address	Field	# of Bytes	Field Description
DITELF-15	157:150	Data Area 1 Event FIFO Size	8	This field shall contain the size of the Event FIFO in in OCP Telemetry Data Area 1 in Dwords. If there is no FIFO in OCP Telemetry Data Area 1 a value of 0h shall be reported.
DITELF-16	165:158	Data Area 2 Event FIFO Start	8	This field shall contain the Dword offset of the start of the Event FIFO in OCP Telemetry Data Area 2. This offset starts from Dword 0 / byte 0 in OCP Telemetry Data Area 2.
DITELF-17	173:166	Data Area 2 Event FIFO Size	8	This field shall contain the size of the Event FIFO in OCP Telemetry Data Area 2 in Dwords. If there is no FIFO in OCP Telemetry Data Area 2 a value of 0h shall be reported.
DITELF-18	511:174	Reserved	338	Shall be cleared to zero.

7.2.11 String Decode Data Structure

The Telemetry Strings Structure is used to hold the information to decode the Telemetry Data Structure.

7.2.12 String Decode Data Requirements

Requirement ID	Description
TELSLG-1	All Statistic Identifiers contained in OCP Telemetry Data Area 1 and OCP Telemetry Data Area 2 shall be contained in the Device String Decode Data Structure.
TELSLG-2	Statistic Identifier String Table entries shall be in numeric order such that the smallest Vendor Unique Statistic Identifier is at the lowest address.
TELSLG-3	All Debug Event Class/ Event Identifier combinations contained in OCP Telemetry Data Area 1 and OCP Telemetry Data Area 2 shall be contained in the String Decode Data Structure.
TELSLG-4	All Debug Event Class/ VU Event Identifier combinations contained in OCP Telemetry Data Area 1 and OCP Telemetry Data Area 2 shall be contained in the Telemetry Strings Structure.
TELSLG-5	Event String Table entries shall be in numeric order such that the smallest Debug event Class is at the lowest address.
TELSLG-6	VU Event String Table entries shall be in numeric order such that the smallest Debug event Class is at the lowest address.
TELSLG-7	Event String Table entries shall be in numeric order such that for a single Debug/Event Identifier combination the smallest Event Identifier is at the lowest address.
TELSLG-8	VU Event String Table entries shall be in numeric order such that for a single Debug/VU Event Identifier combination the smallest VU Event Identifier is at the lowest address.
TELSLG-9	In the ASCII table (see Section 7.2.13 Telemetry Strings Structure), characters which are not valid (beyond the length of the ASCII length) shall be set to <space> (i.e., 20h).
TELSLG-10	The Statistics Identifier String Table shall have no gaps between this table and the Event String Table.
TELSLG-11	The Event String Table entry shall have no gaps between this table and the VU Event String Table.

Requirement ID	Description
TELSLG-12	The VU Event String Table and the ASCII table shall have no gaps between these tables.
TELSLG-13	The Statistics Identifier String Table Entries, the Event String Table Entries and the VU Event String Table Entries may point to the same ASCII Table Entry.

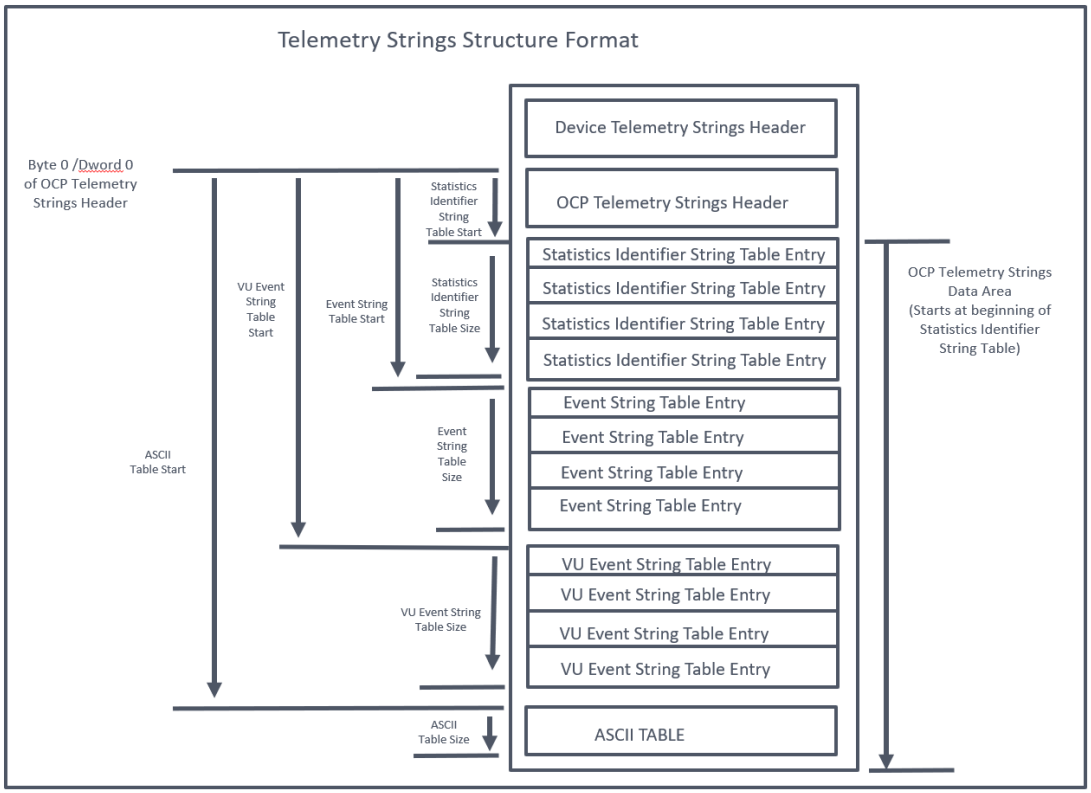
7.2.13 Telemetry Strings Structure

The Telemetry Strings Structure is used to enable decoding the Telemetry Data Structure. The Telemetry Strings Structure consists of the OCP Telemetry Strings Header (bytes 0-431, TELSTR-LOG-1 through TELSTR-LOG-15) followed by the OCP Telemetry Strings Data Area (starting at byte 432, TELSTR-LOG-16 through TELSTR-LOG-19). The mapping of the Telemetry Strings Structure to SAS and SATA is described in the SAS and SATA Specific section.

Requirement ID	Byte Address	Field	# of Bytes	Field Description
TELSTR-LOG-1	0	Log Page Version	1	Shall be set to 01h.
TELSTR-LOG-2	15:1	Reserved	15	Shall be cleared to zero.
TELSTR-LOG-3	31:16	Log Page GUID	16	Shall be set to 4D4406A251294570B4D0CAAD0FE39090h.
TELSTR-LOG-4	63:39	Reserved	32	Shall be cleared to zero.
TELSTR-LOG-5	71:64	Statistics Identifier String Table Start (SITS)	8	Shall be set to the number of Dwords from byte 0 of the start of the OCP Telemetry Strings Header to the start of the Statistics Identifier String Table. This shall be set to 6Ch (i.e., 108 Dwords, which occupy 432 bytes).
TELSTR-LOG-6	79:72	Statistics Identifier String Table Size (SITSZ)	8	Shall be set to the number of Dwords in the Statistics Identifier String Table. If there are no vendor unique defined values this shall be set to 0h.
TELSTR-LOG-7	87:80	Event String Table Start (ESTS)	8	Shall be set to the number of Dwords from byte 0 of the OCP Telemetry Strings Header to the start of the Event String Table.
TELSTR-LOG-8	95:88	Event String Table Size (ESTSZ)	8	Shall be set to the number of Dwords in the Event String Table.
TELSTR-LOG-9	103:96	VU Event String Table Start	8	Shall be set to the number of Dwords from the start of the OCP Telemetry Strings Header to the start of the VU Event String Table.
TELSTR-LOG-10	111:104	VU Event String Table Size	8	Shall be set to the number of Dwords in the VU Event String Table.
TELSTR-LOG-11	119:112	ASCII Table Start (ASCTS)	8	This is the number of Dwords from the start of the OCP Telemetry Strings Header to where the ASCII Table Starts.

Requirement ID	Byte Address	Field	# of Bytes	Field Description
TELSTR-LOG-12	127:120	ASCII Table Size (ASCTSZ)	8	This is the number of Dwords in the ASCII Table.
TELSTR-LOG-13	143:128	DATA AREA 1 EVENT FIFO ASCII String	16	This is the name of DATA AREA 1 EVENT FIFO. If the FIFO has no string name this shall be populated with 0h.
TELSTR-LOG-14	159:144	DATA AREA 2 EVENT FIFO ASCII String	16	This is the name of DATA AREA 2 EVENT FIFO. If the FIFO has no string name this shall be populated with 0h.
TELSTR-LOG-15	431:160	Reserved	272	Shall be cleared to 0h.
TELSTR-LOG-16	(SITSZ *4) + 431:432	Statistics Identifier String Table	SITSZ * 4	The format of the Statistics Identifier String Table can be seen in Section 7.2.14 Statistics Identifier String Table Entry.
TELSTR-LOG-17	(ESTSZ * 4) + (SITSZ * 4) + 431:SITSZ * 4) + 432	Event String Table	ESTSZ * 4	The format of the Event String Table can be seen in Section 7.2.15 Event Identifier String Table Entry.
TELSTR-LOG-18	(VUEDH * 4) + (ESTSZ * 4) + (SITSZ * 4) + 431:ESTSZ * 4) + (SITSZ * 4) + 432	VU Event String Table	VUESTZ * 4	The format of the VU Event String Table can be seen in Section 7.2.16 Vendor Unique Event Identifier String Table Entry.
TELSTR-LOG-19	(ASCTSZ * 4) + (VUEDH * 4) + (ESTSZ * 4) + (SITSZ * 4) + 432: (VUEDH * 4) + (ESTSZ * 4) + (SITSZ * 4) + 432	ASCII Table	ASCTSZ * 4	The format of the ASCII table can be seen in Section 7.2.17 ASCII String Example.

A diagram of the Telemetry Strings Structure is seen below:



7.2.14 Statistics Identifier String Table Entry

The Statistic Identifier String Table enables translating from vendor unique statistic identifiers to ASCII labels. The format of an entry for the Statistic Identifier String Table Entry is in the following table:

Requirement ID	Byte Address	Field	# of Bytes	Field Description
STR-ID-1	1:0	Vendor Specific Statistic Identifier	2	Shall be set the Vendor Unique Statistic Identifier number.
STR-ID-2	2	Reserved	1	Shall be cleared to zero.
STR-ID-3	3	ASCII ID Length	1	Shall be set the number of ASCII Characters that are valid. This is a zero-based value so a value of 0h is 1 character.
STR-ID-4	11:4	ASCII ID offset	8	Shall be set to the offset from Dword 0/Byte 0 of the Start of the ASCII Table to the first character of the string for this Statistic Identifier string. This value is in Dwords.
STR-ID-5	15:12	Reserved	4	Shall be cleared to zero.

7.2.15 Event Identifier String Table Entry

The Event Identifier String Table enables translating from Event Identifier to ASCII labels. The format of an entry for the Event Identifier String Table Entry is in the following table:

Requirement ID	Byte Address	Field	# of Bytes	Field Description
STR-EV-1	0	Debug Event Class	1	Shall be set to the Debug Event Class.
STR-EV-2	2:1	Event Identifier	2	Shall be set to the Event Identifier.
STR-EV-3	3	ASCII ID Length	1	Shall be set to the number of ASCII Characters that are valid. This is a zero-based value so a value of 0h is 1 character.
STR-EV-4	11:4	ASCII ID offset	8	This is the offset from Dword 0/ byte 0 of the start of the ASCII table to the ASCII data for this identifier. This value is in Dwords.
STR-EV-5	15:12	Reserved	4	Shall be cleared to zero.

7.2.16 Vendor Unique Event Identifier String Table Entry

The Vendor Unique Event Identifier String Table enables translating from the VU Event Identifier to ASCII labels. The format of an entry for the Vendor Unique Event Identifier String Table Entry is defined in the following table:

Requirement ID	Byte Address	Field	# of Bytes	Field Description
STR-VUEV-1	0	Debug Event Class	1	Shall be set to the Debug Event Class.
STR-VUEV-2	2:1	VU Event Identifier	2	Shall be set to the VU Event Identifier (see EVC-VU-2).
STR-VUEV-3	3	ASCII ID Length	1	Shall be set to the number of ASCII Characters that are valid. This is a zero-based value so a value of 0h is 1 character.
STR-VUEV-4	11:4	ASCII ID offset	8	This is the offset from Dword 0/ byte 0 of the start of the ASCII table to the ASCII data for this identifier. This value is in Dwords.
STR-VUEV-5	15:12	Reserved	4	Shall be cleared to zero.

7.2.17 ASCII String Example

An ASCII ID Length/Decode example is shown in the following table:

Byte	Data Byte	ASCII Character
400	54h	T
401	45h	E
402	53h	S

Byte	Data Byte	ASCII Character
403	54h	T
404	20h	<space>
405	45h	E
406	58h	X
407	41h	A
408	4Dh	M
409	50h	P
410	4Ch	L
411	45h	E
412	21h	!
413	20h	<space>
414	20h	<space>
415	20h	<space>

In this example the ASCII ID Length is 0Ch, which is 12 decimal. This is a zero-based count and thus there are 13 characters to print. The first 13 characters shall be part of the ASCII output. Thus, the message to print is:

"TEST EXAMPLE!"

The last three ASCII characters of <space> are not part of the ASCII output. They are only used for padding the string to the next Dword boundary. The last three ASCII <space> characters are not part of the next string.

7.3 SATA Specific Telemetry Requirements

Requirement ID	Description
SATAT-1	The Device Internal Status Log (24h) and Saved Device Internal Status Log (25h) shall be supported per this specification.
SATAT-2	The Saved Data Available field shall always be set to 01h in Device Internal Status Log (24h) and Saved Device Internal Status Log (25h).
SATAT-3	The Reason Identifier (bytes 384-511) in the Device Internal Status Log (24h) shall contain the Device Telemetry Reason Identifier.
SATAT-4	The Saved Data Generation Number in Device Internal Status Log (24h) shall be cleared to 0h.
SATAT-5	The Device String Data Size (DSDS) field In Device Internal Status Log (24h) Data Area 1 shall be set to the same value as Device Internal Status Data Area 1 Last Log Page field in Saved Device Internal Status Log (25h).
SATAT-6	The Saved Data Generation Number in Saved Device Internal Status Log (25h) shall be cleared to 0h.

Requirement ID	Description
SATAT-7	The Saved Device Internal Status Log (25h) Reason Identifier value shall be based on the Reason Identifier Requirements in Section 7.2.2 Reason Identifier Requirements.
SATAT-8	The device shall set the DEVICE INTERNAL STATUS DATA AREA 1 LAST LOG PAGE field of the Current Device Internal Status Data log (Log Address 24h) to 0040h as the factory default.

7.3.1 SATA Specific Telemetry Command Requirements

As is the case for all SATA logs, the Device Internal Status Log (24h) and Saved Device Internal Status Log (25h) consist of one or more 512-byte log pages that are accessed as entire log pages (e.g., via the READ LOG DMA EXT command). The Telemetry Data Structure and Telemetry Strings Structure (see Section 7.1 Device Telemetry Overview) are specified in this document using byte ranges and byte offsets, which treats the log pages for an ATA log as concatenated into a single range of bytes. Host code is responsible for translating from byte offsets and ranges to SATA log page numbers.

7.3.2 Telemetry Data Structure Mapping to Device Internal Status Log (24h)

The Telemetry Data Structure is mapped in SATA to the Device internal status Log (24h). A mapping of the data structures can be seen below.

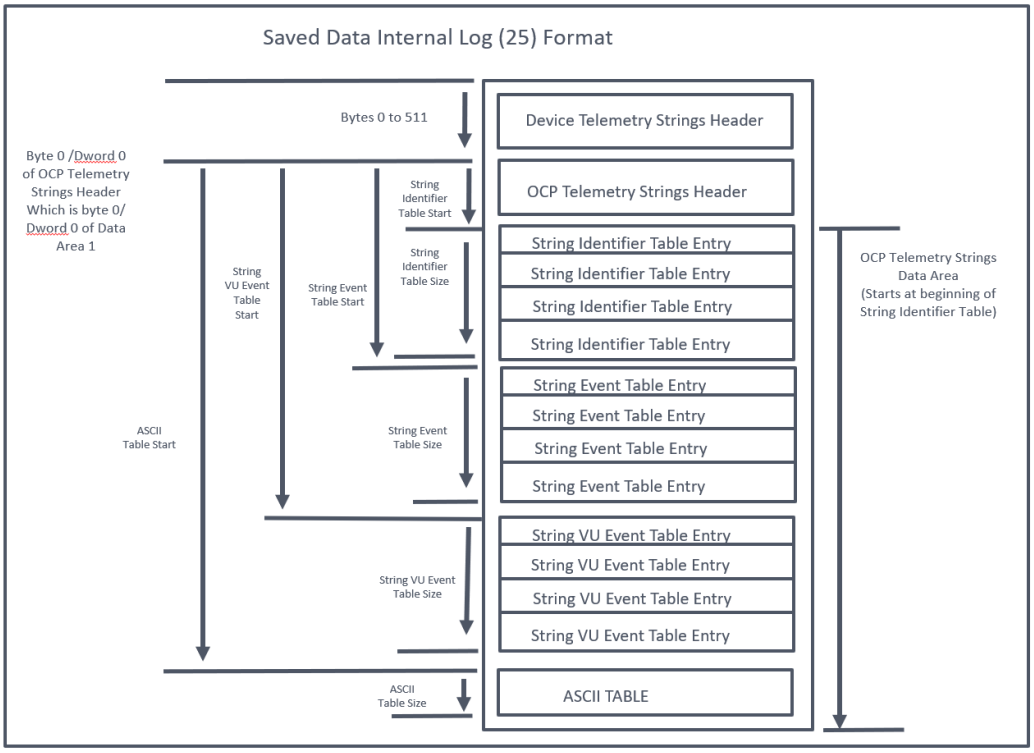
Device Internal Status Log (24h)	Comment
Bytes 511 - 0 (i.e., log page 00h)	This is the Device Telemetry Data Header. This is defined in ACS-5 or newer.
Data Area 1	This contains both the OCP Telemetry Data Header and OCP Telemetry Data Area 1. Byte 0/Dword 0 is the start of the OCP Telemetry Data Header.
Data Area 2	This is mapped to OCP Telemetry Data Area 2. Byte 0/Dword 0 is the start of OCP Telemetry Data Area 2.
Data Area 3	This is mapped to Vendor Telemetry Data Area 3. The format of this data is vendor unique. The start of this data area is defined in the Device Telemetry Data Header.

7.3.3 Telemetry Strings Structure Mapping to Saved Device Internal Log (25)

The Telemetry Strings Structure is mapped to Saved Device Internal Log(25) Data Area 1. A mapping of the data structures can be seen below.

Saved Device Internal Status Log (25h)	Format
Bytes 511 - 0 (i.e., log page 00h)	This is the Device Telemetry Strings header. This is defined in ACS-5 or newer.
Data Area 1	This contains the OCP Telemetry Strings Header. Byte 0/Dword 0 is the start of the OCP Telemetry Strings Header. The OCP Telemetry Strings Data Area starts in this Data Area. The OCP Telemetry Strings Data Area may also use Data Area 2 of this log.
Data Area 2	The OCP Telemetry Strings Data Area may extend into Data Area 2 of this log.
Data Area 3	This area is reserved with a size of 0h.

A figure showing the Saved Device Data Internal Log (25) format can be seen below:



7.4 SAS Specific Telemetry Requirements

Requirement ID	Description
SAST-1	The device shall support one error history buffer than contains Current Internal Status data, which is Telemetry Data as defined by this specification. The device shall support one error history buffer than contains Saved Internal Status data that is Telemetry Strings as defined by this specification.
SAST-2	The New Saved Data Available field shall always be set to 01h in the Telemetry Data Current Internal Status data and in the Telemetry Strings Saved Internal Status data.
SAST-3	The Current Reason Identifier (bytes 384 - 511) in the Telemetry Data Current Internal Status data shall contain the Device Telemetry Reason Identifier.
SAST-4	The Saved Data Generation Number in the Telemetry Data Current Internal Status data shall be cleared to 0h.
SAST-5	The Device String Data Size (DSDS) field in Data Area 1 of the Telemetry Data Current Internal Status data shall be set to the same value as the Saved Internal Status Data Set One Length field in the Saved Internal Status data (see DITELF-7).
SAST-6	The Saved Data Generation Number in the Telemetry Strings Saved Internal Status data shall be cleared to 0h.
SAST-7	The Telemetry Strings Saved Internal Status data Current Reason Identifier value shall be based on the Reason Identifier requirements as described in Section 7.2.2 Reason Identifier Requirements.

7.4.1 SAS Specific Telemetry Command Requirements

In this section, the term “read buffer command” refers to a READ BUFFER (10) command or a READ BUFFER (16) command. Support for both commands is required (see SAS-CMD-1).

Requirement ID	Description
SASTC-1	The device shall support both the READ BUFFER (10) command and the READ BUFFER (16) command in error history mode (i.e., with the Mode field in the command set to 1Ch).
SASTC-2	The device shall support BUFFER ID value 0 for read buffer commands. In response to a read buffer command in error history mode with a BUFFER ID value of 0h, the device shall return an error history directory that contains: • at least one error history directory entry with the BUFFER FORMAT field set to 01h (i.e., Current Internal Status data); and • at least one error history directory entry with the BUFFER FORMAT field set to 02h (i.e., Saved Internal Status data). As specified in SPC-5, the SUPPORTED BUFFER ID field in each of these error history directory entries shall be set to a value in the range 10h to EFh inclusive. Exactly one error history directory entry that has the BUFFER FORMAT field set to 01h (i.e., Current Internal Status data) shall contain a SUPPORTED BUFFER ID value that designates an error history buffer that contains Telemetry Data as defined in this specification. Exactly one error history directory entry that has the BUFFER FORMAT field set to 02h (i.e., Saved Internal Status data) shall contain a SUPPORTED BUFFER ID value that designates an error history buffer that contains Telemetry Strings as defined in this specification. The error history directory may contain additional entries. The presence and use of any such entries is outside the scope of this document. As specified in SPC-5, the use of the same SUPPORTED BUFFER ID value in more than one error history directory entry is prohibited.
SASTC-3	The device shall return Current Internal Status data in response to a read buffer command in error history mode with the BUFFER ID field set to the value of the SUPPORTED BUFFER ID field in the error history directory entry that has its BUFFER FORMAT field set to 01h and that designates an error history buffer that contains Telemetry Data.
SASTC-4	The device shall return Saved Internal Status data in response to a read buffer command in error history mode with the BUFFER ID field set to the value of the SUPPORTED BUFFER ID field in the error history directory entry that has its BUFFER FORMAT field set to 02h and that designates an error history buffer that contains Telemetry Strings.
SASTC-5	Both Current Internal Status data and Saved Internal Status data are intended to be accessed as one or more 512-byte aligned 512-byte units. If a read buffer command as specified in SASTC-3 or SASTC-4 has a BUFFER OFFSET field set to a non-zero value that is not a multiple of 512 or an ALLOCATION LENGTH field set to any value that is not a multiple of 512, then the device may terminate that command with CHECK CONDITION status with the sense key set to ILLEGAL REQUEST and the additional sense code set to INVALID FIELD IN CDB.
SASTC-6	The device shall set the CURRENT INTERNAL STATUS DATA SET ONE LENGTH field of the Current Internal Status data to 0040h as the factory default.
SASTC-7	If the error history directory (see SASTC-2) contains more than one entry with the BUFFER FORMAT field set to 01h or contains more than one entry with the BUFFER FORMAT field set to 02h, then the device shall support the Telemetry VPD page specified in Section 7.4.1.1 SAS Specific Telemetry VPD Page Requirements.

The ability to use SCSI commands to access SATA devices via a SATL (SCSI ATA Translation Layer) depends on the SATL implementation because support for READ BUFFER (10) in error history mode is optional in the SAT standard for SATLs. The SAT standard also does not translate the BUFFER OFFSET field in READ BUFFER (10) – that field is ignored and assumed to be cleared to zero. Hence that field should be cleared to 0h for access to SATA devices via a SATL which results in always reading the telemetry data from the beginning. Alternatives that allow part of the telemetry data to be read without starting at the beginning include using ATA commands directly with the SATA devices and using the ATA PASS THROUGH commands defined by the SAT standard.

7.4.1.1 SAS Specific Telemetry VPD Page Requirements

The device is required to support the 64-byte SCSI VPD (Vital Product Data) page defined in this section if there is any possible error history directory ambiguity about which error history buffers contain Telemetry Data and Telemetry Strings (see SASTC-7).

Requirement ID	Byte	Field	Bits	Field Description
SAST-VPD-1	0	Peripheral Qualifier	7:5	Shall be set to the value used in standard INQUIRY data.
SAST-VPD-2	0	Peripheral Device Type	4:0	Shall be set to the value used in standard INQUIRY data.
SAST-VPD-3	1	Page Code	7:0	Shall be set to DFh.
SAST-VPD-4	3:2	Page Length	15:0	Shall be set to 60 (3Ch) to indicate a total VPD page length of 64 bytes.
SAST-VPD-5	4	Telemetry Data Buffer ID	7:0	Shall be set to the buffer ID of the error history buffer that contains Telemetry Data (see SASTC-2).
SAST-VPD-6	5	Telemetry Strings Buffer ID	7:0	Shall be set to the buffer ID of the error history buffer that contains Telemetry Strings (see SASTC-2).
SAST-VPD-7	47:6	Reserved	All	Shall be cleared to zero.
SAST-VPD-8	63:48	VPD Page GUID	All	Shall be set to C0EF832A9413474EB335E8C562404AB5h.

7.4.2 Telemetry Data Structure Mapping to Current Internal Status Data

The Telemetry Data Structure is mapped in SAS to Current Internal Status data. A mapping of the data structures can be seen below.

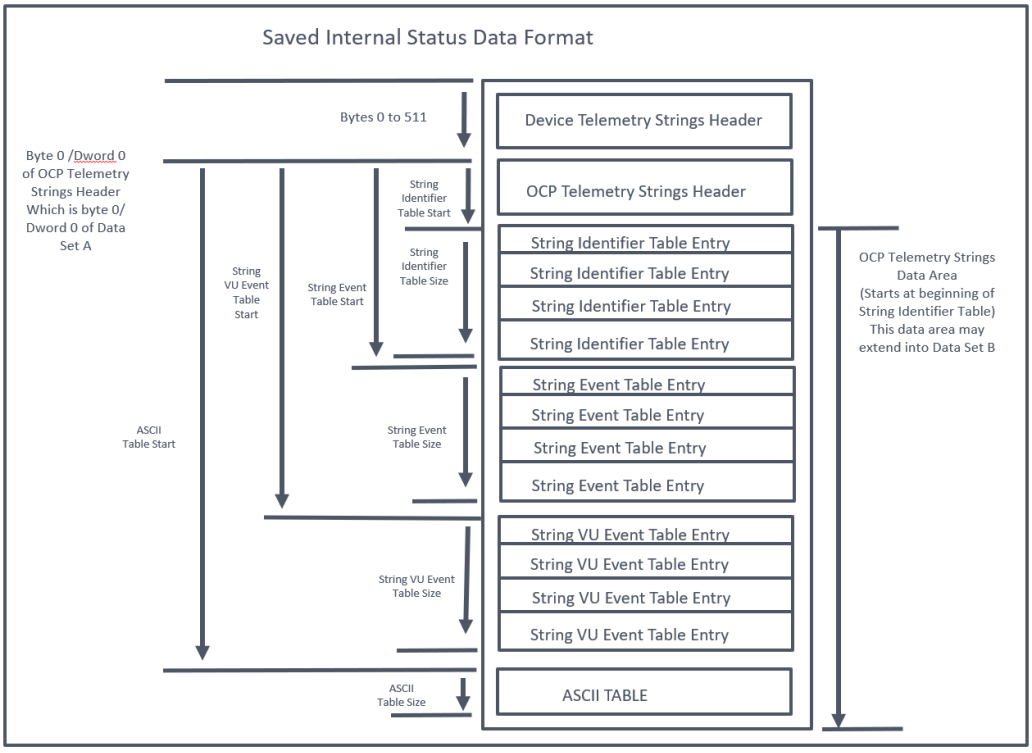
Current Internal Status Data	Comment
Bytes 511 - 0	This is the Device Telemetry Data Header. This is defined in SPC-5 or newer.
Data Set A	This contains both the OCP Telemetry Data Header and OCP Telemetry Data Area 1. Byte 0/Dword 0 is the start of the OCP Telemetry Data Header.
Data Set B	This is mapped to OCP Telemetry Data Area 2. Byte 0/Dword 0 is the start of OCP Telemetry Data Area 2.
Data Set C	This is mapped to Vendor Telemetry Data Area 3. The format of this data is vendor unique. The start of this data set is defined in the Device Telemetry Data Header.
Data Set D	Vendor Telemetry Data Area 3 may extend into Data Set D.

7.4.3 Telemetry Strings Structure Mapping to Saved Internal Status Data

The Telemetry Strings Structure is mapped to Saved Internal Status data. A mapping of the data structures can be seen below.

Saved Internal Status Data	Format
Bytes 511 - 0	This is the Device Telemetry Strings header. This is defined in SPC-5 or newer.
Data Set A	This contains the OCP Telemetry Strings Header. Byte 0/Dword 0 is the start of the OCP Telemetry Strings Header. The OCP Telemetry Strings Data Area starts in this Data Set. The OCP Telemetry Strings Data Area may also use Data Set B.
Data Set B	The OCP Telemetry Strings Data Area may extend into Data Set B.
Data Set C	This data set is reserved with a size of 0h.
Data Set D	This data set is reserved with a size of 0h.

A figure showing the Saved Internal Status Data Set A format can be seen below. This structure may extend into Saved Internal Status Data Set B.



8 Power

<table>
<tr><th>Requirement ID</th><th>Description</th></tr>
<tr><td>PWR-1</td><td>While in Power Saving Mode C, Standby, and Sleep (see PWR-5), the device shall consume < 5W at idle as defined by the COMMISSION REGULATION (EU) 2019/424.</td></tr>
<tr><td>PWR-2</td><td>The device max average power consumption shall be measured in a 1 second window with a sampling rate of 2ms or better. The measurement duration shall be at least 15 minutes.</td></tr>
<tr><td>PWR-3</td><td>The device peak power consumption shall be measured in a 100us window with a sampling rate of 4us or better. The measurement duration shall be at least 15 minutes.</td></tr>
<tr><td>PWR-4</td><td>The Max Power and Current Limits for PWR-5 through PWR-12 shall be measured at 37C° ± 5C° device temperature.</td></tr>
<tr><td>PWR-5</td><td>A 3.5” SAS/SATA LFF HDD device shall consume no more power than the limits in the table below:
<table>
<tr><th>Measurement Condition</th><th>Max Power</th></tr>
<tr><td>Air-filled normal R/W operation, max average
(Max. power access type, workload, queue depth)</td><td>13.5W</td></tr>
<tr><td>Helium-filled normal R/W operation, max average
(Max. power access type, workload, queue depth)</td><td>Single Actuator (SA): 11.0W
Dual Actuator (DA): 14.0W</td></tr>
<tr><td>Power Saving Mode A: Partial electronic power off</td><td>9.0W</td></tr>
<tr><td>Power Saving Mode B: More electronic power off and park heads.</td><td>6.5W</td></tr>
<tr><td>Power Saving Mode C: More electronic power off, park heads and partial RPM spin</td><td>3.5W</td></tr>
<tr><td>Standby/Sleep: Major electronic power off, park heads and motor stop</td><td>Standby: 1.5W
Sleep: 1.5W</td></tr>
</table></td></tr>
<tr><td>PWR-6</td><td>A 3.5” SAS/SATA LFF HDD device shall consume no more current than the limits in the table below:
<table>
<tr><th>Measurement Condition</th><th>5V Current Limit</th><th>12V Current Limit</th></tr>
<tr><td>Spin Up Peak, ImaxPeakSpin < 0.1ms</td><td>1.5A</td><td>2.9A</td></tr>
<tr><td>Spin Up Average, ImaxAvgSpin</td><td>1.2A</td><td>2.0A</td></tr>
<tr><td>Normal R/W operation, peak
(Max. power access type, workload, queue depth)</td><td>1.5A</td><td>2.0A</td></tr>
<tr><td>Air-filled normal R/W operation, max average
(Max. power access type, workload, queue depth)</td><td>1.2A</td><td>1.2A</td></tr>
<tr><td>Helium-filled normal R/W operation, max average
(Max. power access type, workload, queue depth)</td><td>1.2A</td><td>SA: 0.9A
DA: 1.1A</td></tr>
</table></td></tr>
<tr><td>PWR-7</td><td>A 2.5” SAS3 (12G) SFF SSD device shall consume no more power than the limits in the table below:
<table>
<tr><th>Measurement Condition</th><th>Max Power</th></tr>
</table></td></tr>
</table>

<table>
<tr><th>Requirement ID</th><th>Description</th></tr>
<tr><td></td><td>
<table>
<tr><td>Normal R/W operation, max average
(Max. power access type, workload, queue depth)</td><td>11.0W</td></tr>
<tr><td>Idle</td><td>4.0W</td></tr>
<tr><td>Power consumed when Power Disable is asserted</td><td>0.25W</td></tr>
</table>
</td></tr>
<tr><td>PWR-8</td><td>A 2.5” SAS3 (12G) SFF SSD device shall consume no more current than the limits in the table below:
<table>
<tr><th>Measurement Condition</th><th>5V Current Limit</th><th>12V Current Limit</th></tr>
<tr><td>Power Up Peak, ImaxPeakPowerUp < 0.1ms</td><td>1.8A</td><td>3.0A</td></tr>
<tr><td>Power Up Average, ImaxAvgPowerUp</td><td>1.3A</td><td>2.0A</td></tr>
<tr><td>Normal R/W operation, peak
(Max. power access type, workload, queue depth)</td><td>1.7A</td><td>1.8A</td></tr>
<tr><td>Normal R/W operation, max average
(Max. power access type, workload, queue depth)</td><td>1.2A</td><td>0.9A</td></tr>
</table>
</td></tr>
<tr><td>PWR-9</td><td>A 2.5” SAS4 (24G) SFF SSD device shall consume no more power than the limits in the table below:
<table>
<tr><th>Measurement Condition</th><th>Max Power</th></tr>
<tr><td>Normal R/W operation, max average
(Max. power access type, workload, queue depth)</td><td>18.0W</td></tr>
<tr><td>Idle</td><td>5.0W</td></tr>
<tr><td>Power consumed when Power Disable is asserted</td><td>0.25W</td></tr>
</table>
</td></tr>
<tr><td>PWR-10</td><td>A 2.5” SAS4 (24G) SFF SSD device shall consume no more current than the limits in the table below:
<table>
<tr><th>Measurement Condition</th><th>5V Current Limit</th><th>12V Current Limit</th></tr>
<tr><td>Power Up Peak, ImaxPeakPowerUp < 0.1ms</td><td>1.8A</td><td>3.0A</td></tr>
<tr><td>Power Up Average, ImaxAvgPowerUp</td><td>1.3A</td><td>2.0A</td></tr>
<tr><td>Normal R/W operation, peak
(Max. power access type, workload, queue depth)</td><td>2.0A</td><td>2.0A</td></tr>
<tr><td>Normal R/W operation, max average
(Max. power access type, workload, queue depth)</td><td>1.5A</td><td>1.0A</td></tr>
</table>
</td></tr>
<tr><td>PWR-11</td><td>A 2.5” SATA SFF SSD device shall consume no more power than the limits in the table below:
<table>
<tr><th>Measurement Condition</th><th>Max Power</th></tr>
<tr><td>Normal R/W operation, peak
(Max. power access type, workload, queue depth)</td><td>9.0W</td></tr>
<tr><td>Normal R/W operation, max average
(Max. power access type, workload, queue depth)</td><td>9.0W</td></tr>
<tr><td>Idle</td><td>1.5W</td></tr>
</table>
</td></tr>
</table>

<table>
<tr><th>Requirement ID</th><th>Description</th></tr>
<tr><td></td><td>
<table>
<tr><td>Power consumed when Power Disable is asserted</td><td>0.25W</td></tr>
</table>
</td></tr>
<tr><td>PWR-12</td><td>A 2.5” SATA SFF SSD device shall consume no more current than the limits in the table below:
<table>
<tr><th>Measurement Condition</th><th>5V Current Limit</th><th>12V Current Limit</th></tr>
<tr><td>Power Up Peak, ImaxPeakPowerUp < 0.1ms</td><td>1.5A</td><td>0.7A</td></tr>
<tr><td>Power Up Average, ImaxAvgPowerUp</td><td>1.5A</td><td>0.4A</td></tr>
<tr><td>Normal R/W operation, peak
(Max. power access type, workload, queue depth)</td><td>1.5A</td><td>0.4A</td></tr>
<tr><td>Normal R/W operation, max average
(Max. power access type, workload, queue depth)</td><td>1.5A</td><td>0.4A</td></tr>
</table>
</td></tr>
<tr><td>PWR-13</td><td>The device shall operate normally (e.g., no performance or reliability impact) with supply voltage variations up to +/- 5% for 5V and +/- 10% for 12V inputs.</td></tr>
<tr><td>PWR-14</td><td>The vendor shall provide average and peak measurements for power and current during the conditions below:
Spin-up.
Recovery from standby.
Typical workloads (define):
Random read, block size 4k, queue depth 16.
Random write, block size 1m, queue depth 16.
Sequential read, block size 1m, queue depth 16.
Sequential write, block size 1m, queue depth 16.
Idle_A, Idle_B and Idle_C.
Firmware update.</td></tr>
<tr><td>PWR-15</td><td>The device shall report power on hours and power cycle counts during the life of operation. All devices shall report power on hours and power cycle counts in OCP Telemetry (see STATI-COM-4 and STATI-COM-5).

SATA devices shall report power on hours and power cycle count in the ATA Device Statistics Log (04h), General Statistics Log page (01h).

SAS devices shall report power on hours for the last 20 self-tests in the Self-Test Results log page (page code 10h) using bytes 6:7 of a Self-Test Results log parameter for each self-test (parameter codes 01h through 14h) and shall report the total power cycle count in the Start-Stop Cycle Counter log page (page code 0Eh) Accumulated Start-Stop Cycles parameter (parameter code 04h).</td></tr>
<tr><td>PWR-16</td><td>Data integrity shall be guaranteed during emergency or unplanned power off when write cache is disabled.</td></tr>
<tr><td>PWR-17</td><td>The vendors shall provide command sequence guidance for graceful power off or reset operations in the product specification.</td></tr>
</table>

9 Reliability

9.1 MTBF / AFR Requirements

<table>
<tr><th>Requirement ID</th><th>Description</th></tr>
<tr><td>REL-1</td><td>An SSD device shall meet the REL-1 requirements in the OCP Datacenter NVMe SSD Specification version 2.6 or later.

An HDD device shall meet an MTBF of 2.5 million hours (AFR of <= 0.35%) under the following operating environmental conditions:
<table>
<tr><th>Specification</th><th>Environment</th><th>Requirement</th></tr>
<tr><td rowspan="2">Ambient Temperature</td><td>Operational</td><td>• 10˚C to 40˚C (50˚F to 104˚F)
(< 20˚C (68˚F) per hour gradient)</td></tr>
<tr><td>Non-Operational</td><td>• -40˚C to 70˚C (-40˚F to 158˚F)
(< 30˚C (86˚F) per hour gradient)</td></tr>
<tr><td rowspan="2">Humidity</td><td>Operational</td><td>• 10% to 90% non-condensing
(< 20% per hour gradient)
• Yearly weighted average: < 80% RH
• 90% of year: < 80%
• 10% of year: 80% to 90%
• Maximum dewpoint: 29.4°C (85°F)</td></tr>
<tr><td>Non-Operational</td><td>• 5% to 95% non-condensing
(< 20% per hour gradient)
• 29˚C (85˚F) maximum wet bulb temperature</td></tr>
</table></td></tr>
<tr><td>REL-2</td><td>An SSD device shall meet the REL-2 requirements in the OCP Datacenter NVMe SSD Specification version 2.6 or later.

An HDD device shall meet an MTBF of 2.0 million hours (AFR of <= 0.44%) the following environmental conditions:
<table>
<tr><th>Specification</th><th>Environment</th><th>Requirement</th></tr>
<tr><td rowspan="2">Ambient Temperature</td><td>Operational</td><td>• 0°C to 55°C (32°F to 131°F)
(< 20°C (68˚°F) per hour gradient)</td></tr>
<tr><td>Non-Operational</td><td>• -40°C to 70°C (-40˚°F to 158°F)
(< 30°C (86°F) per hour gradient)</td></tr>
<tr><td rowspan="2">Humidity</td><td>Operational</td><td>• 10% to 90% non-condensing
(< 20% per hour gradient)
• Maximum dewpoint: 29.4°C (85°F)</td></tr>
<tr><td>Non-Operational</td><td>• 5% to 95% non-condensing
(< 20% per hour gradient)
• 29°C (85°F) maximum wet bulb temperature</td></tr>
</table></td></tr>
<tr><td>REL-3</td><td>Supplier shall provide the acceleration factor, temperature and humidity conditions used to determine the MTBF based on REL-5 and REL-6.</td></tr>
<tr><td>REL-4</td><td>Obsolete.</td></tr>
</table>

Requirement ID	Description
REL-5	For HDD devices the reliability parameters shall be based on a minimum of 550 TB/ year per actuator for workloads of 100% read, 100% write and 50/50 read/write for a minimum of 5 years.
REL-6	For HDD devices supplier shall provide additional reliability data to the customer for 550 TB/ year per actuator for workloads of 100% read, 100% write and 50/50 read/write based on a 7-year service life.
REL-7	For SSD devices, reliability shall be based on a 5-year service life without degradation and the vendor shall provide additional reliability data to the customer based on a 7-year service life.

9.2 HDD Load / Unload Cycles

Requirement ID	Description
LUL-1	The device shall be able to perform a minimum of 600,000 soft load/unload cycles during the life of the device.

9.3 EPO / Sudden Power Loss

Requirement ID	Description
EPO-1	An HDD device shall be able to perform a minimum of 50,000 hard load/unload cycles during the life of the device based on REL-1 and REL-2.
EPO-2	The device shall support full power-loss protection for all acknowledged user data with volatile write caching disabled.
EPO-3	The device shall support full power-loss protection for all telemetry and internal logging data with or without volatile Write Caching.
EPO-4	Volatile Write caches are prohibited in SSD devices.

9.4 Data Integrity / End to End Data Protection

Requirement ID	Description
E2E-1	All user data and metadata shall be protected using overlapping protection mechanisms throughout the entire read and write path in the device including all storage elements (registers, caches, SRAM, DRAM, NAND, etc.).
E2E-2	At least one bit of correction and 2 bits of detection is required for all memories. This shall be for all memories regardless of function.
E2E-3	If the device contains DRAM, then the entire DRAM addressable space shall be protected with at least one-bit correction and 2 bits of detection scheme (SECDED). This includes but is not limited to the following (if present): • Mapping tables (including metadata related to deallocated LBAs). • Journal entries. • Firmware scratch pad. • System variables. • Firmware code.

Requirement ID	Description
	• Flash translation layer (FTL).
E2E-4	Silent data corruption shall not be tolerated under any circumstances.
E2E-5	The device shall include a mechanism to protect against returning the data from the wrong logical block address (LBA), including previous contents of the same LBA, to the host. It is acceptable if the device stores additional/modified information to provide protection against returning wrong data to the host. Device shall perform host LBA integrity checking on all transfers to and from the media.
E2E-6	All device metadata, firmware, firmware variables, and other device system data shall be protected by at least a single bit detection scheme.
E2E-7	Any memory buffers that are utilized to accelerate data transfer (read-ahead buffers for example) shall follow the protection scheme outlined in E2E-5.
E2E-8	Data path errors that are detected, but not corrected, are considered unrecoverable data errors, and shall be subject to unrecoverable data error rate limits. The total count of all such errors shall be reported in OCP Telemetry (see STATI-COM-12).

9.5 UBER

Requirement ID	Description
UBER-1	An HDD device shall support an Uncorrectable Bit Error Rate (UBER) of < 1 sector per 10^{15} bits read. An SSD device shall support an Uncorrectable Bit Error Rate (UBER) of < 1 sector per 10^{17} bits read.
UBER-2	De-rating curves for the UBER shall be supplied for 5˚C to 50˚C (41˚F to 122˚F).

9.6 ORT Requirement

Requirement ID	Description
ORT-1	The vendor shall provide on-going reliability data for device in production and notify customer immediately if any quality escape or customer excursion scenario is likely.

9.7 Behavior on Assert / Firmware Crash

Requirement ID	Description
CRASH-1	If the device can guarantee data integrity, after a firmware crash, panic or assert in which the device can no longer accept write commands, the device shall allow read access.
CRASH-2	After a firmware crash, panic or assert that is not a hardware failure, the device shall still support Read Log EXT / SMART Read (SATA) and Log Sense (SAS) commands including the ability to read any failure logs from the device to determine the nature of the failure.
CRASH-3	If after a firmware crash, panic or assert there is the possibility of user data corruption, the device shall require a Format to resume operation.

9.8 HDD Head Wear-out / Degradation

Requirement ID	Description
HEADWEAR-1	HDD heads shall maintain the specified UBER for a minimum of 5 years.
HEADWEAR-2	The HDD heads due to wear shall not cause the average or P99 tail latency to degrade more than 5% over 5 years.

9.9 Shelf Life

Requirement ID	Description
SHELF-1	A new device may be kept as a datacenter spare and therefore shall be fully functional even if it sits on the shelf for at least 5 years at 25°C (77°F) before getting installed in the server. The device shall be new in box factory state.
SHELF-2	HDD devices shall have mitigation so that devices stored within the non-operating temperature range will not have issues with stiction among motor and head/disk interface.

10 Performance Requirements

10.1 Time to Ready

Requirement ID	Description
TTR-1	The device shall be ready to successfully process user IO commands within a maximum of 30 seconds after power is applied to the device.

10.2 HDD Sequential Performance Requirements

<table>
<tr><th>Requirement ID</th><th>Description</th></tr>
<tr><td>PERF-SEQ-1</td><td>The HDD device shall match or beat the following criteria for sequential operations for both Reads and Writes based on maximum Sequential Data Rate (SDR) at outer diameter (OD).
<table>
<tr><th>IO Size</th><th>QD</th><th>Performance (MB/s) per Actuator</th></tr>
<tr><td>4K</td><td>1</td><td>62.5 MB/s or 25% of max SDR at OD</td></tr>
<tr><td>4K</td><td>2</td><td>125 MB/s or 50% of max SDR at OD</td></tr>
<tr><td>4K</td><td>4</td><td>187.5 MB/s or 75% of max SDR at OD</td></tr>
<tr><td>4K</td><td>8 and up</td><td>250 MB/s or 100% of max SDR at OD</td></tr>
<tr><td>8K</td><td>1</td><td>125 MB/s or 50% of max SDR at OD</td></tr>
<tr><td>8K</td><td>2</td><td>187.5 MB/s or 75% of max SDR at OD</td></tr>
<tr><td>8K</td><td>4 and up</td><td>250 MB/s or 100% of max SDR at OD</td></tr>
<tr><td>16K</td><td>1</td><td>187.5 MB/s or 75% of max SDR at OD</td></tr>
<tr><td>16K</td><td>2 and up</td><td>250 MB/s or 100% of max SDR at OD</td></tr>
</table></td></tr>
<tr><td>PERF-SEQ-2</td><td>PERFSEQ-1 (OD) shall not degrade by more than 50% at ID. OD and ID are defined by 1% of physical space at the outer and inner portions of the device, respectively. If 1% and 99%</td></tr>
</table>

Requirement ID	Description
	LBA space does not align with physical OD/ID portion of the disk, the vendor shall provide this information for accurate measurement.
PERF-SEQ-3	Newer model HDDs shall not degrade by more than 5% versus previous generations of the same product line.

10.3 HDD Random Read Performance Requirements

Requirement ID	Description
PERF-RD-1	The HDD device shall match or exceed the following criteria for random read operations:

IO Size	QD	Performance (IOPS) per Actuator
4K	1	80
4K	2	100
4K	4	120
4K	8	140
4K	16	170
4K	32	200
4K	64	230 (SAS only)
4K	128	260 (SAS only)

10.4 HDD Random Write Performance Requirements

Requirement ID	Description
PERF-RW-1	The HDD device shall match or exceed the following criteria for random writes.

IO Size	QD	Write Cache Setting	Performance (IOPS) per Actuator
4K / 8K	Any	Disabled	300
4K / 8K	Any	Enabled	350

10.5 HDD Latency Requirements

Requirement ID	Description
LAT-1	The following HDD latency criteria (per actuator) shall not be exceeded with 4K QD1 random reads. The latency measurement test shall take at least 10 minutes ramp time and 60 minutes run time.

Percentile	Latency (ms)
P50	12
P75	15
P99	20
P99.9	25
P99.99	40

10.6 Maximum Command Timeout

Requirement ID	Description
CCTO-1	No read/write IO operation shall take more than 6 seconds regardless of workload including error recovery running in the background.

10.7 Power Mode Transition Times

Requirement ID	Description
PMTT-1	The device shall take no more than 30 seconds to transition from standby to ACTIVE mode.
PMTT-2	The device shall take no more than 10 ms to transition from Idle_A to ACTIVE mode.
PMTT-3	The device shall take no more than 400 ms to transition from Idle_B to ACTIVE mode.
PMTT-4	The device shall take no more than 5 seconds to transition from Idle_C to ACTIVE mode.

10.8 Write Cache Flushing Latency

Requirement ID	Description
SYNC-1	The write cache flushing commands shall take no more than 10 seconds at any conditions if write cache is enabled. The write cache flushing commands refer to FLUSH CACHE / FLUSH CACHE EXT for SATA device and SYNCHRONIZE CACHE for SAS device.

10.9 HDD Multi-actuator Specific Considerations

Requirement ID	Description
MULTIA-1	The performance for devices with multi-actuators is expected to scale linearly. Each actuator in the multi-actuator device shall not lose more than 3% of performance vs baseline due to the adjacent actuators performing activities.

11 Environmental

11.1 Thermal

Requirement ID	Description
THRM-1	The device shall operate normally (e.g., performance, reliability, etc.) within the operating temperature range specified in REL-1 and REL-2.
THRM-2	Obsolete (see REL-1 and REL-2).
THRM-3	Obsolete (see SHELF-1).
THRM-4	Obsolete (see REL-1 and REL-2).
THRM-5	Obsolete (see REL-1 and REL-2).
THRM-6	The device shall update the internal reported temperature with an interval of 5 seconds or less.

Requirement ID	Description
THRM-7	The device's device-to-device reported temperature variation shall be +/- 1 degrees C. Two different devices shall not report a temperature greater than 2 degrees apart under the same environmental conditions, slot location, and workload.
THRM-8	A single device's reported temperature shall not vary by more than +/-1 degrees C (1.8 degrees F) once it is in a steady state under the same environmental conditions, slot location, and workload.
THRM-9	The supplier shall provide the equation, settings, and thresholds used to calculate the reported temperature.
THRM-10	For a device that reports a composite temperature, calculation of the reported temperature shall take into account the temperature of all the major thermal components (e.g., for an SSD: SOC, DRAM, NAND, PMIC, etc.).
THRM-11	A device that has multiple ports shall report the same temperature via all ports.

11.1.1 Thermal Shutdown

Requirement ID	Description
THRMS-1	The device shall shut down to protect itself against data loss or damage due to extreme temperatures. For SSDs, the shutdown temperature value shall be at a reported temperature that is greater than or equal to 85°C (185°F). Shutdown shall preserve all user data.
THRMS-2	Reserved for future use.
THRMS-3	The device shall only come out of thermal shutdown as a result of a main power cycle.

11.2 Altitude

Requirement ID	Description
ALTD-1	The device shall operate normally in data centers located at an altitude between -1,000 feet and 10,000 feet. When considering thermals, there shall be no de-rating up to 6,000 feet above sea level. Above 6,000 feet the derating shall be at most 0.9°C (1.6°F)/1000ft.
ALTD-2	The storage non-operating altitude shall be between -1,000 ft and 40,000 feet

11.3 Humidity

Requirement ID	Description
HUMD-1	The device shall operate normally with relative humidity to be between 10% and 90% with maximum wet bulb temperature of 29°C.
HUMD-2	Obsolete (see REL-1 and REL-2).
HUMD-3	The storage non-operating relative humidity shall be between 10% and 90% with maximum wet bulb temperature of 29°C.
HUMD-4	Obsolete (see REL-1 and REL-2).

11.4 Mechanical

11.4.1 Form Factor Requirements

11.4.1.1 Generic Form Factor Requirements

Requirement ID	Description
GFF-1	A compliance report and gap analysis for each supported form factor (e.g., U3, SFF-8639, SFF-TA-1001) shall be provided to the customer.
GFF-2	A CAD file and/or 2D/3D drawing of each supported form factor shall be provided.
GFF-3	The device shall support the power disable feature (PWDIS). This feature allows the host to disable power to the device circuitry using pin 3.
GFF-4	The default setting for the Power Disable feature shall be enabled.
GFF-5	The device shall support hot installation and removal ("hot-pluggable"). The device shall not implement any form of a "latching" connector mechanism which might inhibit removal and installation of the device from a storage enclosure. Per SFF-8482 specification, connector mate / unmate minimal cycle life is 500 cycles.
GFF-6	Device Activity Signal (Pin 11) shall assert/de-assert (see GFF-7) while processing one or more commands. Once asserted due to command processing, pin 11 shall remain asserted for 50 ms and then shall be de-asserted for 50 ms. This assert/de-assert behavior shall continue while command processing continues (i.e., the LED is usually off, but flashes on and off while commands are processed). Once pin 11 is de-asserted for 50 ms, it shall remain de-asserted as long as there is no command processing activity.
GFF-7	Device Activity Signal (Pin 11) Active State (pin asserted and LED on) is defined as logic "0" (i.e., < 0.3 V) on Pin 11. Inactive State is defined as logic "1" (i.e., > 3.3 V) on Pin 11. A device's internally initiated background activity or maintenance routine that is not commanded to be performed from the host shall not cause the LED to be in the Active State (logic "0" level).

11.4.1.2 3.5" Form Factor Requirements

Requirement ID	Description
FF35-1	The dimensions and connector locations shall conform to small form factor standard defined in SFF-8301 and SFF-8323. The A1 dimension (HDD height) shall meet 26.1 mm max.

11.4.1.3 2.5" Form Factor Requirements

Requirement ID	Description
FF25-1	The dimensions and connector locations shall conform to small form factor standard defined in SFF-8201 and SFF-8223.

11.4.1.4 M.2 Form Factor Requirements

Requirement ID	Description
FFM2-1	The device shall adhere to the latest revision of PCI-SIG M.2 specification.

11.4.2 Weight

Requirement ID	Description
WEIGHT-1	The device shall not weigh more than 750g.

11.4.3 Shock and Vibration

<table>
<tr><th>Requirement ID</th><th>Description</th></tr>
<tr><td>VIBR-1</td><td>The device shall not experience any performance degradation (including hard errors) during normal operating conditions.</td></tr>
<tr><td>VIBR-2</td><td>Vendor shall provide documentation of acoustic sound pressure level and mask recommendations based on the OCP HDD Dynamics Common Language and Tools Phase 2 white paper.</td></tr>
<tr><td>VIBR-3</td><td>Obsolete.</td></tr>
<tr><td>VIBR-4</td><td>The device operation and performance shall not be affected by more than 10% under the test conditions specified in VIBR-5.

The device operation and performance shall not be affected after the testing specified in VIBR-5.</td></tr>
<tr><td>VIBR-5</td><td>The device shall meet the following shock and vibration requirements:
<table>
<tr><th>Test Condition</th><th>Test Description</th></tr>
<tr><td>Non-Operational Shock</td><td>1. a) Square 50 G & 11 ms, 4.2 m/sec b) Half-sine 150 G & 11 ms and 300 G & 2 ms.
2. 2 shocks in each of the 6 directions, total twelve shocks.</td></tr>
<tr><td>Operational Shock</td><td>1. Half-sine 45 G & 2 ms x and y-axis.
2. Half-sine 30 G & 2 ms z-axis.</td></tr>
<tr><td>Operational Vibration</td><td>1. RMS acceleration(g) = 0.7 Grms.
2. 5-500 Hz, random vibration (see Table 1 for PSD profile).
3. 10 minutes in each of the 3 orthogonal axes, for a total of 30 minutes.
<table>
<tr><th colspan="2">Operational Vibration PSD Profile</th></tr>
<tr><th>Frequency (Hz)</th><th>G²/Hz</th></tr>
<tr><td>5</td><td>0.00025</td></tr>
<tr><td>20</td><td>0.00210</td></tr>
<tr><td>200</td><td>0.00210</td></tr>
<tr><td>250</td><td>0.00020</td></tr>
<tr><td>500</td><td>0.00020</td></tr>
</table></td></tr>
</table></td></tr>
</table>

<table>
<tr><th>Requirement ID</th><th>Description</th></tr>
<tr><td></td><td>
<table>
<tr><td></td><td>Table 1: Operational Vibration PSD Profile</td></tr>
<tr><td>Operational Rotational-Vibration</td><td>1. 10 minutes in the z-axis, for a total of 10 minutes.
2. 20 - 1500 Hz, Random 12.5 rad/s^2 with PSD profile (see Tabel 2 for PSD profile).
3. The vendor shall perform both functionality and I/O tests before and after stress.
<table>
<tr><th colspan="2">Operational Rotational-Vibration PSD Profile</th></tr>
<tr><th>Frequency (Hz)</th><th>G^2/Hz</th></tr>
<tr><td>20</td><td>0.0533</td></tr>
<tr><td>200</td><td>0.0533</td></tr>
<tr><td>800</td><td>0.349</td></tr>
<tr><td>1500</td><td>0.000614</td></tr>
</table>
Table 2: Operational Rotational-Vibration PSD Profile</td></tr>
<tr><td>Non-Operational Vibration</td><td>1. RMS acceleration(g) = 2.27 Grms.
2. 5-500 Hz, random vibration (see Table 3 for PSD profile).
3. 10 minutes in each of the 3 orthogonal axes, for a total of 30 minutes.
<table>
<tr><th colspan="2">Non-Operational Vibration PSD Profile</th></tr>
<tr><th>Frequency (Hz)</th><th>G^2/Hz</th></tr>
<tr><td>2</td><td>0.001</td></tr>
<tr><td>4</td><td>0.030</td></tr>
<tr><td>100</td><td>0.030</td></tr>
<tr><td>500</td><td>0.001</td></tr>
</table>
Table 3: Non-Operational Vibration PSD Profile</td></tr>
</table>
</td></tr>
<tr><td>VIBR-6</td><td>Vendors shall review the fixture with the customer and obtain approval.</td></tr>
<tr><td>VIBR-7</td><td>For SSD devices suppliers shall meet the shock and vibration section requirements in the latest OCP Datacenter NVMe SSD Specification.</td></tr>
</table>

12 Sustainability Requirements

The following are the sustainability requirements for devices.

Requirement ID	Description
SUS-1	A Life Cycle Assessment (LCA) aligned with ISO 14040 for the device shall be provided to the customer.
SUS-2	Recycled content information per material type shall be provided to the customer.

13 Security

13.1 Basic Security Requirements

Requirement ID	Description
SEC-1	The device shall support signed firmware binary updates which are checked before firmware is activated. The device firmware shall be authenticated using cryptographic keys on every reboot and during firmware update.
SEC-2	The device shall support XTS-AES-256 (As specified in NIST SP 800-38E, which references IEEE STD 1619) or AES-256-GCM (As specified in NIST SP 800-38D) hardware-based data encryption for performance and to prevent firmware tampering with the encryption mechanism. AES-256-GCM is the preferred mode. The combination of AES-256-CTR (as specified in NIST SP 800-38A) with HMAC-SHA256 (where the HMAC is calculated on the ciphertext) is an acceptable alternative.
SEC-3	The device shall support anti-rollback protection for firmware. The anti-rollback protection shall be implemented with a Security Version Number (see STATI-COM-28) that is different from the Firmware Revision. If the Security Version Number of the firmware being downloaded or activated is greater than or equal to the Security Version Number of the currently running firmware, then the firmware download, or activation is allowed. If the Security Version Number of the firmware being activated is less than the Security Version Number of the currently running firmware, then the firmware download, or activation shall fail.
SEC-4	The device shall support Crypto Erase functionality. The Crypto Erase functionality shall satisfy the Purge requirements as defined in IEEE 2883-2022.
SEC-5	The device shall support Secure Boot.
SEC-6	The device shall have a method of identifying a secure boot failure which does not require physical access to the device.
SEC-7	The device's cryptographic module shall be FIPS 140-3 level 2 capable per CMVP and shall follow the NIST 800-90 (A, B and C) specification.
SEC-8	The device shall implement only FIPS and NIST approved implementations and algorithms and shall be designed to achieve FIPS-140-3 validation (see SEC-7).
SEC-9	The device shall support key revocation allowing a new key to be used for firmware verification. The device shall support at least 4 key revocations, and key revocation shall not make the device unbootable (i.e., the device supports at least 5 keys).
SEC-10	Obsolete (moved to TCG-FEATURE-1).
SEC-11	Obsolete (moved to TCG-FEATURE-2).
SEC-12	Obsolete (moved to TCG-FEATURE-3).
SEC-13	TCG Opal devices shall support the IEEE1667 Probe (100h) and the TCG Storage Transport (104h) silos. See SEC-57 and SEC-58 for additional IEEE1667 requirements. TCG Enterprise devices shall not support IEEE1667 functionality.
SEC-14	Supplier shall follow the Security Development Lifecycle (SDL), and provide a report with the following for each qualification-ready or production-ready firmware revision: • The Threat Model.

Requirement ID	Description
	• Fuzz & Pen Tests. • Static Analysis. • Build Logs and Compiler Security Settings. Additional information about the SDL is available here: *https://www.microsoft.com/en-us/sdl/default.aspx*
SEC-15	A firmware audit of the initial firmware released to customer intended for qualification and any subsequent firmware release intended for production use shall be performed using *OCP SAFE,* Framework V1.0 or later. The device shall have a Scope 1 audit. The device may have a Scope 2 and Scope 3 audit. Each detailed report shall be provided to the customer.
SEC-16	All signing keys shall be stored in a Hardware Security Module (HSM) that is either FIPS 140-2 Level 3 (or greater) certified or FIPS 140-3 Level 3 (or greater) certified.
SEC-17	Access/use of signing keys shall be restricted to a small set of developers, following the principle of least privilege. The number of people with access and their corresponding roles shall be provided.
SEC-18	All debug ports shall be disabled before the device leaves the factory. Alternatively, the ports shall only be accessible in the field after a successful exchange of a challenge-response mechanism using an asymmetric crypto scheme (refer to NIST SP 800-63). The state shall be reset to inaccessible on any reset or power cycle.
SEC-19	All vendor unique commands, log pages or set features that are not explicitly defined in this specification or approved of in writing by the customer shall be disabled before the device leaves the factory. Alternatively, the commands/log pages/set features shall only be accessible in the field after a successful exchange of a challenge-response mechanism using an asymmetric crypto scheme (refer to NIST SP 800-63). The state shall be reset to inaccessible on any reset or power cycle.
SEC-20	Adversarial testing using red teams shall be conducted before the start of customer qualification of the initial product and any major firmware or hardware revision. A report of items attempted, and results shall be provided.
SEC-21	The vendor shall provide timely notification of security issues and delivery of fixes: • The vendor shall document all security fixes with each firmware update. • The vendor shall notify end customer within 7 days of discovering security issues in the device hardware or firmware. • Notification of issues shall include the process and timeline of the vendor's commitment to fix the issue: ○ For privately disclosed vulnerabilities, the duration shall be no longer than 90 days. ○ For publicly disclosed vulnerabilities, the duration shall be no longer than 7 days. ○ Vendors shall notify the customers about the known CVEs and security issues and provide security-related updates before public announcement.
SEC-22	All Telemetry and debugging logs shall be human readable.
SEC-23	The device shall not include user data, passwords, keys or any secret or sensitive information in any Telemetry or debug logs.

Requirement ID	Description
SEC-24	All public keys shall be revocable.
SEC-25	Secure Boot Flow shall be based upon hardware root of trust. All mutable key(s), certificate(s) and/or firmware shall be cryptographically bound to the hardware root of trust (see Section 12.2 Secure Boot).
SEC-26	The device shall only update firmware via a secure firmware update flow that shall be based upon immutable public keys.
SEC-27	Obsolete (moved to TCG-CRYPTO-1).
SEC-28	The device shall purge all symmetric keys and all private asymmetric keys from volatile memory as soon as they are no longer needed for operation during the current power on state.
SEC-29	The device shall only store, even if temporarily, host provided passwords, host provided keys, or any host provided secret information in non-volatile memory in an encrypted form. The encryption key(s) for this protection shall not be stored in non-volatile memory. Device shall not store plaintext/cleartext secrets in any non-volatile memories. The device shall protect all such stored passwords, keys, and secrets from unauthorized modification.
SEC-30	The supplier must deliver key management and encryption flow diagram details (not source code) which includes: • Encryption algorithms and modes (e.g., RNG, wrapping function, key derivation function). • Key size and password length. • Any crypto information that is stored in nonvolatile memory. • Critical Security Parameters. Complete details about algorithm inputs (e.g., Initialization vector source and size, salt source and size, unique per device/product line/vendor).
SEC-31	The supplier shall provide industry certification reports once they are available, such as FIPS and OCP SAFE for device and device components such as TRNG, RNG, Crypto engine etc.
SEC-32	Log data and user data (data written to the device by the host in write commands) shall be stored on physically separate areas on the device. For example, in the system area and the user data area, respectively.
SEC-33	The vendor shall provide a comprehensive list of what is included in the logs covering both the standard logs and vendor-unique logs. The list shall identify any defined optional events or items that are not logged by the device.
SEC-34	Obsolete (see SATA-SDC-10 and SAS-SDC-10).
SEC-35	All telemetry and debug logs shall only be writable by device firmware after a successful secure boot.
SEC-36	Obsolete (moved to TCG-AUTH-3).
SEC-37	Obsolete.
SEC-38	Obsolete (moved to TCG-FEATURE-1).
SEC-39	Obsolete (moved to TCG-AUTH-1).
SEC-40	Obsolete (moved to TCG-AUTH-2).
SEC-41	Obsolete (moved to TCG-FEATURE-5).

Requirement ID	Description
SEC-42	Obsolete (moved to TCG-FEATURE-1).
SEC-43	Obsolete (moved to TCG-PSID-1).
SEC-44	Obsolete (moved to TCG-PSID-3).
SEC-45	Obsolete (moved to TCG-PSID-4).
SEC-46	Obsolete (moved to TCG-PSID-5).
SEC-47	Obsolete (see Section 13.4 TCG Implementation Requirements).
SEC-48	The device shall not allow the host to read firmware or boot code from the device.
SEC-49	All device firmware packages shall be encrypted based on a symmetric encryption and follow SEC-2 and SEC-29 for key storage.
SEC-50	Device shall only support the activation of encrypted firmware images.
SEC-51	The device shall only store firmware in encrypted format when stored in non-volatile memory to protect against physical disassembly of device to extract stored content from media.
SEC-52	For any command that changes the media encryption key(s) (e.g., a SATA CRYPTO SCRAMBLE EXT command, a SAS SANITIZE command with a CRYPTOGRAPHIC ERASE service action (3h), TCG Revert, TCG RevertSP, etc.), the device shall purge, as defined in IEEE 2883-2022, all copies of the previous key(s) before completing the command and the results of purging the previous key(s) shall satisfy the requirements for purging user data as defined in IEEE 2883-2022. If the device cannot guarantee that all copies of the previous media encryption key(s) have been purged or cannot meet the IEEE 2883-2022 requirements for purging user data, then the device shall fail the command.
SEC-53	Any optional security features supported by the device and not described in this document shall be clearly documented and disclosed to the customer.
SEC-54	The Cryptographic Module Boundary shall be defined to minimize the size and scope of the Cryptographic Module, and in particular the Cryptographic Module Boundary shall not encompass the entire physical device. This enables FIPS certification of the Cryptographic Module to be leveraged across multiple device configurations and versions. To facilitate a timelier NIST certification approval process, any revalidations made to the Cryptographic Module Boundary should be Non-Security Relevant. The Cryptographic Module Boundary shall include all relevant cryptographic functions within the boundary itself, such as the data at rest encryption algorithms per FIPS 197 and digital signature algorithms for firmware authentication per FIPS 186-5.
SEC-55	At a minimum, the device shall support the following signature and hashing algorithms for secure boot, firmware update, and SPDM implementations: • ECDSA P384 or other asymmetric key signature algorithms that meet at least 192-bits of security strength as documented in NIST SP 800-57 Part 1 Revision 5 (Table 2). The device may also support RSA 3072 instead. • SHA-384, SHA3-384 or other hashing algorithms that meet at least 192-bits of security strength as documented in NIST SP 800-57 Part 1 Revision 5 (Table 3).
SEC-56	The device shall not reveal secrets (including passwords, asymmetric private keys, symmetric keys, and any other secret information) even in encrypted form via programmatic methods, logging operations (including debug outputs), and within externally accessible configuration files or settings.

Requirement ID	Description
SEC-57	After power on of a TCG Opal device, for a given ComID, any IEEE1667 functionality that is different than what is specified by the TCG specifications (e.g., Command Sequencing Reset differences) shall only take effect if IEEE1667 support is enabled (see G1667-2) and the device receives a Device Security Protocol[1] command with the Security Protocol field set to a value of EEh on that ComID only (i.e., IEEE1667). Once the device receives a Device Security Protocol command with the Security Protocol field set to a value of EEh on a given ComID, then the device shall behave as specified by the IEEE1667 Specification for that ComID until the next Controller Level Reset when the device shall switch back to TCG-compliant behavior for all ComIDs. This requirement does not apply to TCG Enterprise devices (see SEC-13).
SEC-58	If IEEE1667 support is enabled on a TCG Opal device and the host completes taking ownership of the device (refer to the TCG Storage Application Note: Encrypting Drives Compliant with Opal SSC) via Device Security Protocol[1] commands with the Security Protocol field set to a value of 01h to 06h (i.e., TCG), then the device shall go into a mode where Device Security Protocol commands with the Security Protocol field set to a value of EEh are prohibited and result in a TCG Data Protection Error, i.e., in this mode for any prohibited Device Security Protocol command: • a SATA device shall complete that command with error, with the STATUS field set to 0x51 and the ERROR field set to 0x04 in the Error Outputs. • a SAS device shall terminate that command with CHECK CONDITION status, with the sense key set to DATA PROTECT (i.e., 7h), and the additional sense code set to ACCESS DENIED - NO ACCESS RIGHTS (i.e., 20h/02h). If the device is in the mode where Device Security Protocol commands with the Security Protocol field set to a value of EEh are prohibited, then a Revert shall cause the device to switch back to a mode where Device Security Protocol commands with the Security Protocol field set to a value of EEh are allowed. This requirement does not apply to TCG Enterprise devices (see SEC-13).

Notes:

[1]For SEC-57 and SEC-58, a Device Security Protocol command is one of the following commands:

- SATA devices: TRUSTED NON-DATA, TRUSTED RECEIVE, TRUSTED RECEIVE DMA, TRUSTED SEND, or TRUSTED SEND DMA (see SATA-16).
- SAS devices: SECURITY PROTOCOL IN or SECURITY PROTOCOL OUT (see SAS-CMD-1).

13.2 Secure Boot

The device shall support Secure Boot. There are two fundamental things to address for secure boot:

- Secure boot rooted in hardware.
- Core Root of Trust Measurement.

The vendor should follow the recommendations in the *TCG Publication for Hardware Requirements for a Device Identifier Composition Engine* and the guidelines in NIST SP 800-193. DICE coupled with *TCG Implicit Identity Based Device Attestation* and Source for *RIoT Reference Architecture* can help implement Cryptographic Identity with implicit attestation.

Requirement ID	Description
SBT-1	The device shall comply with the *FIPS 186-5 Digital Signature Standard (DSS)* or later and the *OCP Hardware Secure Boot V1.0*.
SBT-2	For Core Root of Trust measurement, each device shall implement a Cryptographic Device Identity.
SBT-3	The *TCG Hardware Requirements for a Device Identifier Composition Engine V1.0 R0.91 or later* standard or an alternative hardware based cryptographic identity shall be implemented.
SBT-4	The device shall implement at least one of the Digital Signature PQC algorithms as specified in the *Commercial National Security Algorithm Suite 2.0* no later than 2025.
SBT-5	Secure boot flow shall be immutable for exploitation and use immutable public keys.

13.3 SPDM Storage Binding

The next version of this specification is expected to require SAS and SATA support for SPDM via the DMTF SPDM Storage Binding specification (DSP0286).

13.4 TCG Implementation Requirements

Devices that support TCG functionality (i.e., TCG devices) shall implement the specifications in accordance with the additional requirements in this section.

Feature/Parameters		Opal Value	Enterprise Value
SSC Feature Descriptor		Opal 2 (0203h)	Enterprise (0x0100)
TCG Storage Architecture Core Specification		Version 2.01, Revision 1.00	Version 1.0, Revision 0.9 - Draft
SID	# of SIDs	1 (See TCG-FEATURE-1)	1 (See TCG-ENT-FEAT-1)
	TryLimit	10 (See TCG-AUTH-3)	
	Persistence	FALSE (See TCG-AUTH-5)	
Number of Locking SP	Admin authorities	4 (min) (See TCG-LOCKING-6)	N/A
	User authorities	8 (min) (See TCG-LOCKING-2)	N/A
	EraseMaster authority	N/A	1 (See TCG-LOCKING-6)
	BandMaster authorities	N/A	4 (min) (See TCG-LOCKING-2)
	Global	1	1

Feature/Parameters		Opal Value	Enterprise Value
Number of Locking Ranges		(See TCG-FEATURE-1)	(See TCG-ENT-FEAT-1)
	Additional	8 (min) (See TCG-LOCKING-1)	3 (min) (See TCG-LOCKING-1)
DataStore size		10 MiB (min) (See TCG-FEATURE-1)	1kiB (min) (See TCG-ENT-FEAT-1)
MBR Table		Mandatory	
Configurable Access Control		Mandatory (See TCG-FEATURE-1)	N/A
Life Cycle support		Mandatory (See TCG-FEATURE-1)	N/A
Repurpose (Revert/ Revert SP)		Mandatory (See TCG-FEATURE-1)	Mandatory[1] (See TCG-ENT-FEAT-3)
Shadow MBR		Mandatory (and disabled by default) (See TCG-FEATURE-8)	Prohibited (See TCG-ENT-FEAT-8)
Feature Sets	Single User Mode	Mandatory (See TCG-FEATURE-2)	N/A
	PSK Secure Messaging	Optional (See TCG-FEATURE-9)	Optional[1] (See TCG-ENT-FEAT-9)
	PSID	Mandatory (See TCG-FEATURE-1)	Mandatory[1] (See TCG-ENT-FEAT-2)
	Block SID	Mandatory (See TCG-FEATURE-1)	Mandatory[1] (See TCG-ENT-FEAT-4)
	Additional Data Store Tables	Mandatory (See TCG-FEATURE-7)	N/A

Notes:

[1]This TCG Enterprise requirement is based on the applicable TCG Opal specification.

13.4.1 TCG Version and Features

13.4.1.1 TCG Opal Version and Features

TCG version and feature requirements for devices that support TCG Opal (see DP-CFG-2) are listed in the following table:

Requirement ID	Description
TCG-FEATURE-1	TCG Opal devices shall support TCG Storage Security Subsystem Class: Opal Version 2.02, Revision 1.0 or later, and all mandatory feature sets (i.e., the Additional DataStore, Block SID Authentication, and PSID feature sets). See Section 13.4.8 TCG PSID Feature Set for additional PSID requirements.
TCG-FEATURE-2	TCG Opal devices shall support TCG Opal Single User Mode feature set Version 1.00, Revision 2.00 or later.

Requirement ID	Description
TCG-FEATURE-3	TCG Opal SAS devices shall support TCG Configurable Locking for NVMe Namespaces and SCSI LUNs feature set Version 1.02, Revision 1.16 or later and shall support the Namespace Global Range Locking object. The Namespace Non-Global Range Locking object may be supported. This requirement does not apply to TCG Opal SATA devices.
TCG-FEATURE-4	Obsolete (see TCG-FEATURE-1).
TCG-FEATURE-5	The current TCG feature set configuration settings, if any, of a TCG Opal device shall persist through a download and activation of firmware code (e.g., TPer Lock State, Single User Mode configuration, no C_PIN Reset). This is required regardless of the "default setting state" of the new code being downloaded.
TCG-FEATURE-6	TCG Opal devices shall support the Data Removal Mechanism Feature Descriptor (Feature Code 0404h). The Data Removal Time Format bit shall be set to 0 to indicate the Data Removal Time value is stored in (value x 2) seconds. The Data Removal Time shall be less than or equal to 10 seconds per CTO-1.
TCG-FEATURE-7	TCG Opal devices shall support at least 8 Additional DataStore Tables, each with a minimum size of 128 KiB. The required size of these tables is in addition to the 10 MiB minimum required for the DataStore Table.
TCG-FEATURE-8	TCG Opal devices shall disable MBR Shadowing by default (i.e., the default value of the Enable column of the MBR_Control table shall be False).
TCG-FEATURE-9	TCG Opal devices may support TCG Opal PSK Secure Messaging feature set Version 1.00 Revision 1.00 or later.

13.4.1.2 TCG Enterprise Version and Features

TCG version and feature requirements for devices that support TCG Enterprise (see DP-CFG-2) are listed in the following table:

Requirement ID	Description
TCG-ENT-FEAT-1	TCG Enterprise devices shall support TCG Storage Security Subsystem Class: Enterprise Version 1.01, Revision 1.0 or later, and all mandatory features (e.g., Locking).
TCG- ENT-FEAT-2	TCG Enterprise devices shall support TCG Storage Opal SSC Feature Set: PSID Version 1.00 Revision 1.00 or later. This requirement is in addition to TCG Enterprise SSC support (see TCG-ENT-FEAT-1). See Section 13.4.8 TCG PSID Feature Set for additional PSID requirements.
TCG- ENT-FEAT-3	TCG Enterprise devices shall support the Revert and RevertSP methods specified in Section 5.1.2 and Section 5.1.3 of TCG Storage Security Subsystem Class: Opal Version 2.02, Revision 1.0 or later. This requirement is in addition to TCG Enterprise SSC support (see TCG-ENT-FEAT-1).
TCG- ENT-FEAT-4	TCG Enterprise devices shall support TCG Storage Opal SSC Feature Set: Block SID Version 1.01 Revision 1.00 or later. This requirement is in addition to TCG Enterprise SSC support (see TCG-ENT-FEAT-1).
TCG- ENT-FEAT-5	The current TCG feature set configuration settings, if any, of a TCG Enterprise device shall persist through a download and activation of firmware code (e.g., TPer Lock State, no C_PIN Reset). This is required regardless of the "default setting state" of the new code being downloaded.

Requirement ID	Description
TCG- ENT-FEAT-6	Reserved for future use.
TCG- ENT-FEAT-7	Reserved for future use.
TCG- ENT-FEAT-8	TCG Enterprise devices shall not support MBR Shadowing (i.e., the MBR_Control table shall not be supported).
TCG-ENT-FEAT-9	TCG Enterprise devices may support TCG Opal PSK Secure Messaging feature set Version 1.00 Revision 1.00 or later. This optional support is in addition to TCG Enterprise SSC support (see TCG-ENT-FEAT-1).

13.4.2 TCG Communication Layer

TCG communication layer requirements for both TCG Opal devices and TCG Enterprise devices are listed in the following table:

Requirement ID	Description
TCG-COMM-1	The device shall support ComID management, and the Level 0 Discovery TPer Feature Descriptor shall report ComID Mgmt supported.
TCG-COMM-2	The device shall support at least 2 Static ComIDs.
TCG-COMM-3	The device should support at least 2 Dynamic ComIDs.
TCG-COMM-4	The MaxComIDTime value shall be at least 120 seconds.
TCG-COMM-5	The MaxComIDTime value shall be greater than the DefSessionTimeout value.
TCG-COMM-6	The MaxComPacketSize value shall be at least 2048 bytes. The MaxResponseComPacketSize value shall be at least 2048 bytes.
TCG-COMM-7	The MaxPacketSize value shall be at least 2028 bytes.
TCG-COMM-8	The MaxIndTokenSize value shall be at least 1992 bytes.

13.4.3 TCG Session Layer

TCG session layer requirements for both TCG Opal devices and TCG Enterprise devices are listed in the following table:

Requirement ID	Description
TCG-SES-1	A single Read-Write Session shall be supported per SP and shall be indicated by MaxSessions. It is not required to support an active RW session on the AdminSP at the same time as sessions on other SPs, because this would violate the TCG Storage Architecture Core Specification.
TCG-SES-2	At least 2 Read-Only Sessions shall be supported per SP and shall be indicated by MaxReadSessions. SET methods and all other TCG commands that modify the TCG configuration state shall be rejected during a Read Only session.
TCG-SES-3	DefSessionTimeout and MinSessionTimeout shall be included in the Properties Method response from the device.
TCG-SES-4	For a TCG Opal device, the DefSessionTimeout value shall be 90,000 ms.

Requirement ID	Description
	For a TCG Enterprise device, the DefSessionTimeout value shall be 0. This value indicates that there is no limit.
TCG-SES-5	The MaxSessionTimeout value shall be 0. This value indicates that there is no limit.
TCG-SES-6	The MinSessionTimeout value should be 500 ms.
TCG-SES-7	The SessionTimeout parameter of the StartSession method call shall be used to configure the session timeout.
TCG-SES-8	Support for setting the SPSessionTimeout column of the SPInfo table is recommended.

13.4.4 TCG Transactions

TCG transactions requirements for both TCG Opal devices and TCG Enterprise devices are listed in the following table:

Requirement ID	Description
TCG-TRAN-1	TCG devices shall support TCG Transactions.
TCG-TRAN-2	TCG Nested Transactions may be supported. If TCG Nested Transactions are supported, all supported transactions including any nested transactions shall be reflected in the MaxTransactionLimit property.

13.4.5 TCG Authentication

TCG authentication requirements for both TCG Opal devices and TCG Enterprise devices are listed in the following table:

Requirement ID	Description
TCG-AUTH-1	TCG devices shall enforce a minimum password length (C-PIN PinLength of 16) of 16 bytes for all TCG authorities.
TCG-AUTH-2	TCG devices shall allow a password length (C-PIN PinLength of 32) of at least 32 bytes for all TCG authorities.
TCG-AUTH-3	TCG devices shall set the default TryLimit to 10 for all TCG authorities. The value of the Tries column shall only be cleared to 0 for the mandatory reasons specified in the TCG Storage Architecture Core Specification (e.g., do not clear the Tries column via the Set method, do clear it upon successful authentication or power cycle).
TCG-AUTH-4	Authentication of each C_PIN shall be delayed limiting how fast the next authentication of that C_PIN can occur: • The delay shall be at least 100 ms and should not exceed 250 ms and I/O shall not be blocked during this time. • The delay shall apply regardless of whether the authentication is successful. • The device manufacturer shall supply documentation with the implemented delay value. • The time limit should be enforced cryptographically (e.g., by tuning the number of PBKDF iterations to reach the target time).
TCG-AUTH-5	The Persistence column shall be set to false for all C_PIN rows.
TCG-AUTH-6	The MaxAuthentications value shall be at least 2.

13.4.6 TCG Crypto

TCG crypto requirements for both TCG Opal devices and TCG Enterprise devices are listed in the following table:

Requirement ID	Description
TCG-CRYPTO-1	For a TCG Opal device, if the TCG Locking SP has been activated and the host has supplied an authority password, then the device shall use that password as an added source of randomness for its Deterministic Random Number Generator (DRNG). For a TCG Enterprise device, if the TCG Locking SP has been enabled (i.e., the LockingEnabled bit is set to 1b in the Level 0 Discovery Locking Feature Descriptor), and the host has supplied an authority password, then the device shall use that password as an added source of randomness for its Deterministic Random Number Generator (DRNG). For example, the device may use KDF (PBKDF2) on the user provided locking range password to generate an output of equal length to the required locking range key length plus DRNG seed length. This output can then be utilized in parts, one part as the locking range key and the other part as an additional input to the DRNG. If the DRNG implementation does not allow the use of additional input, then an alternative option would be to XOR this output (e.g., from PBKDF2 operation) with initial DRNG seed.

13.4.7 TCG Locking Security Provider

TCG Locking SP requirements for both TCG Opal devices and TCG Enterprise devices (unless otherwise stated) are listed in the following table:

Requirement ID	Description
TCG-LOCKING-1	The number of Additional Locking Ranges supported (in addition to the Global Locking Range) shall be: • at least 8 for a TCG Opal device; and • at least 3 for a TCG Enterprise device. The MaxRanges field in the LockingInfo Table shall report the number of Additional Locking Ranges supported. This includes ranges that do not have a row in the Locking Table and includes ranges for which no locking is enabled.
TCG-LOCKING-2	The LockingSP of a TCG Opal device shall support at least one User authority per Locking Range. The LockingSP of a TCG Enterprise device shall support the BandMaster0 authority for the Global Locking Range and at least one additional BandMaster authority for each Additional Locking Range. Each BandMaster authority shall be initialized with Enabled = True.
TCG-LOCKING-3	The K_AES_256 Table shall be supported.
TCG-LOCKING-4	The K_AES_256 Table's Key columns shall not be readable by the host or be accessible outside the device in any manner.
TCG-LOCKING-5	Any attempt to remove the Power Cycle reset type from a LockOnReset column set value shall be rejected. The LockOnReset column value of "[ ]" (i.e., the empty set) is prohibited.
TCG-LOCKING-6	The LockingSP of a TCG Opal device shall support at least 4 Admin authorities. The Locking SP of a TCG Enterprise device shall support the EraseMaster authority.

Requirement ID	Description
TCG-LOCKING-7	The LockingSP shall support Media Encryption. The Level 0 Discovery Locking Feature Descriptor shall report that Media Encryption is supported. The EncryptSupport field shall be set to 1 in the LockingInfo Table.

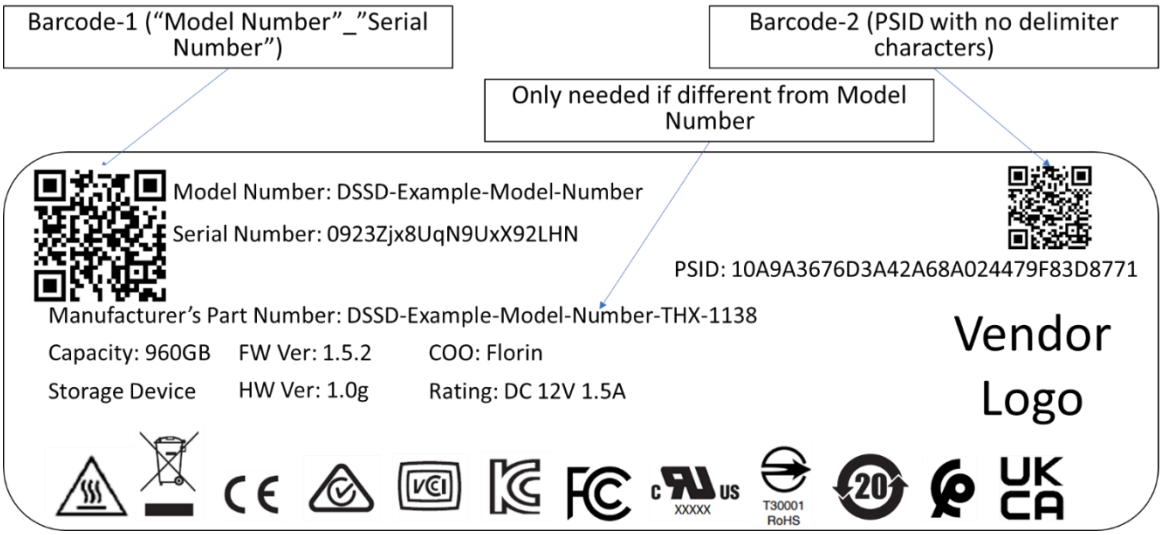
13.4.8 TCG PSID Feature Set

PSID requirements listed in the following table apply to the TCG Opal PSID Feature Set, which both TCG Opal devices and TCG Enterprise devices are required to support:

Requirement ID	Description
TCG-PSID-1	The PSID shall not be deterministic, shall comprise at least 160 bits of entropy. The entropy shall be unique for each device. The PSID shall not be derived from any information that is available without physical access to the device. For example, the PSID shall not be the same as the MSID, the PSID shall not be derived from the device serial number or the date of manufacture because these can be read via the device interface or obtained from other documentation. The PSID shall not be the serial number of the device hashed with a secret string shared with any other device or among all devices, because the entropy shall be unique for each device. The presence of the PSID on the device label does not violate this requirement.
TCG-PSID-2	The PSID shall consist of 32 ASCII characters. The allowed ASCII characters are upper case letters, lower case letters and the digits 0-9. The characters 'I', 'l', 'O' and 'o' (i.e., 49h, 6Ch, 4Fh and 6Fh) and the digits '0' and '1' (i.e., 30h and 31h) shall not be used to limit possible visual confusion.
TCG-PSID-3	The generation method for the PSID shall be documented and provided to the customer.
TCG-PSID-4	After the PSID is provisioned in manufacturing, the PSID shall not be persistently stored in a form that makes retrieval or reconstruction of the PSID by firmware possible, either on the device or outside the device (e.g., it is acceptable to store the PSID on the device hashed via a cryptographic hash or KDF such that the hashed PSID can only be used to verify a PSID input from the host). The presence of the PSID on the device label does not violate this requirement.
TCG-PSID-5	Once manufacturing is complete, the PSID shall not be stored anywhere other than on the physical device label.

14 Labeling

The following sample label is meant to be used to refer to the label requirements in Section 14.1 Top Cover Label Requirements. It is not a model label and any markings on it are informative only.



14.1 Top Cover Label Requirements

Requirement ID	Description

The following fields specify the required information that shall be placed on the label:

Item	Format	Text Required	Barcode Required	Barcode Type
Barcode-1	'Model Number' 'Underscore' 'Serial Number' \n.	No	Yes	2d
Model Number	See LABL-11 (Model Number must match).	Yes	No	N/A
Serial Number	See LABL-12 (Serial Number must match), LABL-15 (certification logos), and LABL-17 (Serial Number format).	Yes	No	N/A
Manufacturer's Part Number	Number used for ordering.	Yes, if different from Model Number	No	N/A
Capacity	Number of GB or TB.	Yes	No	N/A
STORAGE DEVICE	Text shall be "STORAGE DEVICE".	Yes	No	N/A
PSID	TCG Storage Security Subsystem Class: Opal Specification	Yes	No	N/A
Barcode-2	'PSID' \n	No	Yes	2d
Firmware Name & Revision		Yes	No	N/A

LABL-1

<table>
<tr><th>Requirement ID</th><th>Description</th></tr>
<tr><td></td><td>
<table>
<tr><td>Regulatory Mark</td><td>See LABL-15 (certification logos).</td><td>No</td><td>N/A</td></tr>
<tr><td>Country Certification Numbers</td><td>See LABL-15 (certification logos).</td><td>No</td><td>N/A</td></tr>
<tr><td>Certification Logos</td><td>See LABL-15 (certification logos).</td><td>No</td><td>N/A</td></tr>
<tr><td>RoHS/ Green</td><td>See LABL-15 (certification logos).</td><td>No</td><td>N/A</td></tr>
</table>
</td></tr>
<tr><td>LABL-2</td><td>The Model Name on the shipping label shall match the model’s name used during qualification.</td></tr>
<tr><td>LABL-3</td><td>The minimum font size shall be 3 points, and the typical size shall be 6 points.</td></tr>
<tr><td>LABL-4</td><td>For the Capacity field, if there are space constraints, the manufacturer may remove “Capacity:” and just show “XXXGB” or “XXXTB”.</td></tr>
<tr><td>LABL-5</td><td>To distinguish Model Number and Serial Number, Barcode-1 shall have an underscore “_” between the Model Number portion and the Serial Number portion.

Example:
Model Number: OCP-Example-Model-Number
Model Number: DSSD-Example-Model-Number
Serial Number: 0923Zjx8UqN9UxX92LHN
Barcod-1 Readout: DSSD-Example-Model-Number_0923Zjx8UqN9UxX92LHN</td></tr>
<tr><td>LABL-6</td><td>There shall be a line with the text “STORAGE DEVICE”.</td></tr>
<tr><td>LABL-7</td><td>The following fields are optional information that can be placed on the label at the discretion of the device maker. Placement is also at the device maker’s discretion if such information does not interfere with the mandatory information above. No additional barcode shall be present.
<table>
<tr><th>Item</th><th>Format</th><th>Text Required</th><th>Barcode Required</th><th>Barcode Type</th></tr>
<tr><td>Processor Code (BA)</td><td></td><td>Optional</td><td>No</td><td>N/A</td></tr>
<tr><td>Maker Logo</td><td></td><td>Optional</td><td>No</td><td>N/A</td></tr>
<tr><td>Rated Voltage & Current</td><td></td><td>Optional</td><td>No</td><td>N/A</td></tr>
<tr><td>Production Date</td><td>DDMMYYYY: DD (Date), MM (Month), YYYY (Year)</td><td>Optional</td><td>No</td><td>N/A</td></tr>
<tr><td>Weekly Code</td><td>YYWW: YY (Year), WW (Week)</td><td>Optional</td><td>No</td><td>N/A</td></tr>
<tr><td>Warranty VOID IF REMOVED</td><td></td><td>Optional</td><td>No</td><td>N/A</td></tr>
<tr><td>Makers Own Label Material Number</td><td></td><td>Optional</td><td>No</td><td>N/A</td></tr>
</table>
</td></tr>
</table>

<table>
<tr><th>Requirement ID</th><th>Description</th></tr>
<tr><td></td><td>
<table>
<tr><td>Website, Company Address</td><td></td><td>Optional</td><td>No</td><td>N/A</td></tr>
<tr><td>HDD</td><td></td><td>Optional</td><td>No</td><td>N/A</td></tr>
<tr><td>Product Series Name</td><td></td><td>Optional</td><td>No</td><td>N/A</td></tr>
<tr><td>SA: Stocking ID Value used within manufacturing</td><td></td><td>Optional</td><td>No</td><td>N/A</td></tr>
<tr><td>PBA: Physical Board Address (identifies the physical configuration of the device)</td><td></td><td>Optional</td><td>No</td><td>N/A</td></tr>
<tr><td>WWN: World Wide Number (unique for each device)</td><td></td><td>Yes</td><td>No</td><td>N/A</td></tr>
</table>
</td></tr>
<tr><td>LABL-8</td><td>To ensure that datacenter operations personnel can quickly and easily identify devices that have been ticketed for field replacement, it is mandatory to have the proper identifying fields on the label(s), in the format specified in in this table.</td></tr>
<tr><td>LABL-9</td><td>The label shall not degrade over the standard device lifetime under standard operating conditions.</td></tr>
<tr><td>LABL-10</td><td>For each form factor, the label shall be placed as specified below:
2.5” and 3.5”: the label shall be placed on the top side of the device.</td></tr>
<tr><td>LABL-11</td><td>The Model Number in Barcode-1, the Model Number printed on the label and the Model Number returned in Identify Device (SATA), or Inquiry (SAS) shall all match and shall have no leading spaces.</td></tr>
<tr><td>LABL-12</td><td>The Serial Number in Barcode-1, the Serial Number printed on the top and side labels and the Serial Number returned in Identify Device (SATA) or Inquiry (SAS) shall all match and shall have no leading spaces.</td></tr>
<tr><td>LABL-13</td><td>The Hardware Revision printed on the label and returned by STATI-COM-42 shall match.</td></tr>
<tr><td>LABL-14</td><td>All other electronically readable information shall also match their counterparts printed on the label.</td></tr>
<tr><td>LABL-15</td><td>The following certification logos and their corresponding certifications are required:
<table>
<tr><th>Regulatory Mark/Text</th><th>Description</th></tr>
<tr><td>Regulatory Model Number</td><td>Unique regulatory Identifier.</td></tr>
<tr><td>Made in XXXX</td><td>Country of Origin.</td></tr>
</table>
</td></tr>
</table>

<table>
<tr><th>Requirement ID</th><th>Description</th></tr>
<tr><td></td><td>
<table>
<tr><td>Manufacturer or Brand name</td><td>Identification of the party responsible for placing the device into the market.</td></tr>
<tr><td>Address of the Manufacturer</td><td>Required for devices with the CE mark or UKCA mark.</td></tr>
<tr><td>Date of Manufacture</td><td>Not needed if embedded in the Serial Number.</td></tr>
<tr><td>Serial Number</td><td>Alpha-Numeric, 12-20 digits.</td></tr>
<tr><td>  </td><td>[Europe] Compliance with EU WEEE directive 2010/19/EU.</td></tr>
<tr><td>  </td><td>[Europe] Compliance with EU EMC directive 2014/30/EU and RoHS directive 2011/65/EU.</td></tr>
<tr><td>  </td><td>[Australia, New Zealand] Compliance with requirements of the relevant Australian ACMA Standards, under the Radiocommunications Act 1992 and the Telecommunications Act 1997.</td></tr>
<tr><td>  </td><td>[Japan] Compliance with Japan VCCI requirements.</td></tr>
<tr><td> 
XXXX-XX-XX</td><td>[Korea] Compliance with requirements of the Radio Research Laboratory Ministry of Information and Communication Republic of Korea.</td></tr>
<tr><td>CAN ICES-3(*)/NMB-3(*)</td><td>[Canada] Compliance with Canada standard ICES 003. Where * is either A or B.</td></tr>
<tr><td>  </td><td>[USA] Optional. Compliance with United States Federal Communications Commission requirement.</td></tr>
<tr><td> 
XXXXX</td><td>[USA] Compliance with UL standards and Canadian Safety Standards.</td></tr>
<tr><td> 
XXXXX
RoHS</td><td>[Taiwan] Compliance with Taiwan EMC and RoHS.</td></tr>
<tr><td>  </td><td>[China] Compliance with Chinese environmental requirements. The number inside the circle is usually 10 or 20.</td></tr>
<tr><td>  </td><td>[Morocco] Compliance with Moroccan EMC standards.</td></tr>
<tr><td>  </td><td>[United Kingdom] Mandatory after 1/1/2022. Guidance to UKCA marking.</td></tr>
</table>
</td></tr>
<tr><td>LABL-16</td><td>If the surface of any component or casing will reach a temperature of 70°C (158°F) or greater the following warning logo shall be either printed on the label or placed separately on the device:</td></tr>
</table>

Requirement ID	Description
	 In addition, instructional safeguards shall be provided in accordance with IEC 62368-1 3rd Edition or later.
LABL-17	The format of the Serial Number shall have no leading spaces (e.g., 1234567890123456).
LABL-18	The Model Number shall have no leading spaces.
LABL-19	Barcodes shall be printed using Datamatrix ECC200 or Model 2 QR code only.
LABL-20	QR codes shall use a minimum of ECC Level M (15%).
LABL-21	The density of the barcode shall be 10 mil or larger.
LABL-22	The label shall only be printed on Polyester or Plastic labels using a Wax/Resin ribbon.
LABL-23	The PSID shall be printed on the label in its direct 32-character alphanumeric representation without any ancillary delimiting characters (e.g., underscore, dash, backslash, forward slash, etc.) and shall exactly match the readout of Barcode-2 (see the label example at the top of this section).

14.2 Edge Label Requirements

Requirement ID	Description
LBLE-1	The edge label shall contain the QR code for HDD serial number.
LBLE-2	The edge label shall be placed within and towards the center of the label area defined in the diagram below: 

15 Compliance

15.1 ROHS Compliance

Requirement ID	Description
ROHS-1	The Supplier shall provide component-level reporting on the use of listed materials by concentration (ppm) for all homogenous materials.

15.2 ESD Compliance

Requirement ID	Description
ESD-1	The device shall support an ESD Human Body Model (HBM) immunity level in accordance with IEC-61000-4-2.